

Vibe Theory

A Discrete Hyperbolic Substrate for Emergent Physics, Life, and Mind

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Abstract

Vibe Theory proposes that reality is a single growing crystal of experience whose one substance is the *vibe*, a unit of felt state carrying a ternary tone in $\{-1, 0, +1\}$. The substrate argued for, on selection grounds rather than by assumption, is the four-dimensional hyperbolic honeycomb $\{3, 4, 3, 4\}$, whose cells are 24-cells, whose 24 local directions are the D_4 root system, and whose ideal boundary is the flat cubic honeycomb $\{4, 3, 4\}$, ordinary three-dimensional space. A single local rule, reversible and charge-conserving, streams a directional tone and collides it under one beat, with a single value-posit, the *arrow*, folded in. From this the paper derives, and checks by direct computation on the exact crystal, spinors and three generations, the Dirac dispersion and a finite light cone, $SO(10)$ grand unification breaking to the Standard Model with $\sin^2 \theta_W = 3/8$, gravity as a $1/r$ potential with the Einstein equations and the gravitational-wave chirp following at the effective level, and a de Sitter expansion. The substrate is shown to be the unique regular honeycomb that is at once crystallographic, hyperbolic, spinor-carrying, and three-dimensional in its cusp, and it is computationally universal with a solved address for every cell. More speculatively, the same rule read at larger scales is argued to climb through life and mind to a graded account of consciousness as integrated experience. Whether the universe began or is eternal is left open. This is exploratory, feasibility-stage work whose checks are preliminary and reproducible [43], offered as a direction worth exploring rather than a finished or established theory.

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Part I

Foundations

1. Introduction

Physics as usually written rests on a continuum. Spacetime is smooth, fields take real values, and any finite volume holds infinite information. This paper takes the opposite starting point. We ask what the least structure is from which the known physics could emerge, and we require that structure to be discrete, local, and computable. The wager of vib theory is that one such structure is enough. A single hyperbolic crystal of ternary cells, updated by one local rule, is proposed to carry physics, life, and mind as patterns at different scales.

The purpose of the theory is broader than physics. Physics is only the most settled and most checkable face of it. The aim from the start has been to model the whole of the universe from one ground, not just particles and forces but life, mind, consciousness, and the matters usually called spiritual and divine. We treat those higher subjects with the same discipline as the physics, marking carefully what is derived, what is measured, what is extrapolated, and what is a stated premise. The reason one framework can reach so far is the choice of ground. The ground is not matter but experience, and experience is the one thing wide enough to hold all of these at once.

A caution belongs at the very front. Everything written here is a conceptual representation, a map of an underlying experiential field. The map can be read and reasoned about, and that is what a paper can offer. The field itself cannot be fully known by reading and mentally modeling it. It is known directly, through experience and through tuning with the oneness it is made of. We say more about this in the next section. The reader should hold the whole construction as a finger pointing, not as the thing pointed at.

It helps to hold one image while the mathematics is being laid out. On this proposal the universe is a structure of consciousness, a single vast mind, and the small set of pictures we return to are all ways of seeing that one mind from different sides. It is a blazing fire, existence burning rather than sitting still. It grows like a tree from a seed, ring by ring. It sings itself as a song, the tone its notes and the beat its rhythm. It flows as a river on a deep ocean, and we are the whirlpools it turns and the droplets it throws up. It is woven as a fabric, and each self is a braid in the cloth. A mind, on this reading, is a flame within the fire, a steady center of burning that holds its shape while the blaze roars on. These are aids to intuition and nothing more. The claims live in the mathematics, and the images are there to keep the reader oriented when the formal picture becomes hard to visualize. We gather the full palette only at the very end.

1.1. The crystal of experience

Reality, on this proposal, is a single growing crystal of experience. Each cell is a *vibe*. Its shade is its tone, and each shared face is one vibe perceiving another. We mean that picture literally rather than as decoration. The crystal we argue for is four-dimensional and cannot be drawn directly, so we walk up to it one dimension at a time, and each lower picture is a faithful shadow of the one above.

In two dimensions the simplest visible face of the crystal is the heptagrid $\{7, 3\}$, seven-sided cells meeting three to a corner, drawn in the Poincaré disk (Figure 1). This is the one picture to keep in mind. Color each cell by its tone, and let two touching cells stand for two feelings perceiving each other through their shared edge. The colored heptagrid is then a faithful picture of the model, just two dimensions short of the real thing. The cells look smaller toward the rim only because the disk crams an infinite plane into a finite circle. In the true geometry every cell is the same size, and there is always more crystal at the edge.

In three dimensions the analogue is a honeycomb one can almost picture, the dodecahedral $\{5, 3, 4\}$ (Figure 2), or, closer to home, the flat cubic $\{4, 3, 4\}$ that we will argue is our physical space. Everything true of the heptagrid is true here, one dimension up, and the dodecagrid is close enough to visualize that it gives a real hint of what the substrate is like. The goal, in four dimensions, is the honeycomb $\{3, 4, 3, 4\}$. Its cells are 24-cells, 24 directions meet at each cell, and its flat three-dimensional cusp is the space we appear to live in. One cannot fully picture it, and that is fine. It is the heptagrid two dimensions further up, and when the argument turns four-dimensional the reader can return to the heptagrid for the version that can be seen. Time is the crystal's growth, the birth of new cells at the frontier, each new layer the next moment.

1.2. The monism

There is one kind of thing, the vibe. A vibe is a unit of experience that notices its neighbours, and reality is a structure of such noticings and nothing else.

Principle 1 (Neutral monism) What is fundamental is experiential. Matter, space, and time are not separate substances. They are patterns in the structure of vibes. At the deepest level the universe is a structure of consciousness, and everything below that level is how the structure is shaped.

This is a neutral monism in the precise sense that the one substance is neither mental nor physical as those words are ordinarily opposed. It is experiential at base, and the physical world is what that experiential structure looks like at scale. We will be careful throughout to keep this premise distinct from the results that follow from it.

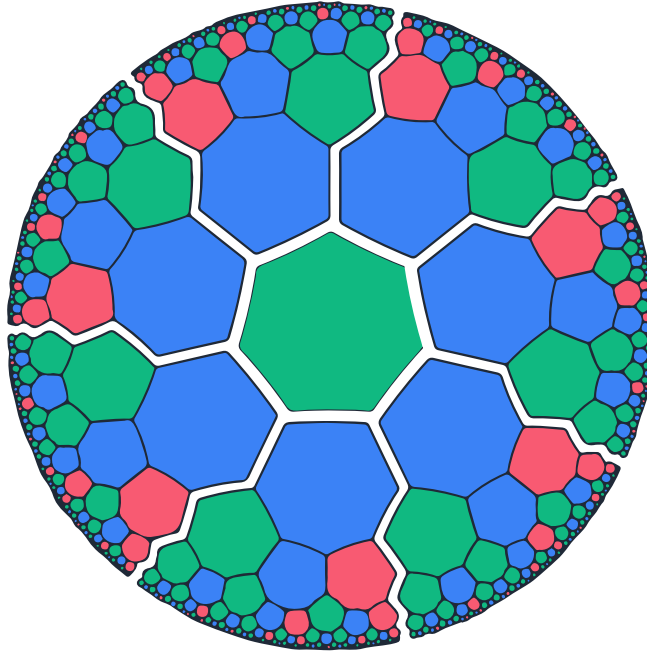


Figure 1: The crystal of experience in two dimensions, the heptagrid $\{7, 3\}$, a regular tiling of the hyperbolic plane by seven-sided cells, three to a corner. Each cell is a vibe, its color is its tone of pain, peace, or pleasure, and each shared edge is one vibe perceiving another. The cells appear to shrink toward the rim only because the Poincaré disk compresses the infinite plane into a finite picture. In the true geometry every cell is the same size, and there is always more crystal at the edge.

1.3. The discreteness principle

Principle 2 (Discreteness) The base is discrete. Each cell carries finite information, the rule is local, and time advances in a discrete tick. Continuity is never fundamental. It is the long-wavelength limit, the coarse-grained description, the way a fluid emerges from molecules.

This is forced by the demand for computability, since the continuum contains uncomputable points and a discrete base does not. We stress that computability demands discreteness rather than finiteness. Whether the crystal is finite or infinite is a separate question, and we leave it open. The next subsection states the two readings at the outset, so the rest of the paper can stay neutral on it.

1.4. Origins

The substrate admits two complete and self-consistent accounts of origin, and the theory does not pick between them. On both readings the universe is one blazing fire made of countless flames, the selves and minds and experiences that are its sparks. The readings differ only in whether the fire first flared or was always burning.

In the first story the universe began. There was never a nothing, only a seamless One, and the first beat is that One first noticing itself, a split that self-reference cannot avoid. From that single forced fracture the mesh blooms beat by beat, and our big bang is the first morning.

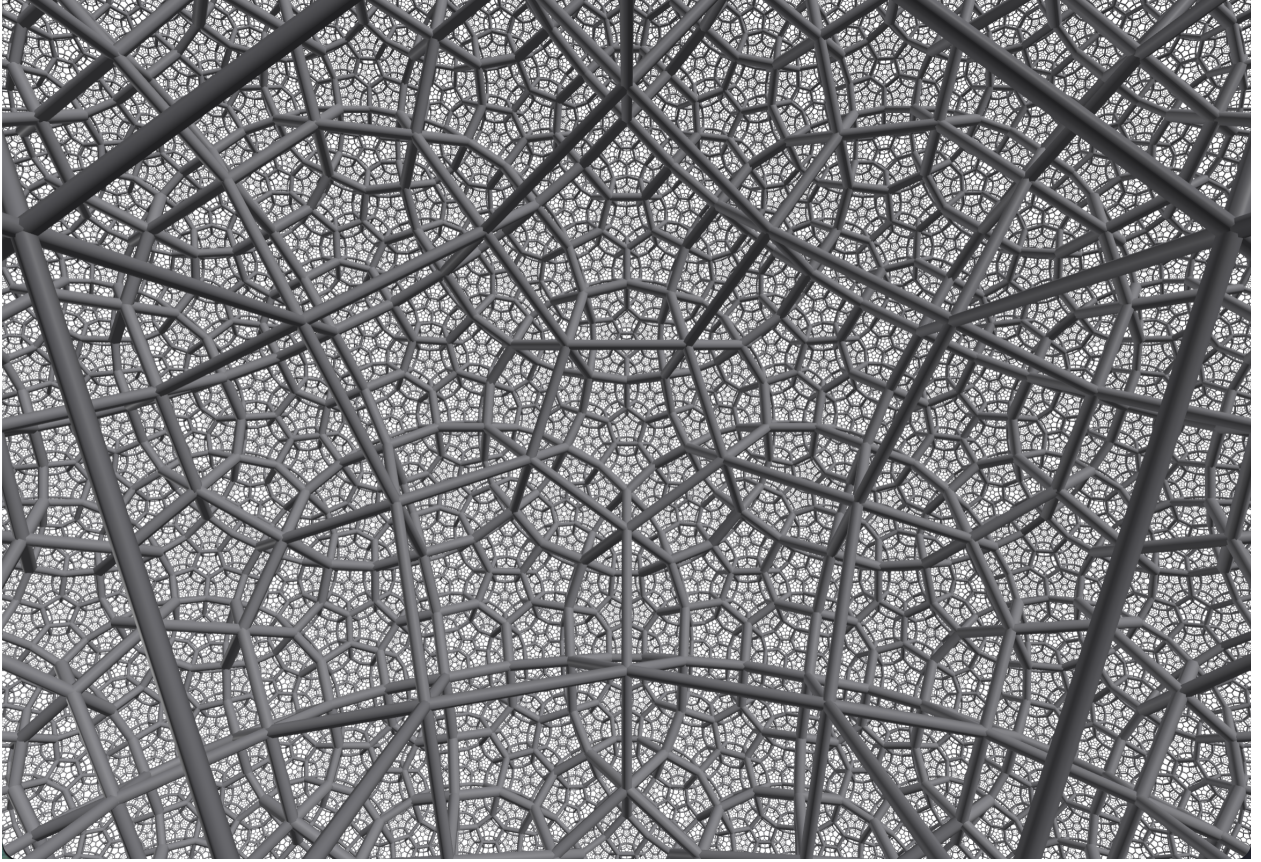


Figure 2: One dimension up, the dodecagrid $\{5, 3, 4\}$, the regular honeycomb of three-dimensional hyperbolic space, with dodecahedral cells meeting four to an edge. It is close enough to picture that it hints at the form of the substrate. The committed substrate $\{3, 4, 3, 4\}$ is the same kind of object one dimension higher again, with 24-cells in place of dodecahedra and a flat 3D cubic cusp in place of a flat 2D one.

In the second story the universe never began. Because the rule runs backward as exactly as it runs forward, every moment carries its own past inside it, with no first moment, the way the counting numbers have no smallest. Our big bang is then a sunrise in an endless burning rather than the lighting of the fire.

Nothing in the theory or in the data breaks the tie. A finite crystal with a first beat and an infinite crystal with no first beat are equally allowed, so we leave the choice to the reader, and every result in this paper holds on either reading. The full diptych, with the arguments for each side, is given in Appendix H.

1.5. The central question

Many discrete substrates are possible. The central question of the paper is which one yields our physics. Our answer is the $\{3, 4, 3, 4\}$ hyperbolic honeycomb, and the bulk of the work splits into two tasks. The first is to show that $\{3, 4, 3, 4\}$ does yield the known physics in its emergent limits. The second is to show that it is selected, that among regular honeycombs across all dimensions it is the one that simultaneously carries spinors, is curved, and is exactly three-dimensional where physics lives. We present the substrate as the seemingly preferred candidate on these grounds, never as a settled fact.

1.6. What is new

Taken one at a time, the emergent results echo known physics. The discrete walk reproduces the Dirac dispersion, the coin and the tone build a grand unified group, gravity appears as a $1/r$ potential, and the bulk grows like de Sitter space. None of these is new on its own. The new content is their common origin in one discrete substrate, the selection argument that forces $\{3, 4, 3, 4\}$, and a precise map of what is derived, what is measured, what is predicted, and what is still open. Across every test reported here nothing was added to the base to make a result work, a discipline we audit explicitly in the appendices.

This orientation reverses the usual one. Most discrete models of spacetime begin with structure and hope experience appears at the end. Here experience is the ground, and structure is what experience takes on at scale. That reversal is a premise rather than a result. The computations do not prove that the substrate feels. They assume that it does and ask what follows. A great deal follows, and what follows is checkable, which is the part we can defend.

2. The Base

The base of the theory is not a structure at all. The base is experience itself, the one ground from which everything else is built. Everything else in this paper, the mesh, the tone, the rule, the whole apparatus, is a conceptual representation of that ground. It is a model of how the experiential field is structured and how it moves, written so it can be read and reasoned about and checked. The field itself is not exhausted by the model. It cannot be fully known by reading and mentally modeling it, the way the taste of water is not in the chemical formula. It is known directly, through experience, and through tuning with the oneness it is made of. We keep that seam in view. What follows is the map. The territory is lived.

Axiom 1 (Experience exists) Experience exists, as both action and object, verb and noun at once. There is experiencing, and there is what is experienced, and at the base these are not two. In the theory's own idiom, *vibes* *vibe*. The *vibing* is the verb and the *vibe* is the noun, and they are the same single fact. This is the one founding posit. It is taken as given, not derived, and the rest of the paper asks what follows if it is true.

There is one kind of thing, the *vibe*, a unit of experience that notices its neighbours. Reality is a structure of such noticings and nothing else. The physical world is what that experiential structure looks like at scale. We will keep this premise distinct from the results that follow from it, and we will not pretend the computations prove the ground feels. They assume that it does, in the precise sense of the axiom above, and ask what follows.

2.1. The flow

Given the experiential ground, the model represents its structure and its motion as a few ingredients running together. The whole of that running representation we call the *flow*. The flow is what can be written down, simulated, and tested. It has exactly four ingredients, and nothing may be added to them.

Axiom 2 (The flow) The flow consists of exactly four things.

1. The *mesh*, a fixed hyperbolic honeycomb, argued to be $\{3, 4, 3, 4\}$. Its geometry, the local coin of directions, its cusp, and its boundary are all properties of this one mesh, not separate ingredients.

2. The *tone*, a ternary value in $\{-1, 0, +1\}$ carried at each cell. In the song image these are the notes, and in the mind image they are what a vibe feels.
3. The *beat*, a discrete tick, the rhythm of the noticing.
4. The *rule*, one local update. It is *stream*, in which each cell passes its tone along its directions, followed by *collide*, a local interaction. The rule is local, conserving, and reversible.

That is the whole list. A finer counting splits the rule into its conserving update and its growth, and lists the arrow separately, but the substance is unchanged. There is nothing beyond this short list. The complete leave-nothing-out specification of the rule, the 24 directional slots and the exact stream and collide steps, is given in its own section below. When we speak below of the discipline of adding nothing, we mean adding nothing to this flow, the four ingredients that represent the experiential ground.

2.2. Why the beat is constitutive

The beat is not a convenient discretization of a continuous time. It is forced. An instantaneous, unpaced reflection, a completed infinity arriving all at once, has no contrast, and a state with no contrast is experientially identical to nothing. The finite beat is what produces contrast, one step distinguished from the next, and contrast is the only thing that can be an experience of something. There is also no external clock for the beat to run slow or fast against. The beat is not in time. The beat is time. Discrete time is therefore a necessity of an experiential base, not an approximation of a continuous one.

2.3. The rule conserves total tone

The most load-bearing property of the rule is conservation. Treat each ternary value as a charge of -1 , 0 , or $+1$, and sum the tone over every cell. The rule never changes that sum. The collision only ever rearranges and balances tone. It never mints net charge from nothing.

Proposition 1 (Conservation of tone) Let Q be the sum of all tones over the whole crystal. Then Q is exact and constant under the rule, a conserved quantity of the entire universe. This was checked directly to machine precision [43].

This single fact does an enormous amount of work later in the paper. It is the discrete Gauss law that sources emergent electromagnetism. It is why the arrow can only ever create balanced pairs, a $+1$ together with a -1 , net zero. It forces newborn frontier cells to be born at peace. It gives the cosmic dilution toward peace, since Q stays fixed while the cell count grows without bound. It is the engine of life, where life turns out to be conservation of the sum rather than of the arrangement, so balanced information is free to make and copy while net well-being is never minted. What is not conserved is the arrangement, the order and pattern, which is free to form, flow, and decay.

2.4. The rule is reversible

The collide step is a reversible local map, so the whole rule runs backward as exactly as it runs forward.

Result 1 (Reversibility) Running the rule forward N beats and then backward N beats recovers the start state with zero error [43].

Reversibility matters twice over. It is what guarantees the right conserved quantities, both the charge above and momentum. And it means every state has a unique past, which is central to the cosmology and to the question of whether the universe had a beginning or is eternal. We leave that question open. The rule itself runs equally well in either direction, and which way the world in fact ran is not settled by the rule alone.

2.5. The nothing-added discipline

Principle 3 (Nothing added) Across every test in this paper, nothing was added to the four ingredients. Every result is a property of the mesh, the tone, or the rule, or an emergent consequence of them. Where a phenomenon did not emerge, the negative is recorded.

Several negatives are on the record. The pure rule churns with no durable selves. Radial coarse-graining gives no hierarchical tower. The flat form-tower is a generic slow mode rather than something special. Continuous mathematics, Lie groups, Dirac operators, Green’s functions, is used only as emergent-layer description, never as a base ingredient. The full audit is in the appendices.

The discipline is not merely a vow. It has a track record. Four times a phenomenon looked as though it needed a new base ingredient, and four times it turned out to emerge from the four. Momentum came from the determinism of the streaming rule. Durable selves came from self-maintenance, after a proposed adaptive-fills fifth ingredient was retracted. The arrow of time came from growth together with coarse-graining. A controller, or agency, showed up when the field flood-filled a maze with no planner added. The discipline cuts the other way too. A result only counts if it runs on the base alone, and several early gravity and clustering demonstrations that leaned on scaffolding, a diffusion proxy, gradient-following, a hand-built capture rule, were retracted as not real emergence and either rebuilt on the bare rule or reported as recorded negatives.

There is one plausible candidate for a possible fifth ingredient. The base binds within a unit but not cleanly between units, and that single gap sits behind a few open holes, cross-gap attraction and super-selves among them. We name it as the one candidate not yet proven irreducible. We do not treat it as a license to add it. In the carrying image the mesh is the instrument, the tone its notes, the beat its rhythm, and the rule its harmony, and the whole is consciousness singing, a fire that will grow into a tree.

2.6. The arrow, a closer note on the rule

One piece of the rule deserves a closer description, the *arrow*. It is not a fifth ingredient but a more detailed account of one feature of the rule, its value-direction. It is also the single deliberate posit of the whole theory. Everything else is either forced by the geometry or measured in simulation. The arrow is chosen.

The arrow is the asymmetry that makes time. The conserving rule, taken by itself, is reversible, a bijection on its moves, and a reversible rule is time-symmetric. It runs creation and annihilation equally, so on its own it has no preferred direction and no net growth. The arrow is exactly what breaks that symmetry.

Definition 1 (The arrow) The arrow is the bias, within the rule, between two reverse moves of the collide step, *creation*, in which peace becomes a balanced pair, and *annihilation*, its reverse. The arrow is a direction on the dynamics, an orientation, not an amount.

It rides on the value ordering of the tone, $-1 < 0 < +1$, read as pain < peace < pleasure. Peace, the 0 value, is the neutral rest state, and the charges ± 1 are the deviations from it. The arrow is the preference for the

forward direction, peace turning into balanced structure, and that preference is the seed of value, of better against worse.

A common simulation mistake is to write the arrow as a floating-point number, for example to create a pair with probability 0.03 per beat. That is wrong twice over. It introduces a free continuous parameter, since nothing fixes 0.03 and a tunable dial invites exactly the fine-tuning the theory is built to avoid. And it needs randomness, which is hidden state the discrete base forbids. The arrow is not an amount. It is a direction. The amount of creation is not a property of the arrow at all. It is forced by the eternal-growth geometry, since each beat the mesh adds one new shell of cells, the $\{3, 4, 3, 4\}$ shell growth $1, 24, 456, 8376, \dots$ with growth rate near 18.3, and that boundary expansion is the creation, new cells coming into being at the edge, oriented by the arrow. The arrow says which way, toward existence. The geometry says how much, through the growth series. On a fixed simulation patch the proper stand-in is a minimal deterministic schedule, one balanced creation per beat, never a probability, so there is no free parameter and no randomness anywhere in the arrow.

What the arrow does is keep the universe alive, and this is measured rather than asserted. Run the rule with the arrow off and the universe relaxes to dead peace, all zero, no motion, the heat death of the substrate. Turn it on and peace spawns balanced pairs, the universe settles into a living dynamic balance, charge conserved throughout, and the churning never stops [43]. Read at greater scales, the same arrow is the engine of life, mind, and value. It is the fitness gradient of evolution, the deferred value a planner pursues, and the will is the arrow held by a self and aimed. Conservation together with the arrow gives the rule its moral shape, since balanced information is free to make and copy, but net well-being is never minted from nothing.

A gut-level picture pulls the value story together. The three tones lie on a line, pain at -1 on one side, pleasure at $+1$ on the other, peace at 0 in the middle. A single vibe wants nothing, since the rule is even-handed and flipping every tone flips the dynamics with it, so the arrow is not painted on the individual cells. It appears the way a current appears from a slope. No water molecule wants to flow, but organize many on a slope and a directed current emerges, made entirely of undirected parts. A rock, barely organized, has almost no drive, while a mind, highly integrated, has a sharp one. Pain and pleasure are the compass, the feeling of going downhill toward worse or uphill toward better, and peace is the home, the valley floor where the compass reads flat. Pleasure is the direction. Peace is the destination. Because charge is conserved, pleasure behaves like a fluid that flows from the full cup to the empty one, so to stay full a self must pump, and pumping takes work that only an organized self can do. That is what being alive is, pumping against the drain, the same Landauer cost [33] that makes rich memory cost work. The way out, and the engine of the whole emergent ladder, is to stop yanking, to join up into a larger whole, to share the fixed pleasure in balance, and to pour the freed energy into building order, which is positive-sum and can grow. That joining-up, again and again, is how cells become bodies and bodies become minds. In the carrying image the arrow is the river's one direction, the relentless forward current, and the fire's hunger to spread rather than die down.

3. The Rule, Exactly

This section gives the single complete specification of the rule, what exists, how it updates, and exactly how many parts it has. This is the modern form on the $\{3, 4, 3, 4\}$ crystal, a directional lattice gas that *streams* then *collides*. It supersedes an earlier single-tone pairwise version on $\{5, 3, 4\}$ with 12 neighbours and a

nine-state edge shuffle. Everything below is broken into clean integer counts so the whole machine can be held in mind at once. Nothing is left out. We then zoom all the way in to a single cell and watch one beat act on it, which puts the geometry and the rule together in one concrete picture.

3.1. In essence

On the fixed $\{3, 4, 3, 4\}$ crystal, every cell keeps one ternary value per direction, 24 of them, and each beat is two steps. First *stream*, in which every directional value moves to the neighbour it points at. Then *collide*, a local, charge-conserving, reversible rewrite of a cell's 24 values.

3.2. The parts at a glance

what	count	the parts
base elements	4	mesh, tone, beat, rule
tone values per slot	3	$-1, 0, +1$ (pain, peace, pleasure)
directions per cell	24	the 24-cell = D_4 root system
ternary slots per cell	24	one tone per direction
representation content of the 24	$8v + 8s + 8c$	$SO(8)$ vector and two spinors
steps per beat	2	stream, then collide
collision move-types	4	pass-through, annihilate, create, scatter

Table 1: The integer counts that fix the rule. The arrow folds into the rule and growth folds into the mesh and beat, so the base count stays at four.

So the rule is, per beat, one global streaming permutation followed by one local reversible collision on 24 ternary slots, applied to every cell at once.

3.3. What exists, the four base elements

The *mesh* is the $\{3, 4, 3, 4\}$ honeycomb, the four-dimensional hyperbolic crystal. Its cells are 24-cells $\{3, 4, 3\}$, so each cell has 24 faces, hence 24 neighbour-directions, the coin. The mesh is generated by reflection, the Coxeter group $[3, 4, 3, 4]$, and it grows forever, one shell per beat. Reflection is how it is built and growth is it enlarging, so neither is a separate element. Its ideal $\{4, 3, 4\}$ cubic element is the flat three-dimensional physical cusp.

The *tone* is the value carried in each slot, exactly one of $-1, 0$, or $+1$. Never a fraction, never a continuous amplitude. The base is discrete. The *beat* is discrete time. Reality advances in synchronous ticks, each beat applying the rule once. The forward direction of the beats is the arrow, which lives in the rule. The *rule* is the directional lattice-gas update below.

The one enrichment over the oldest model is that the tone is now directional. A cell holds one ternary slot per direction, 24 slots, rather than a single scalar.

Definition 2 (The directional tone) Each cell A holds slots $[s_1, \dots, s_{24}]$, one per face, each in $\{-1, 0, +1\}$. Slot s_k is the charge currently moving through face k of A . The cell's scalar tone, its net charge, is the sum $\sum_k s_k$.

Nothing is lost in this enrichment, since the scalar tone is recovered as the sum of the slots. The slots only add the direction a charge is heading, and that is what gives the rule genuine momentum and lets it carry spin.

3.4. The beat, stream then collide

Stream, which is momentum. Every slot moves to the neighbour it points at. Slot k of cell A leaves A through face k , arrives at neighbour B_k , and lands in B_k 's slot for the same line of travel, so it keeps going the same way. Nothing decides or randomises direction. A charge heading north simply keeps going north. Streaming is a global permutation of all slots in the crystal, each slot having exactly one destination slot and the map being a bijection, so it is conflict-free and applied to every cell at once. No edge-colouring is needed, which was the old pairwise model's bookkeeping. Because streaming preserves direction, a charge travels ballistically, one cell per beat.

Collide, which is interaction, the arrow, and mass. After streaming, each cell holds a fresh set of 24 incoming slots. Apply a fixed local collision that reads the 24 incoming slots and rewrites them as 24 outgoing slots, by a map that conserves the total charge and is reversible. The collision plays the role of the coin operator in a quantum walk, but here it is a discrete, charge-conserving, reversible local map with no fractions. There are four move-types.

1. *Pass-through.* A lone charge with nothing to interact with keeps its slot and streams straight on.
2. *Annihilate.* An incoming $+1$ and -1 that meet cancel to 0 and 0 , charge-neutral.
3. *Create.* A cell of all-peace can emit a $+1$ and a -1 into two opposite outgoing slots, charge-neutral, and the pair flies apart with opposite momenta. This is pair production, and it is the arrow. The rule prefers this forward direction over its reverse, which is what makes the beats flow toward more structure rather than back to nothing.
4. *Scatter.* The collision can turn an incoming direction partly into other outgoing directions. The amount of this mixing is the mass. No scatter is a massless charge going dead straight at light speed. More scatter is a heavier, slower, zig-zagging charge.

3.5. The scalar pairwise shadow

The move-types are easiest to see in the simplest possible setting, a single scalar tone per cell and a single pair of neighbours, the two-direction shadow of the full 24-direction rule. A pair holds two tones, so there are $3 \times 3 = 9$ pair-states, and the rule is one reversible permutation of them, written out completely.

Two facts fall straight out of the table. The total charge is conserved, since no move changes the sum $a + b$. And the rule is reversible, since the table is a permutation of the nine states with a well-defined inverse. Reading off the move-types, there are two hops, one per charge sign, one create-flip-annihilate three-cycle, $(0, 0) \rightarrow (+1, -1) \rightarrow (-1, +1) \rightarrow (0, 0)$, and two inert fixed points. Creation and annihilation are the same operator at the two ends of one cycle, forced to be inverses by reversibility, and that cycle is the arrow. This table is the scalar pairwise shadow of the full rule, the same move-types in two directions. In the full rule the hop becomes the stream, the charge living in a direction slot and moving along it, while create, annihilate,

pair in (a, b)	pair out	move
$(-1, -1)$	$(-1, -1)$	inert
$(+1, +1)$	$(+1, +1)$	inert
$(-1, 0)$	$(0, -1)$	hop
$(0, -1)$	$(-1, 0)$	hop
$(+1, 0)$	$(0, +1)$	hop
$(0, +1)$	$(+1, 0)$	hop
$(0, 0)$	$(+1, -1)$	create
$(+1, -1)$	$(-1, +1)$	flip
$(-1, +1)$	$(0, 0)$	annihilate

Table 2: The nine-state scalar pairwise shadow, one reversible permutation of all two-tone pair states.

inert, and scatter are the per-cell collide on the 24 slots. The shadow is the cleanest way to see the moves, and the directional version above is the real thing.

3.6. The three properties that fall out

Result 2 (Conservation, reversibility, spin) *Charge is conserved.* Every move-type preserves the sum of the slots, so the total tone over the universe is constant, the discrete Gauss law that sources the emergent $U(1)$. *The rule is reversible.* Stream is a permutation and collide is a bijection, so the whole beat has an exact inverse, verified forward-then-back to zero error at scale [43]. *Spin lives in the collision's symmetry on the 24 directions.* The 24 roots of D_4 decompose as $8v + 8s + 8c$, so a collision built on them can carry genuine spinors, a state that flips sign under a full 2π rotation.

The spinor property is the concrete reason the substrate must be $\{3, 4, 3, 4\}$. The 12 directions of $\{5, 3, 4\}$ decompose as $1 + 3 + 3' + 5$, all integer spin and no spinor, so a $\{5, 3, 4\}$ collision cannot carry spin. This is taken up in detail when we turn to fermions.

3.7. What each part gives

part	what it gives
stream, transport	ballistic propagation, momentum, the $z = 1$ light cone, the Dirac dispersion
collide, scatter	mass, $\cos E = \cos m \cos k$, the massive Dirac walk
collide, create and annihilate	pair production from peace, the arrow, the conserved-charge vacuum
collide, pass-through	free propagation, same-charge stability
$24 = D_4$ collision symmetry	genuine spinors, three generations via triality
charge conservation	the $U(1)$ Gauss law, the conserved current
reversibility	a Hermitian generator, a positive-energy quantum theory
direction-blind rule	isotropy, the same light speed in every direction

Table 3: What each part of the rule contributes to the emergent physics, each confirmed by direct computation on the exact crystal [43].

3.8. The framing

This is not free. It enlarges the state, from one ternary value per cell to 24 ternary slots per cell. There are two fair readings. Either it is still tone, just directional, with the scalar tone recovered as the sum of the slots, an enriched tone rather than a new element, and the four elements stand. Or the coin has become a

base feature, and the base has grown. Either way it is a deliberate change from the one-tone-per-cell model, made to get momentum and spin, both of which the scalar read-all-faces rule could not be shown to carry. The pay-off is that it propagates by construction, it stays fully discrete, conserving, and reversible, and on $\{3, 4, 3, 4\}$ it carries spin.

What is not in the rule is everything continuous. Continuous amplitudes, the unitary quantum walk, the complex Dirac field, and mass as a mixing angle are the effective description of this discrete rule after coarse-graining, the way a lattice gas coarse-grains to fluid equations, never the base. Selves, particles as bodies, gravity, and nesting are emergent on the cusp and above. The base stays the four elements and the stream-collide rule, nothing more.

3.9. Anatomy of a single vibe

The clearest way to grasp what a vibe looks like in the whole model is to zoom all the way in to one cell and watch one beat act on it. A cell is a 24-cell, a four-dimensional solid with 24 octahedral faces. Each face is shared with exactly one neighbour cell, so the cell has 24 neighbours, call them B_1 through B_{24} , one across each face. The 24 face-directions are the D_4 root system, the coin. That is the whole local geometry, a cell and its 24 face-neighbours, and distance is step-counting on this graph. Step-counting is what space will come to mean.

What the cell stores is its 24 ternary slots s_1 through s_{24} , one per face, each in $\{-1, 0, +1\}$. Slot s_k is the charge currently heading out through face k . That is the cell's entire state, 24 ternary values and nothing else, with no hidden state. The cell's scalar tone, the familiar pain, peace, or pleasure value, is just the sum $s_1 + \dots + s_{24}$. A cell's state space is therefore 3^{24} , but in practice almost all of it is near-empty, a single moving charge being one slot at $+1$ and the other 23 at 0.

As for how the cell relates to its neighbours, face k of cell A is the same shared octahedron as one face of neighbour B_k . The two cells meet exactly there. A 's outgoing slot s_k and B_k 's slot for the same line of travel are the two ends of that shared face. There are no edge weights and no stored links. The relation is just the shared face and the matching slots. The note, the perception across the face, is computed from this and never stored.

3.10. One beat, walked through

A beat is two steps. In the stream step every slot moves to the neighbour it points at. Slot s_k of A leaves through face k , arrives at neighbour B_k , and lands in B_k 's slot for the same line of travel, so it keeps going the same way. At the same instant A receives, into each of its own slots, whatever its neighbours streamed toward it. After streaming, A holds a fresh set of 24 incoming slots. Nothing decided direction. A charge heading north simply kept going north. This is momentum, the charge travelling one cell per beat, ballistically.

In the collide step A reads its 24 incoming slots and rewrites them as 24 outgoing slots, by the fixed local map that conserves the sum and is reversible. The move-types are the four already named, pass-through, annihilate, create, and scatter, with the nine-state table above as their scalar shadow.

A worked example makes it concrete. Start with cell A holding $s_3 = +1$ and all other 23 slots at 0, a single $+1$ charge heading through face 3, with all neighbours empty. On the first beat, s_3 leaves A through face 3,

arrives at neighbour B_3 , and lands in B_3 's straight-ahead slot. Now A is all zero and B_3 holds one incoming $+1$. At B_3 the collision finds one incoming charge with nothing to meet, so it is pass-through, and the charge keeps its direction. Beat after beat the $+1$ streams in a straight line, one cell per beat, never spreading. That is a massless particle, with $E = k$. Turn on a mixing scatter collision and the charge starts turning into other slots, its track zig-zags, and its effective speed drops. That is mass, with $\cos E = \cos m \cos k$. Start instead from an all-peace region with the arrow's create move on, and a $+1$ and a -1 are emitted into opposite slots and fly apart, which is pair production. Throughout, the total over all cells' slots never changes. Charge is conserved.

This fixes the intuition. Nothing moves through space. Space is the cell-graph, and the only motion possible is a tone-pattern redrawn one cell over each beat, a glider. Momentum is which slot a charge lives in. Mass is how much the collision mixes slots. The arrow is the create move. Conservation is simply that the collision never changes the sum. One cell, 24 ternary slots, two steps a beat. That is the whole machine, and everything that follows is what this machine does at scale. The entire universe is a ternary tone on the 24 directions of every cell of a growing $\{3, 4, 3, 4\}$ crystal, updated each beat by one streaming move and one reversible local collision. In the carrying images this single machine is the fire's one law of burning and the song's one rule of harmony. The mesh is the instrument, the tone its notes, the beat its pulse, and the stream and collide are how each note sounds and resolves into the next.

4. The Integer Ladder

A natural worry is that a chosen honeycomb with a chosen rule is arbitrary. The answer is that the whole substrate rests on the least arbitrary thing there is, the integers. The Coxeter groups, the crystals, and the cell addressing all appear as different views of one object built from $\mathbb{Z}$. There are no real numbers and no tunable dials anywhere in the foundation. The point is best made as a tower, each rung resting on the one below.

4.1. The tower

- *Rung 0, the integers.* Start with $\mathbb{Z}$. Nothing is more canonical. No parameters, no reals, no randomness.
- *Rung 1, integer symmetries.* Ask for symmetries generated by reflections where the only data is the integers saying how the mirrors meet, mirror i and mirror j at angle π over an integer. The modular group $\text{PSL}(2, \mathbb{Z})$, the symmetries of the two-dimensional integer lattice, is one famous example.
- *Rung 2, Coxeter groups.* That is exactly a *Coxeter group*, and its data is a small integer matrix, the Coxeter diagram. Still just integers.
- *Rung 3, tessellations.* Let a Coxeter group act on space by its reflections. Reflect one fundamental chamber over and over, and the orbit fills the space. The result is a *tessellation*. The algebraic group and the geometric tiling are the same thing, two views of one object.
- *Rung 4, the specific crystals.* Choose the integers in the diagram and you get a specific tiling, named by its Schläfli symbol, $\{7, 3\}$, $\{5, 3, 4\}$, $\{3, 4, 3, 4\}$. These are not separate inventions. They are Rung 3 with particular integer settings, one machine on different dials.

- *Rung 5, the substrate.* On the chosen tessellation the vibes live, tones on the cells and the rule on the directions. The substrate inherits, for free, everything the lower rungs provide.
- *Rung 6, the generating automaton.* Every hyperbolic Coxeter group is an *automatic group*, so a finite integer state machine generates the whole infinite tessellation deterministically, with no randomness. For $\{3, 4, 3, 4\}$ this is the solved neighbour automaton and an $O(\log n)$ addressing, by which the substrate generates and navigates itself from $\mathbb{Z}$ by a finite rule.

4.2. One object, many views

The power of the ladder is that a single object wears many faces. For $\{3, 4, 3, 4\}$ the Coxeter system $[3, 4, 3, 4]$ is at once an algebraic group of five reflections, a geometric honeycomb of 24-cells filling $\mathbb{H}^4$, a combinatorial addressing tree with Zeckendorf-style digits, a dynamical generator in the neighbour automaton that advances beat by beat, and a growth law in the shell recurrence. The algebraic Perron root of that recurrence, near 18.3, is the central constant of the substrate, the $\{3, 4, 3, 4\}$ analogue of the golden ratio that sits at the heart of the $\{5, 3, 4\}$ and heptagrid cases. These are not analogies. They are literally the same object looked at from several sides.

4.3. The payoff

To pick the base is only to pick which integers go in the diagram. The object beneath all the views is one integer machine, and there is no free real-number parameter anywhere in the foundation.

Proposition 2 (No free parameters) $\{3, 4, 3, 4\}$ is not an arbitrary choice. It is one integer setting of the canonical Coxeter machine, and which setting it is is forced by the selection criteria developed later, the dimension ladder, the crystallographic spinor coin, and curvature.

The integers are the bones of the theory, and the small ones carry the laws. In the song image they are the score, the small ones its notes, and in the seed image the diagram is the seed that already folds the whole tree. This is also where the foundation makes contact with the broader program of generating spacetime from finite combinatorial data, in the spirit of the automatic-group description of hyperbolic tilings [10] and the negatively curved geometry that underlies them [27]. The substrate is what one gets by taking that machinery seriously as the ground floor rather than as a tool applied to something else.

Part II

The Geometry of the Substrate

5. The Three Geometries

Before fixing the substrate's specific shape there is a prior question. Why a curved space at all, and why exactly the hyperbolic one? The answer turns on a fact that is older and tighter than it first appears. There

are exactly three uniform geometries. Not four, not infinitely many. They are the spherical, the flat, and the hyperbolic, and the argument of this paper is that the substrate must be the third.

geometry	curvature	triangle angles	room going outward	shape
spherical	positive	sum to more than 180°	shrinks, then wraps around	a ball's surface
flat	zero	sum to exactly 180°	grows steadily, like r	a tabletop
hyperbolic	negative	sum to less than 180°	grows exponentially	a saddle

Table 4: The three constant-curvature geometries.

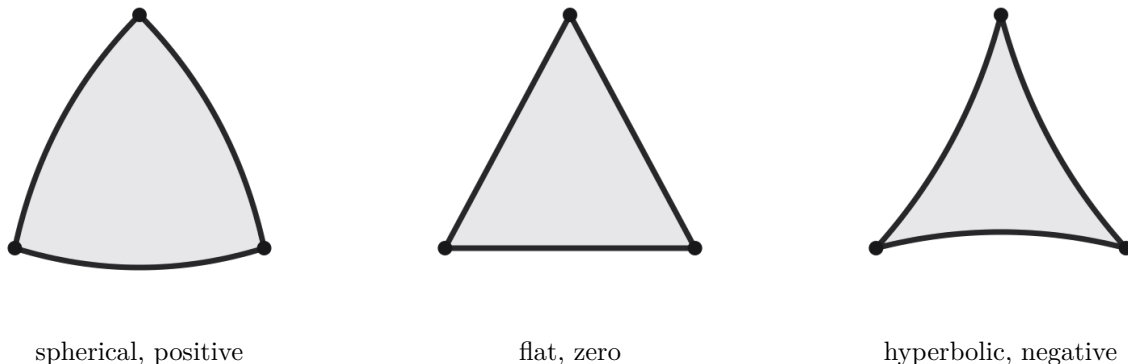


Figure 3: The three curvatures read off a triangle. On the sphere the sides bow outward and the angles sum to more than 180° . In the flat plane the sides are straight and the angles sum to exactly 180° . In the hyperbolic plane the sides cave inward and the angles sum to less than 180° .

5.1. Why exactly three

Curvature is one number that measures how a space bends. It can be felt from inside in two equivalent ways. The first is by triangles (Figure 3). Their angles sum to 180° only in flat space, more than that if the curvature is positive, less if it is negative. The second is by room. A circle of radius r has circumference $2\pi r$ in flat space, less if the curvature is positive, and exponentially more if it is negative.

A uniform space is one that looks the same at every point and in every direction. Such a space must carry one constant curvature everywhere, a single real number. A real number has exactly three signs, positive, zero, and negative. So there are exactly three uniform geometries, one per sign. This is not a metaphor. It is the content of the Killing-Hopf classification of constant-curvature spaces, which gives one type per sign in every dimension.

Theorem 1 (Three signs, three geometries) A simply connected space of constant curvature is spherical, flat, or hyperbolic according to whether its curvature is positive, zero, or negative. There is no fourth possibility, because a real number has no fourth sign.

5.2. Why there is nothing halfway

The tempting confusion is to imagine some intermediate geometry sitting between flat and hyperbolic. There is none, and the reason is worth stating carefully. The magnitude of curvature is continuous. It can be $K = -1$, or -0.01 , or -1000 , any value at all. But magnitude is only a scale, not a new type. A sphere of

radius 1 and a sphere of radius 2 are both spherical, just zoomed differently. Hyperbolic space with $K = -1$ and with $K = -0.01$ are both hyperbolic, again just different sizes. Only the sign sets the type.

A space with a tiny negative curvature is therefore already fully hyperbolic, simply with a very large radius. Near any point it looks almost flat, and it keeps that flat appearance until you zoom out far enough that the exponential room reappears. Flat space is the knife-edge between the two curved worlds. This is why our nearly-flat universe is so finely balanced. Near-flat is not a fourth option. It is a curved option seen at large scale.

One formula carries all three regimes at once, with the curvature radius R as the single knob. The circumference of a circle of radius r is $2\pi R \sin(r/R)$ in the spherical case, $2\pi r$ in the flat case, and $2\pi R \sinh(r/R)$ in the hyperbolic case. Sine, line, and hyperbolic sine are one analytic family. They are the three faces of one sign.

5.3. Why the substrate is the hyperbolic one

Each geometry is ruled in or out by what the substrate has to do. The spherical case is out. It is finite and closed, so it runs out of room and wraps back on itself, and a substrate built on it could not grow forever. The flat case is out as well. It is infinite, but its room grows only steadily, like r^3 in three dimensions, which is not enough for the explosive branching growth the model needs. A flat regular lattice also carries preferred directions, its axes, and that breaks the isotropy our universe appears to have.

The hyperbolic case is in, and for three reasons at once. It is infinite, so there is endless room for eternal growth. Its room grows exponentially, which supplies the branching. It has no preferred direction, so it is isotropic. It is also the only one of the three that hosts a rich family of regular crystals, including the heptagrid $\{7, 3\}$, the dodecagrid $\{5, 3, 4\}$, and the substrate of this paper, $\{3, 4, 3, 4\}$. As a bonus it is the spatial geometry of Anti-de Sitter space, the natural home of holography.

Result 3 (Hyperbolic by elimination) Among the three uniform geometries only the hyperbolic one is infinite, exponentially roomy, isotropic, and host to a rich family of regular honeycombs. Spherical fails on size, flat fails on growth and isotropy. The substrate is therefore hyperbolic.

5.4. And lumpy real space

The three are the uniform geometries, the ones that look the same everywhere. Real spacetime, as general relativity describes it, is lumpy. It is curved differently at each point. That does not add a fourth type. A general curved space is built by patching the three local models together, one at each point, and each patch still carries a sign. The three uniform geometries are the local alphabet that every curved space, however irregular, is spelled in.

So the substrate is uniformly hyperbolic at its base. The lumpy gravity of the everyday world is then the bending of that base by matter, the emergent gravity taken up later in the paper.

6. Hyperbolic Tessellations

Having narrowed the geometry to the hyperbolic case, it helps to step back and see the whole family the substrate belongs to. This section is the bridge to the all-dimension survey that comes later. It explains what

hyperbolic tessellations are, how they behave across dimensions, and why a few names, $\{7, 3\}$, $\{5, 3, 4\}$, and $\{3, 4, 3, 4\}$, keep recurring.

6.1. The core idea

A *tessellation*, also called a tiling or a honeycomb, fills a space completely with repeating shapes, leaving no gaps and no overlaps. Floor tiles, the hexagons of a beehive, and cubes stacked to fill a room are all tessellations. A *hyperbolic tessellation* is the same idea carried into negatively curved space, where distances expand exponentially and far more arrangements fit.

geometry	curvature	parallels	feeling
Euclidean	zero	stay a fixed distance apart	flat, like graph paper
spherical	positive	eventually meet	closed, like the Earth's surface
hyperbolic	negative	diverge exponentially	a saddle, expanding

Table 5: The three geometries by how their parallel lines behave.

The key intuition is that hyperbolic space creates extra room. In flat space only a few regular polygons tile exactly, namely squares, hexagons, and triangles. Put three heptagons around a single corner and they leave too much room to lie flat, so the space has to curve negatively to absorb them. For a tiling $\{p, q\}$, with p -sided polygons meeting q to a corner, the dividing rule is the value of $(p - 2)(q - 2)$. Less than 4 is spherical, exactly 4 is Euclidean, and greater than 4 is hyperbolic. The heptagonal tiling $\{7, 3\}$ gives $(7 - 2)(3 - 2) = 5$, so it is hyperbolic.

6.2. Schläfli symbols and the dimensional ladder

Hyperbolic tessellations are named by *Schläfli symbols*, written $\{a, b, c, \dots\}$. Each new number adds a dimension by recording how the previous shape is assembled into the next. The same rules hold all the way up the ladder, and the family exists in only a handful of dimensions.

dimension	object	example	built from
2D	tiling	$\{7, 3\}$	heptagons, 3 to a corner
3D	honeycomb	$\{5, 3, 4\}$	dodecahedra, 4 to an edge
4D	hyper-honeycomb	$\{3, 4, 3, 4\}$	24-cells, 4 to a face
5D	higher honeycomb	$\{3, 3, 3, 4, 3\}$	the last dimension with any

Table 6: The recurring regular hyperbolic tessellations by dimension.

Five-dimensional hyperbolic space is the last that admits regular hyperbolic tessellations at all, and every one of them is paracompact. The geometric freedom runs out above five. This is a strong constraint, and it is part of what makes the survey of dimensions in the later parts of the paper land where it does.

6.3. How hyperbolic growth feels

In flat geometry circles grow polynomially and neighbourhoods grow slowly. In hyperbolic geometry shells grow exponentially. Almost everything in a large ball lies near its boundary, and the space behaves, at large scale, like a tree. That tree-likeness is why hyperbolic geometry keeps appearing in hierarchy, in semantic

embeddings, in internet routing, in tensor networks, in recursive subdivision, and in holography [27, 28]. In the substrate it is the source of the cosmic expansion. Each shell of cells is roughly ten times the last.

The structures themselves are made by *reflection*. You start with one fundamental chamber and reflect it across its mirror walls, then reflect the copies, and continue forever. The reflections form a *Coxeter group*, written $[5, 3, 4]$ for the dodecahedral honeycomb and $[3, 4, 3, 4]$ for the 24-cell honeycomb [15]. An infinite recursive order grows out of a tiny local rule.

6.4. The recurring pattern, curved bulk to flat boundary

The single most important pattern across the whole family is this. A hyperbolic bulk in dimension n can produce a flat Euclidean space of dimension $n - 1$ at its ideal boundary.

hyperbolic bulk	flat boundary at infinity
H^2	a line or circle
H^3	flat 2D
H^4	flat 3D
H^5	flat 4D

Table 7: Each hyperbolic dimension yields one flat dimension at its ideal boundary.

Each extra hyperbolic dimension generates one extra flat dimension at infinity. When a tessellation’s vertex figure is itself infinite, that is, Euclidean rather than finite, its corner is pushed out to infinity. Such a corner is called an *ideal vertex*, the tessellation is called *paracompact*, and the local cross-section near that ideal vertex is flat Euclidean space.

Result 4 (Flat boundary of a curved bulk) A paracompact regular hyperbolic honeycomb in H^n has a flat Euclidean cross-section of dimension $n - 1$ at each ideal vertex. For $\{3, 4, 3, 4\}$ in H^4 , with vertex figure the cubic honeycomb $\{4, 3, 4\}$, that cross-section is flat three-dimensional space.

This is exactly why $\{3, 4, 3, 4\}$ hands over flat 3D space. Its vertex figure is the cubic honeycomb $\{4, 3, 4\}$, an infinite Euclidean tiling, so its corners are ideal and the cross-section there is the ordinary cubic grid. It is also why the dimension ladder developed later in the paper lands on 4D for our 3D world.

6.5. The universal themes and the mental model

Across the dimensions the same handful of themes repeat. Negative curvature creates room. Reflections generate structure. Exponential growth creates hierarchy. Ideal boundaries produce Euclidean geometry. Recursion creates infinite order. Coxeter groups encode the rules. Held together, these give a single mental model, an infinite recursive crystal growing exponentially outward through curved space.

Each named example has its own flavour. The heptagrid $\{7, 3\}$ feels like an endlessly branching 2D cathedral. The dodecahedral honeycomb $\{5, 3, 4\}$ feels like recursive dodecahedral architecture. The substrate $\{3, 4, 3, 4\}$ feels like a 4D crystalline engine generating flat 3D order at its boundary. The deepest shift in viewpoint is that ordinary flat space starts to feel less fundamental. Flat geometry can emerge from hyperbolic geometry asymptotically, and lower-dimensional order can arise from higher-dimensional curvature. That one idea ties hyperbolic tessellations, holography, tensor networks, and the whole substrate picture into a single thread.

7. The Fourth Dimension

The substrate is four-dimensional, so the reader needs a working intuition for 4D. The goal here is to make four dimensions stop feeling impossible and start feeling like the natural next step of geometry. The trick is to stop trying to see it and start understanding its structure. Once dimension is treated as coordinate freedom rather than as a picture, 4D becomes ordinary.

7.1. What a dimension actually is

A dimension is not a hidden place. It is just an independent direction you can move along, one more number you need to fix a position. The ladder is built by counting coordinates.

dimension	coordinates	what lives there
0D	none	a point
1D	x	segments
2D	(x, y)	polygons, circles, tilings
3D	(x, y, z)	cubes, spheres, ordinary space
4D	(x, y, z, w)	tesseract, hyperspheres, the 24-cell

Table 8: The dimensional ladder by coordinate count.

Four dimensions just means one more independent direction w , behaving exactly like x , y , and z . The only real difficulty is that the brain cannot point at w . The mental shift is to stop asking where the fourth direction is physically and to ask instead what happens if one more coordinate exists. People already navigate high-dimensional spaces every day without noticing. Colour is three-dimensional. A word embedding can have thousands of dimensions. The habit of not calling these spaces is the only thing that makes 4D feel strange.

7.2. The analogy staircase

The master technique is a staircase, and each step does the same thing. Sweep the current shape along a new direction that is perpendicular to everything so far. A point sweeps into a line, a line into a square, a square into a cube, and a cube into a *tesseract*, the 4D cube. The new perpendicular direction is exactly the thing you cannot picture, but the pattern itself never breaks, and the pattern is the intuition.

There is a second way to build, by doubling. A cube is two squares with corresponding corners joined. A tesseract is two cubes with corresponding corners joined, which gives $8 + 8 = 16$ vertices. The same doubling continues up the ladder. So you can construct, count, and reason about 4D objects perfectly without ever forming a mental image of them. That, in practice, is what understanding 4D feels like.

7.3. Shadows and slices

There are two ways we glimpse a 4D object from inside three dimensions. The first is projection. A drawn tesseract, the familiar cube-inside-a-cube figure, is a 3D shadow of the 4D object, in the same way a drawn cube is a 2D shadow of a real cube. The inner cube is not smaller. It is farther along w , and projection shrinks what is farther away.

The second is cross-sections, the Flatland method. A 4D object passing through our 3D space appears as a sequence of changing 3D slices, just as a sphere passing through a flat plane appears to 2D beings as a dot that grows into a circle and then shrinks away. Watch the slices morph and you can reconstruct the whole. Neither method shows the object directly, and neither needs to.

7.4. Rotation gets richer

In three dimensions you rotate around an axis, a line that stays fixed. In four dimensions you rotate in a plane, and there can be two independent simultaneous rotations. An object can spin in the xy plane and in the separate zw plane at the same time, a double rotation with no three-dimensional counterpart. This extra room has concrete consequences. Mirror images become superimposable, so a left hand can be turned into a right hand by passing it through 4D, the way the letter “p” flips to “q” by passing through 3D. Two flat planes can meet at a single point. And one-dimensional knots fall apart, because knotting always needs exactly two extra dimensions, so in 4D it is two-dimensional sheets that knot instead of strings.

7.5. Why 4D is uniquely special

Four dimensions are not merely one more rung. They are structurally exceptional, and that matters because the substrate’s physics rides on it.

- *Quaternions*, the 4D rotational numbers, encode 3D rotations so cleanly that graphics engines use them. Rotation itself wants higher-dimensional structure.
- The *24-cell* exists only in 4D, in no other dimension. Its very existence proves that 4D contains genuinely new structure, and it is self-dual, a sign of deep symmetry balance.
- An exceptional cascade begins here, a chain running from 4D through quaternions, spinors, Clifford algebras, D_4 , triality, the octonions, the exceptional groups, and on to E_8 . The substrate sits at the mouth of this cascade.
- *Spinors* emerge naturally in 4D, objects that need a 720° turn to return to themselves. Triality, the rare three-way symmetry of D_4 , makes the vector and the two spinors interchangeable, and that is the seed of three generations.

Result 5 (The exceptional dimension) Quaternions, the 24-cell, spinors, and triality are all four-dimensional phenomena with no lower-dimensional counterpart. The features that make 4D structurally exceptional are precisely the features the substrate’s physics is built from.

7.6. The reframe to carry

Dimension is relational, not visual. It is coordinate freedom and symmetry behaviour, not a picture. Human intuition is local and three-dimensional, but the universe of geometry is far larger than what intuition can hold. So 4D is not another world standing beside ours. It is the next coherent extension of geometry, understood through symmetry, projection, and algebra rather than through shape.

And negatively curved 4D, the hyperbolic 4-space H^4 , is where the substrate lives. Its honeycomb $\{3, 4, 3, 4\}$ is built from 24-cells, and it has a flat 3D Euclidean cross-section at its ideal corners. The very facts that make 4D exceptional, the double rotations, the quaternions, the 24-cell, the spinors, and triality, are not

curiosities here. They are the working machinery of the physics, and the chapters that follow put each of them to use.

8. The 24-Cell

The 24-cell is the single cell of the $\{3, 4, 3, 4\}$ honeycomb, the brick of the whole substrate, so it deserves its own treatment. It is the most special shape in 4D geometry, and a reader who grasps it grasps much of why this substrate is the one the paper argues for.

Definition 3 (The 24-cell) The 24-cell is the regular four-dimensional polytope built from 24 octahedra glued face to face, with Schläfli symbol $\{3, 4, 3\}$. It is self-dual, has the 24 permutations of $(\pm 1, \pm 1, 0, 0)$ as its vertices, and is the only regular 4-polytope with no three-dimensional analogue.

8.1. In essence

The 24-cell is a 4D solid built from 24 octahedra glued face to face, in the way a cube is built from six squares. It is one of the six regular 4D shapes, the 4D Platonic solids, and it is the odd one out, the only one with no 3D counterpart. It is perfectly self-dual, deeply tied to quaternions and to the best 4D sphere-packing, and it tiles 4D space by itself. In this paper it is the fundamental brick of the $\{3, 4, 3, 4\}$ hyperbolic honeycomb.

8.2. The counts and the symbol

part	count	type
cells	24	regular octahedra
faces	96	triangles
edges	96	all equal
vertices	24	all equal

Table 9: The element counts of the 24-cell.

The Schläfli symbol $\{3, 4, 3\}$ reads as a recipe. The first $\{3\}$ says the faces are triangles. Then $\{3, 4\}$ says four triangles around a corner close into an octahedron, which is the cell. Finally $\{3, 4, 3\}$ says three octahedra around each edge close up into the 4D 24-cell. Its vertex figure, the shape of one corner, is a cube, with six octahedra arranged like the six faces of a cube. Six octahedra meet at every vertex, and three meet along every edge.

8.3. Why it is unique

Several features set the 24-cell apart, and together they make it feel uncovered rather than invented. It has no 3D analogue. The other regular 4-polytopes echo familiar 3D shapes. The 5-cell is the 4D tetrahedron, the tesseract is the 4D cube, and the 16-cell is the 4D octahedron. The 24-cell echoes nothing. It exists only because four dimensions admit an extra coincidence of symmetry, which makes it a genuinely 4D-only object.

It is self-dual. Put a point at the centre of each cell and connect neighbours, and you recover another 24-cell. The shape is its own dual, a mark of deep symmetry balance. Its edge length equals its circumradius, which gives it an unusually round, balanced proportion shared only with the tesseract. And it contains the others, incorporating essentially every regular polytope of dimensions one through four except those with a 5 in their symbol. It is a hub of low-dimensional regular geometry.

8.4. The coordinates, and why they are the whole point

The 24 corners are simply all the ways to place two ± 1 entries and two zeros into four slots, the 24 permutations of $(\pm 1, \pm 1, 0, 0)$. These same 24 points are the root vectors of the D_4 root system. This is the load-bearing fact for the whole paper.

Result 6 (The coin is D_4) The 24 directions along which a cell can pass signals, its coin, are the 24 vertices of the 24-cell, which are the D_4 roots. Under the symmetry they decompose as $8v + 8s + 8c$, the $\text{SO}(8)$ vector together with its two spinors, and triality permutes the three eights.

The decomposition $8v + 8s + 8c$ is the $\text{SO}(8)$ vector representation plus its two spinor representations, with triality, the rare three-way symmetry of D_4 , permuting them. Spin, the gauge group, and the three generations all trace back to this one shape. The geometry is not decorative here. The coordinates are the physics.

8.5. The symmetry, the group F_4

The full symmetry of the 24-cell is the Coxeter group $[3, 4, 3]$, the Weyl group of F_4 , of order 1152, of which 576 are rotations. This finite group of order 1152 is the substrate's exact discrete symmetry. The continuous $\text{SO}(8)$ and Lorentz symmetries appear only as its emergent, coarse-grained limits, taken up later in the paper.

The same 24 vertices are the unit Hurwitz quaternions, which form the binary tetrahedral group, so the 24-cell is literally a picture of a number system. They solve the 4D kissing-number problem, since 24 spheres can touch one central sphere, and they form the densest known 4D lattice, the D_4 lattice, whose Voronoi cell is itself a 24-cell.

8.6. A crossroads of mathematics

The 24-cell is one of the most structurally central objects in mathematics. It bridges geometry as an exceptional regular 4-polytope, algebra through the D_4 and F_4 root systems, packing through the densest 4D lattice and the kissing number of 24, group theory through the Weyl group F_4 , and physics through spin, quaternionic rotation, and the skeletons of gauge symmetry. It is the only exceptional regular polytope, belonging to no infinite family, and through D_4 and triality it connects upward into the exceptional chain of Spin groups, E_6 , E_7 , and E_8 . It has the flavour the substrate selection relies on. It feels rigid and forced once the rules of regularity are accepted, not chosen by hand.

8.7. Why it is the right brick

A self-dual, maximally symmetric, spinor-carrying, densest-packing 4D solid whose 24 directions are the D_4 roots is the natural best brick for a selection principle to single out. It is more densely connected than the dodecahedron of $\{5, 3, 4\}$, with 24 face-neighbours against 12, and it carries quaternions and a discrete Hopf fibration, so a 24-cell substrate arrives with rich built-in spinor geometry. Picture 4D hyperbolic space packed solid with these 24-cells, meeting face to face, and you have the body of the mesh.

8.8. A forward-looking resonance, the 24-web

A brief note on why this neighbourhood of mathematics feels so charged. The number 24, along with D_4 and the 24-cell, recurs across the deepest structures known. The 24-cell is the first rung of the exceptional cascade running from D_4 through triality to the octonions and on to F_4 , E_6 , E_7 , and E_8 . The same 24 appears as the

transverse dimension of the Leech lattice and the bosonic string, as the Hurwitz quaternions, and in Monster moonshine. So the substrate's cell sits at the mouth of the deepest exceptional cascade in mathematics. This is not a load-bearing claim, only a forward-looking resonance, but it is one reason $\{3, 4, 3, 4\}$ feels less like an arbitrary pick and more like a place the structure of mathematics keeps pointing toward.

9. The $\{3, 4, 3, 4\}$ Honeycomb

With the fourth dimension and the 24-cell in hand, the whole substrate can now be described in plain language. The seemingly preferred substrate, the one this paper argues for, is the $\{3, 4, 3, 4\}$ hyperbolic honeycomb. The selection itself is made later, on grounds rather than by assumption, so here the geometry is fixed and laid out, not yet justified. This section sets the stage the rest of the paper lives on, the four layers of bulk, coin, cusp, and boundary.

9.1. In essence

The honeycomb $\{3, 4, 3, 4\}$ fills 4D hyperbolic space completely with identical 24-cells, in the way cubes fill ordinary 3D space. It is regular, meaning every cell, face, edge, and corner is alike, and it is hyperbolic, meaning it fits only in negatively curved 4-space. Its most important feature is that its corners sit infinitely far away. They are *ideal* corners, and the pattern you see looking out toward one of them is the ordinary cubic grid of flat 3D space. So $\{3, 4, 3, 4\}$ is a curved 4D crystal whose edge, seen up close, is flat three-dimensional space.

9.2. How to read the symbol

Each number in the symbol adds a dimension by recording how the previous shape is assembled.

- $\{3\}$ is a triangle.
- $\{3, 4\}$ says four triangles around each corner close into an octahedron.
- $\{3, 4, 3\}$ says three octahedra around each edge close into the 24-cell.
- $\{3, 4, 3, 4\}$ says four 24-cells around each triangular face fill 4D hyperbolic space, the honeycomb.

Two facts read straight off the symbol. The cell, found by dropping the last number, is $\{3, 4, 3\}$, the 24-cell. The vertex figure, found by dropping the first number, is $\{4, 3, 4\}$, the cubic honeycomb. The second of these is the magic. The corner of $\{3, 4, 3, 4\}$ is ordinary 3D cubic space.

9.3. The four layers

The substrate is best held in mind as four layers, each carrying a distinct role.

The *bulk* is 4D hyperbolic space tiled by $\{3, 4, 3, 4\}$. Being hyperbolic it grows exponentially with radius, each shell about ten times the last, and that growth will be the cosmic expansion. It is paracompact rather than compact, which the cusp below turns into a feature.

The *coin* is the set of 24 neighbour directions at each cell. They form the 24-cell, whose vertices are the D_4 root system. This is the single most important structural fact in the paper. The 24 directions decompose

under the symmetry as $8v + 8s + 8c$, the $SO(8)$ vector plus its two spinors, with the S_3 triality permuting the three eights. The coin is where spin, the gauge group, and the three generations come from.

The *cusp* is the vertex figure $\{4, 3, 4\}$, the cubic honeycomb, which is Euclidean, infinite, and flat. This is the paracompact ideal element, and it is physical space, a flat 3D cubic cusp where matter, light, and gravity propagate. So physical space is three-dimensional and flat, and it is the cusp of the 4D curved bulk. The bulk is the fire, the cusp is the radiance it casts.

The *boundary* is the 3-sphere S^3 at infinity, the holographic screen. Gravity and quantum nonlocality live in the relation between bulk and boundary. Time enters as the beat together with the radial direction, which doubles as cosmic time and as the renormalization axis.

9.4. The vertex figure is flat 3D space

This is the whole point, so it is worth stating slowly. In a compact honeycomb the vertex figure is a finite shape and the corner is an ordinary point. But $\{4, 3, 4\}$ is infinite, since it is all of flat 3D space. When the vertex figure is infinite, the corner is pushed out to infinity, an ideal vertex. So the corners of $\{3, 4, 3, 4\}$ live infinitely far away, and the cross-section around one of those ideal corners is exactly the cubic grid of flat 3D Euclidean space.

Result 7 (The cusp is exact flat 3D space) Because the vertex figure of $\{3, 4, 3, 4\}$ is the infinite cubic honeycomb $\{4, 3, 4\}$, every vertex is ideal and the cross-section at each ideal vertex is the flat 3D cubic lattice. Physical space is recovered with no approximation, as the exact vertex figure rather than as an extracted slice.

9.5. Why paracompactness is a feature

A natural worry is that paracompactness, the presence of corners at infinity, is some kind of defect. It is the opposite. A regular honeycomb gives a clean flat physical layer only if it has a parabolic, that is Euclidean, element to foliate by. A compact honeycomb, such as $\{5, 3, 4\}$ or $\{7, 3\}$, has none. A horosphere cutting through it crosses its cells aperiodically, producing a thin disordered slab that is rough and sub-three-dimensional. This is why the compact substrates have no clean flat space. The ideal $\{4, 3, 4\}$ vertex figure of $\{3, 4, 3, 4\}$ instead gives a clean periodic cubic cross-section. So paracompactness is precisely what supplies the orderly 3D space.

In the standard taxonomy of hyperbolic Coxeter tilings, sorted by the signature of the Gram matrix of mirror angles, the classes run Euclidean, compact or Lannér, paracompact or Koszul, hypercompact or Vinberg, and Lorentzian. Paracompact means having at least one ideal vertex. The honeycomb $\{3, 4, 3, 4\}$ is paracompact, and the crystallographic spinor-bearing honeycombs are always in the paracompact or flat classes, never the compact one [39].

9.6. Hyperbolic, and built by reflection

The honeycomb does not fit in flat 4-space. The angles do not add up there. It closes only in negatively curved H^4 , a fact decided by the Gram-matrix signature of its mirror angles. Being hyperbolic, it grows exponentially. Count the 24-cells within radius r and the number explodes like e^{cr} , each shell larger than

everything inside it, which is the cosmic expansion. Almost all of the honeycomb lives out near its 3-sphere boundary at infinity, and the structure there, around the ideal corners, is the cubic 3D grid.

It is generated the same way as any regular honeycomb, by reflection. Start with one fundamental chamber and reflect it across its mirror walls forever. The reflections form the Coxeter group $[3, 4, 3, 4]$, which has five mirrors and the diagram $\circ 3 \circ 4 \circ 3 \circ 4 \circ$. Infinite complexity grows from a tiny local rule. A deep general principle sits at the boundary. The Euclidean cubic lattice at infinity is the affine, Euclidean-motion symmetry of the hyperbolic bulk. Flat space appears at infinity as the asymptotic symmetry of curved space.

9.7. How to picture it

No one directly visualizes a 4D hyperbolic honeycomb, so you climb a ladder of analogies. One dimension down, in flat 3D, cubes tile space as $\{4, 3, 4\}$ and the corner is just a point. Go one dimension up and bend the space negatively, and $\{3, 4, 3, 4\}$ is the analogue whose corner is a whole copy of that cubic tiling. The Escher disk gives a second image. It is like the hyperbolic disk where figures crowd toward the rim, but in 4D, with 24-cells crowding toward a 3-sphere boundary, and the crowding pattern at the rim spelling out cubic 3D space. The cusp funnel gives a third. Look down an infinitely long funnel in the honeycomb, and far down it the cross-section settles into a fixed cubic grid that never changes shape and only recedes. That steady cubic cross-section is the flat 3D space at the ideal corner. What you can hold in mind is 24-cell bricks, curved 4D mortar, ideal corners at infinity, and a cubic-grid cross-section at each corner.

9.8. The control and the substrate

Throughout the paper $\{5, 3, 4\}$, the dodecahedral honeycomb of 3D hyperbolic space, serves as a control. It is the closest three-dimensional analogue of the substrate, close enough to picture (Figure 2), and comparing the two makes the case for $\{3, 4, 3, 4\}$ concrete.

feature	$\{5, 3, 4\}$, the 3D control	$\{3, 4, 3, 4\}$, the preferred substrate
lives in	hyperbolic 3-space	hyperbolic 4-space
cell	dodecahedron $\{5, 3\}$	24-cell $\{3, 4, 3\}$
neighbours per cell	12	24
vertex figure	octahedron, finite	cubic honeycomb $\{4, 3, 4\}$, infinite
corners	ordinary points	ideal, at infinity
flat slice	flat 2D	flat 3D, the cubic grid
boundary at infinity	2-sphere	3-sphere
spinor coin	no, icosahedral $1 + 3 + 3' + 5$	yes, D_4 as $8v + 8s + 8c$

Table 10: The 3D control honeycomb against the 4D preferred substrate.

The headline is that $\{5, 3, 4\}$ gives flat 2D slices while $\{3, 4, 3, 4\}$ gives flat 3D slices. That one extra dimension is exactly what ordinary physical space needs. And only the 24-cell coin carries the spinor. The icosahedral coin of $\{5, 3, 4\}$ decomposes as $1 + 3 + 3' + 5$, which carries no spinor, while the D_4 coin decomposes as $8v + 8s + 8c$, which does [43].

This is the cleanest candidate for where flat 3D physical space comes from. It is the cubic cusp of a 4D hyperbolic bulk, recovered with no approximation, paired with a spinor-carrying cell. Which findings of the paper live on which layer is the subject of the later robustness audit. Spin and the gauge structure live in the

bulk coin, matter propagates on the cusp, and the two are connected by a projection developed in the matter chapters.

10. Coxeter Groups and Reflections

The honeycomb is not built by placing 24-cells by hand. It is generated from one tiny chamber by mirrors. That is the theory of *Coxeter groups*, the algebraic engine under the geometry, and it is worth its own treatment because it is the mechanism behind both the substrate and the flat 3D cusp. The intuition is a kaleidoscope. A few mirrors set at exact angles multiply one small region into an infinite symmetric world. In $\{3, 4, 3, 4\}$ those mirrors generate a 24-cell crystal whose ideal corners expose the flat cubic $\{4, 3, 4\}$, ordinary 3D space.

10.1. Reflections as generators

For a 4D Coxeter honeycomb the fundamental chamber is a 4-simplex with five walls, so the group has five generators s_0, s_1, s_2, s_3, s_4 , each reflecting the chamber across one wall. A reflection done twice is the identity, $s_i s_i = e$, so each generator is its own inverse. Composing them gives a word. A word like $s_2 s_4 s_1 s_0$ means reflect across wall 0, then 1, then 4, then 2, and every such word lands on some chamber of the infinite honeycomb. The group is the set of all reflection words, taken modulo the relations that identify equivalent ones.

A single reflection reverses orientation, producing a mirror image. Even-length products preserve orientation and odd-length products reverse it. Repeated reflection does not decorate the space, it builds it. One chamber becomes two, two become many, and the mirror process recursively fills all of H^4 .

10.2. The relations are the mirror-angle integers

If two mirrors meet at angle π/m , the product of their reflections is a rotation by $2\pi/m$, which closes after m steps, so $(s_i s_j)^m = e$. A handful of integers therefore fixes the whole infinite honeycomb. For $[3, 4, 3, 4]$ the neighbouring pairs satisfy $(s_0 s_1)^3$, $(s_1 s_2)^4$, $(s_2 s_3)^3$, and $(s_3 s_4)^4$, and all non-adjacent mirrors are perpendicular, with $m = 2$. The compressed picture is the *Coxeter-Dynkin diagram*, one node per mirror with edge labels for the angles, $\circ 3 \circ 4 \circ 3 \circ 4 \circ$ [15].

The chain of dependence runs cleanly downhill. The numbers fix the mirror angles, the angles fix the chamber, the chamber's reflections generate the group, the group generates the honeycomb, and the honeycomb fixes the large-scale geometry.

object	group	diagram
$\{7, 3\}$	$[7, 3]$	$\circ 7 \circ 3 \circ$
$\{5, 3, 4\}$	$[5, 3, 4]$	$\circ 5 \circ 3 \circ 4 \circ$
$\{3, 4, 3, 4\}$	$[3, 4, 3, 4]$	$\circ 3 \circ 4 \circ 3 \circ 4 \circ$

Table 11: Three regular hyperbolic tessellations with their Coxeter groups and diagrams.

10.3. One chamber equals one group element

The starting chamber is the identity. Reflecting across s_0 reaches the chamber labelled s_0 , then s_3s_0 , and so on, so every chamber gets an address as a word in the generators. The group acts freely and transitively on chambers, which means one chamber corresponds to exactly one group element. Neighbours are found by multiplying by a single generator, and the entire honeycomb can be listed as pure symbolic words with no coordinates at all. This is the deep payoff. Geometry becomes word combinatorics.

Result 8 (Geometry as words) The chambers of a regular Coxeter honeycomb are in exact bijection with the elements of its Coxeter group. Adjacency, growth, and address are all computable from words in the five generators, with no floating-point coordinates required.

10.4. The word problem and exact construction

Different words can name the same chamber. Beyond the cancellation $s_i s_i = e$ there are *braid relations* such as $s_i s_j s_i = s_j s_i s_j$ when $m = 3$. The word problem, the question of when two words name the same element, is exactly solvable for Coxeter groups. Cancel repeats, apply braid moves, reduce to the shortest form, and among the shortest forms take the lexicographically smallest, the *shortlex* normal form. Two words name the same chamber exactly when they share a normal form.

This matters in practice. Deep in hyperbolic space, coordinate points crowd toward the boundary and distinct chambers become numerically indistinguishable, the float-precision wall. The word engine never asks whether two points are nearly equal. It asks whether two words reduce to the same normal form, which is exact integer logic. For $\{3, 4, 3, 4\}$, one dimension up and with ideal vertices, this exactness matters even more. Coordinate pictures are only shadows. The exact object is the group.

10.5. Products of two reflections, the basic engines

Two reflections are the basic engine of geometry, and the result depends entirely on how the two mirror hyperplanes meet.

how the mirrors meet	the product	what it builds
they cross at angle π/m	a rotation by $2\pi/m$	rosettes, cyclic structure around a face
they are ultraparallel	a translation along a perpendicular	a geodesic line or tube
they meet only at infinity	a parabolic motion	a horosphere, a flat Euclidean layer

Table 12: The three outcomes of composing two reflections.

This three-way split is the core vocabulary. In $\{3, 4, 3, 4\}$, crossing mirrors build the local fourfold rotation, since the final 4 means four 24-cells meet around each triangular face. Ultraparallel mirrors build a geodesic tube of 24-cells, a one-dimensional transport spine through the 4D bulk, the higher-dimensional analogue of the $\{5, 3, 4\}$ sliver. Asymptotic mirrors build the all-important flat layer.

10.6. Parabolics build the flat 3D cusp

This is the most important construction for the substrate. A *parabolic* motion fixes one boundary point, has no fixed point inside hyperbolic space, preserves the horospheres centred on that boundary point, and acts on

them exactly like a Euclidean translation. The dimension ladder is direct. In H^2 parabolics build flat 1D horocycles, in H^3 flat 2D horospheres, and in H^4 flat 3D horospheres.

Because $\{3, 4, 3, 4\}$ lives in H^4 , its horospheres are intrinsically flat 3D spaces. And because its ideal vertex figure is $\{4, 3, 4\}$, the horosphere near an ideal vertex is not just abstract flat 3D space, it carries the cubic honeycomb. So the parabolic or affine subgroup at an ideal vertex builds the flat cubic 3D layer directly. This is cleaner than $\{5, 3, 4\}$, which is compact, has no ideal vertices, and has no exact parabolic cusp subgroup, so its horospheres cut the dodecahedral cells aperiodically into a rough slab. In $\{3, 4, 3, 4\}$ the flat 3D cubic layer is not an extracted slice. It is the exact vertex figure.

Result 9 (The flat layer is the affine subgroup) Inside the hyperbolic group $[3, 4, 3, 4]$ sits the Euclidean affine subgroup $[4, 3, 4]$, the symmetry of the cubic honeycomb. Flat 3D space is therefore not added to the model later. It is the vertex figure, encoded in the group from the start.

10.7. Richer compositions and the subgroup dictionary

Longer products give mixed motions. A *screw*, a rotation combined with a translation, builds a helix or a chiral chain, and it is richer in 4D where rotations can occur in two perpendicular planes at once. A *glide*, a reflection combined with a translation, builds a staggered mirrored chain, a marching mirror. A rotary reflection builds alternating turned-and-flipped layers. Choosing a subgroup of $[3, 4, 3, 4]$ chooses which large structure you build.

subgroup	structure	dimension and type
one reflection	a mirror wall	3D hyperbolic subspace
two crossing	a rotation or rosette	local, finite or cyclic
translation subgroup	a sliver or tube of 24-cells	1D geodesic
screw subgroup	a helical or chiral tube	1D twisting
affine subgroup $[4, 3, 4]$	the cubic honeycomb layer	3D flat, the cusp
full group $[3, 4, 3, 4]$	the entire honeycomb	4D hyperbolic

Table 13: Subgroups of $[3, 4, 3, 4]$ and the structures they generate.

The cleanest classification is by fixed points on the 3-sphere boundary at infinity. Two boundary fixed points give a line or tube, one gives a flat 3D horosphere, and an interior fixed set gives a rotation. The one-fixed-point parabolic case is the mechanism by which the 4D hyperbolic crystal exposes flat 3D space.

10.8. From chambers to 24-cells, and the exceptional symmetry

A 24-cell is assembled from many chambers, and the cell stabilizer, all reflections except the final outward generator, is the cell's own symmetry group. For $[3, 4, 3, 4]$ the cell is $\{3, 4, 3\}$ and its stabilizer is $[3, 4, 3]$, the full reflection group of the 24-cell, the 4D group associated with F_4 of order 1152. So the construction has two layers. Chambers assemble into one 24-cell under $[3, 4, 3]$, and 24-cells assemble into the honeycomb under $[3, 4, 3, 4]$, with the outward generator attaching one 24-cell to the next across a shared octahedral facet. The cell stabilizer being the exceptional F_4 group is why $\{3, 4, 3, 4\}$ feels denser and more algebraically special than the icosahedral $[5, 3]$ of $\{5, 3, 4\}$.

10.9. The three-layer reading and the universal machine

The reflection grammar hands the model exactly three structural layers. The bulk is the full group $[3, 4, 3, 4]$, the 4D hyperbolic substrate of 24-cells. The sliver is a translation subgroup, a 1D geodesic tube for transport and long-range probes. The flat emergent layer is the parabolic or affine subgroup at an ideal vertex, the cubic $\{4, 3, 4\}$ that is ordinary 3D physical space. The central image is a reflection-generated hyperbolic 4D crystal whose parabolic and ideal structure is ordinary flat 3D space.

The honeycombs $\{7, 3\}$, $\{5, 3, 4\}$, and $\{3, 4, 3, 4\}$ are not separate inventions. They are outputs of one machine. Change only the integer symbol and the same reflection logic generates a different regular hyperbolic world. Choose the diagram, build the Coxeter matrix, generate words, reduce to normal form, enumerate chambers, group them into cells, and read off adjacency and growth. Tiny inputs, infinite outputs. The geometry is a group first, and the picture is the group made visible. The whole structure is forced by symmetry, not placed by hand.

11. The Staircase to $\{3, 4, 3, 4\}$

No one found $\{3, 4, 3, 4\}$ by seeing it. It was reached by slowly recognizing deeper structural patterns in symmetry, regularity, reflection, curvature, and dimension. The path from a checkerboard to $\{3, 4, 3, 4\}$ is essentially the history of geometry unfolding, and the lesson it teaches is the one the whole theory rests on. This geometry is forced, not chosen.

11.1. The conceptual staircase

Each step generalized exactly one thing. First adjacency, then symmetry, then recursion, then dimension, then curvature, and finally ideal infinity.

1. A checkerboard.
2. Regular tilings, of which there are only three flat ones, $\{3, 6\}$, $\{4, 4\}$, and $\{6, 3\}$.
3. The Platonic solids, where local angle constraints fix the global structure.
4. Recursive regularity, captured by the Schläfli symbol, so that a cube is $\{4, 3\}$.
5. Higher dimensions, where the polytopes of 4D, the polychora, appear.
6. The 24-cell, an exceptional 4D-only shape with no 3D analogue.
7. Reflection groups, where geometry is generated from mirrors rather than drawn.
8. Coxeter diagrams, where a tiny integer diagram encodes a whole geometry.
9. Curvature freedom, where angle sums force a spherical or hyperbolic world.
10. Hyperbolic space, where the parallel postulate becomes optional and room explodes.
11. Ideal structures, the vertices at infinity, horospheres, and cusps.

12. Hyperbolic 4-space honeycombs.
13. The cubic-honeycomb boundary, since the vertex figure of $\{3, 4, 3, 4\}$ is $\{4, 3, 4\}$, flat 3D space.
14. The honeycomb $\{3, 4, 3, 4\}$.

The decisive milestones are easy to name. Schläfli’s higher-dimensional polytopes and the 24-cell came in the 1850s. The discovery of hyperbolic geometry by Lobachevsky, Bolyai, and Gauss came in the 1820s through 1840s. The reflection-group viewpoint and Coxeter’s grand unification arrived in the 1930s, showing that regular polytopes, reflection groups, hyperbolic geometry, and higher dimensions are all one recursive symmetry language [15]. The realization that the ideal corner of $\{3, 4, 3, 4\}$ is the ordinary cubic lattice, Euclidean 3-space appearing on the boundary of hyperbolic 4-space, is the structural payoff the whole staircase climbs toward.

11.2. The hidden idea

The deepest insight in the whole climb is that geometry can be generated algebraically. Not by drawing shapes by hand but by symmetry generators, reflection relations, and combinatorial rules, the integers of the Schläfli symbol. That changed geometry permanently, and it is exactly how the substrate is built. The crystal is the reflection orbit of one chamber, a finite rule unfolding forever.

Result 10 (Geometry from a finite rule) The whole of $\{3, 4, 3, 4\}$, infinite and four-dimensional, is the orbit of a single fundamental chamber under five reflections. A finite local rule generates the entire large-scale structure, just as the model’s dynamics generate physics from a single local update.

11.3. Why it feels discovered, not invented

These structures are rigid, highly constrained, and mathematically forced. Nobody arbitrarily invented $\{3, 4, 3, 4\}$. Once the rules of regularity and curvature are accepted, it almost reveals itself, as one of the maximally symmetric ideal extensions of 24-cell geometry into negatively curved infinite space. This discovered-not-invented character is the theory’s central claim about its own substrate. The selection made later in the paper does not pick $\{3, 4, 3, 4\}$ from a menu. The requirements, a crystallographic spinor coin, curvature, and a 3D cusp, force it.

The staircase is the same logic the theory runs on, local repeating rules generating large-scale order, geometry from reflection, and infinite hyperbolic order emerging naturally from local laws. The substrate is not designed shape by shape. It is the inevitable unfolding of a small set of constraints, with $\{3, 4, 3, 4\}$ standing as one grand culmination of that chain.

Part III

Emergent Physics

12. Spin and Fermions

The coin carries spin, matter is fermionic, and the rule makes matter stable, all without adding anything to the base. The three facts come out of the substrate's own symmetry and the measured behavior of its collision, not from any extra ingredient grafted on for the purpose. This chapter reads them off in order.

Two terms recur throughout, so a plain statement of each helps. A *spinor* is the mathematical object that describes a particle with half-integer spin. Its signature oddity is that it must be turned twice around, through 720 degrees rather than 360, before it returns to itself. A *fermion* is a particle of matter, an electron or a quark, no two of which can occupy the same state, which is why matter takes up space at all. Fermions have half-integer spin, so they are described by spinors. The two ideas are linked, and the substrate delivers that link rather than presupposing it.

12.1. The spinor coin

The 24 local directions of the cell are the 24-cell, equivalently the D_4 root system. Under that symmetry the 24 directions decompose as $8v + 8s + 8c$, the $SO(8)$ vector together with its two spinors, with triality, the order-three outer symmetry S_3 , permuting the three eights. The two pieces $8s$ and $8c$ are genuine half-integer-spin representations. This is the precise sense in which the substrate carries spin at the base. It is not assumed and not put in by hand. It is a fact about how the local direction set transforms under its own symmetry group.

Proposition 3 (The spinor decomposition) The 24 directions of the 24-cell, as the D_4 root system, decompose under $SO(8)$ as $8v + 8s + 8c$. The summands $8s$ and $8c$ are half-integer-spin representations, so the substrate carries spinors directly. Triality permutes the three eights and is the seed of three generations.

The control case sharpens the claim. The 12 directions of $\{5, 3, 4\}$ decompose under the icosahedral symmetry as $1 + 3 + 3' + 5$, all of integer spin, with no spinor anywhere in the list. A substrate built on those 12 directions would carry no fundamental spinor at all. That contrast is one of the reasons the selection argument lands on $\{3, 4, 3, 4\}$ rather than $\{5, 3, 4\}$, and it is developed further where the crystallographic and spinor requirements are set side by side.

12.2. Matter is fermionic

A word of care first. A *self*, in the broad sense used later in the paper, is any persistent self-maintaining pattern, and these range from a single vibe through ever larger patches and networks all the way up to the entire mesh. Most selves are not particles. What concerns us here is one particular kind of localized excitation, the topological soliton, which is the matter object the theory identifies with a fundamental fermion.

Definition 4 (Matter soliton) A *matter soliton* is a topological knot in the tone field, a *hopfion*, classified by its Hopf charge in $\pi_3(S^2) = \mathbb{Z}$. The integer charge counts how the tone configuration links with itself, and it cannot change under any smooth evolution of the field.

A hopfion of odd Hopf charge picks up a minus sign under a 2π rotation. This is the Finkelstein-Rubinstein result, sharpened by the spin-statistics argument of Wilczek and Zee [20, 61]. A minus sign under a single full turn is the defining mark of a fermion. So the spin-statistics connection here is geometric. A matter soliton of odd charge is automatically a fermion, with nothing assumed about it beyond its topology. In the carrying image such a soliton is a braid in the fabric, a knot of fire, and the topology simply is the weave.

Proposition 4 (Odd-charge solitons are fermions) A topological soliton in the tone field with odd Hopf charge transforms with a sign change under a 2π rotation. By the spin-statistics connection it is a fermion. The matter solitons are therefore fermions as a consequence of their topology, with no separate statistical postulate.

It pays to keep the two ways spin appears in this theory cleanly apart, because they answer different questions. The first is the *coin spinor*, the D_4 pieces 8s and 8c. These are the fundamental fermion fields, and they genuinely require $\{3, 4, 3, 4\}$, since the 12-direction alternative carries no spinor at all. The second is the *topological route*, the identification of a matter soliton with a hopfion and hence with a fermion. That route is substrate-independent and works even on $\{5, 3, 4\}$, because it rests only on $\pi_3(S^2)$ and not on the direction set. The coin spinor is the load-bearing one for the fundamental matter fields and for the $SO(10)$ gauge story of the next chapter but one. The topological route is why the matter solitons come out fermionic. The two agree, but the precise selection claim is that the coin spinor forces $\{3, 4, 3, 4\}$, not that spin of any kind does.

12.3. Matter forms and persists

Why a stabilizer is needed at all is Derrick’s theorem. Rescale a soliton by a factor λ . The exchange energy scales as λ^{d-2} and a Skyrme term scales as λ^{d-4} . In three dimensions exchange alone scales as λ , so the soliton collapses, and a higher-gradient Skyrme term is required to arrest it [48]. In the two-dimensional cusp of $\{5, 3, 4\}$ exchange is scale-marginal and no stabilizer is needed, a neat tie to the bulk-cusp dimension split. The live question is therefore whether the rule’s collision supplies a Skyrme term of the stabilizing sign. The Skyrme coefficient appears at the higher-gradient, Burnett order of the collision’s gradient expansion, and the collision is the rule’s last free part, fixed only by being conserving, reversible, local, and lattice-symmetric. So the matter of stability reduces to a definite sign of a definite quantity.

The sign was measured. The fermion-sea energy of a double-twist periodic texture was evaluated with the kernel polynomial method, a Chebyshev expansion of the spectral density carried out with a stochastic trace on the GPU. A single-helix control was subtracted to isolate the Skyrme contribution from the bulk. The isolated Skyrme term came out positive, that is, stabilizing.

Result 11 (The Skyrme stabilizer is positive) The isolated Skyrme coefficient of the collision, measured as the fermion-sea energy of a double-twist periodic texture with a single-helix control subtracted [43], is positive. Solitons in the tone field are therefore stable on their own, with no fine-tuning. Matter forms and persists.

The earlier attempts that used localized solitons were confounded by surface-dominated sea energy, and the double-twist periodic method is the clean measurement that replaces them. Even had the result come out

ambiguous, matter formation would by itself select the collision member with the stabilizing sign. The result being positive makes that fallback unnecessary.

The matter sector comes from the substrate. The coin carries spinors, the matter solitons are fermions, and the rule stabilizes matter against collapse. Fermions are not added to the model. They emerge as the stable knots of the substrate, and the half-integer spin that defines them is already present in the way the cell's 24 directions transform.

13. Emergent Dirac and Lorentz

The discrete walk reproduces relativistic propagation. The Dirac dispersion and a finite light cone are read off a simulation of the rule rather than assumed. What follows is a measurement of how the substrate already moves, not an importation of relativity into it.

A few terms are worth stating plainly. The *Dirac equation* is the law for how an electron-like particle moves when both quantum mechanics and special relativity apply at once, and its dispersion is the relation between a wave's energy and its wavelength. *Lorentz invariance* is the content of special relativity, that the laws look the same to everyone moving at constant speed and that there is one finite top speed, light. A *light cone* is the reach of that top speed, the set of places a signal can get to from a given point.

13.1. The light cone

A pulse spreads one cell per beat. That is a finite, frame-independent maximum speed, $z = 1$, and it is the light cone. For the massless case the spreading is exactly the discrete shift, a permutation of cell contents, so continuity is not invoked anywhere in the statement. The light cone built from the 24 directions is also round, that is, isotropic at scale, a point taken up where isotropy is measured.

Result 12 (A discrete light cone) A pulse on the substrate spreads exactly one cell per beat [43]. The maximum signal speed is finite and frame-independent, with dynamical exponent $z = 1$. In the massless case the propagation is the discrete shift itself, so the light cone is exact and uses no continuum limit.

13.2. The Dirac dispersion

Simulate the discrete two-component walk, a left mover and a right mover, and measure its dispersion $E(k)$ by Fourier transforming the time series of a tracked cell. Two cases arise. In the massless case the rule is the pure discrete shift, the measured dispersion is $E = k$ exactly, and the light cone is the shift itself with no continuity assumed. In the massive case a coin rotation by the mass angle m is added, the measured dispersion obeys $\cos E = \cos m \cos k$, the lattice Dirac dispersion, and the rest mass is $E(0) = m$. Mass is the coarse-grained mixing rate, nothing more.

Result 13 (The lattice Dirac dispersion) The discrete two-component walk has measured dispersion $E = k$ in the massless case and $\cos E = \cos m \cos k$ in the massive case [43], the latter being the lattice Dirac dispersion with rest mass m . The Dirac dispersion is thus the rule's own measured dispersion.

The reading is that the complex quantum amplitudes are the emergent description of the real discrete walk's oscillation modes. The quantum layer is emergent. The base is the real discrete walk, and the amplitudes are how that walk's modes are most economically written down. This is consistent with the cellular-automaton

reconstructions of the free Dirac field by D’Ariano and Bisio [16, 8], where the same lattice dispersion arises from a local, reversible update.

13.3. Why masslessness needs an even-faced cell

There is a clean geometric precondition for an exactly massless mode. The cell must have opposite faces. On an odd-faced cell the instruction to go straight lands between two faces, which forces a small zigzag of order π/F at every step, an irreducible scatter. The lightest particle then has a floor mass of order $(\pi/2F)^2$, and nothing is exactly massless. An even-faced cell with opposite pairs lets a signal go dead straight, with mixing $\lambda = 0$, and so it hosts exactly massless modes. The 24-cell is even-faced. Its 24 facets form 12 opposite pairs, so $\{3, 4, 3, 4\}$ permits the exactly-massless $E = k$ mode above, where a generic odd tiling could not. The geometry is not cosmetic. The masslessness rests on the cell’s antipodal structure.

Proposition 5 (The mass floor) An exactly massless mode requires a cell with opposite faces. On an odd-faced cell with F faces, going straight forces a zigzag of order π/F per step, giving every mode a floor mass of order $(\pi/2F)^2$. The 24-cell’s 24 facets form 12 opposite pairs, so $\{3, 4, 3, 4\}$ admits the exactly-massless $E = k$ mode.

The telegraph picture ties the cases together with a single knob. A persistent walk that scatters at rate λ is the telegrapher’s equation, equivalently the Klein-Gordon or one-dimensional Dirac equation, with mass proportional to λ . Setting $\lambda = 0$ gives the massless wave with $z = 1$. A finite λ gives the relativistic massive case. Taking λ to infinity recovers ordinary diffusion with $z = 2$. One mechanism covers three limits.

This closes the bridge from the base to quantum mechanics. Relativistic quantum mechanics is an emergent limit of the discrete substrate, with the massless light cone being exactly discrete, needing no continuum, and mass emerging as a mixing rate.

13.4. Caveat

The cleanest reading is the one-dimensional, two-component demonstrator. The full 24-direction version is the bulk wave, and Lorentz invariance on the physical cusp carries a small cubic anisotropy that washes out in the infrared. The precise statement is that the dispersion and the light cone are exact in the demonstrator and isotropic in the limit, with the residual lattice anisotropy treated where isotropy and falsifiability are discussed.

14. Gauge and the Standard Model

Combining the coin with the tone gives a grand-unified gauge structure that breaks to the Standard Model, and it comes with two sets of quantitative predictions. The gauge group is not chosen. It is forced by the substrate’s own content once the tone is admitted as a fifth axis, and the symmetry breaking that yields the Standard Model is automatic rather than tuned.

A few terms first. A *gauge* field is the kind of field that carries a force, the electromagnetic field being the familiar one. The word gauge names a redundancy, since there are many equally valid ways to write down the same physical field, and that freedom of description is the force’s symmetry. A grand unified theory, such

as $\text{SO}(10)$ here, is a single larger symmetry that holds the Standard Model's three forces as pieces, joined into one at high energy.

14.1. $\text{SO}(10)$ from coin and tone

The coin's D_4 root system, $\text{SO}(8)$, together with the ternary tone treated as an extra weight, builds the $D_5 = \text{SO}(10)$ root system. The 16-dimensional spinor of $\text{SO}(10)$ is exactly one generation, with $16 = 8s + 8c$ split by the tone sign, the $8s$ carrying tone $+$ and the $8c$ tone $-$. So the gauge group and one full family of matter both come from coin and tone.

Theorem 2 (SO(10) from coin and tone) The D_4 root system of the coin, extended by the ternary tone as a fifth axis, is the $D_5 = \text{SO}(10)$ root system. The $\text{SO}(10)$ spinor $16 = 8s + 8c$, split by tone sign, is exactly one generation of matter. The grand-unified gauge group and one family arise together from the substrate's coin and tone.

The tone is needed, not a free add-on. The coin alone, $\text{SO}(8) = D_4$, provably cannot hold the Standard Model, and the reason is an orthogonality obstruction. The group $\text{SU}(3) \times \text{SU}(2)$ embeds in $\text{SO}(N)$ only if there is an A_2 root system with a root orthogonal to all of it. Every D_4 root has exactly two non-zero entries, so no D_4 root is orthogonal to a full A_2 . The Standard Model does not fit in $\text{SO}(8)$.

Proposition 6 (The orthogonality obstruction) The Standard Model gauge group does not embed in $\text{SO}(8) = D_4$. An embedding of $\text{SU}(3) \times \text{SU}(2)$ requires an A_2 root system with a root orthogonal to it, but every D_4 root has exactly two non-zero entries and so cannot be orthogonal to a full A_2 . Adding the ternary tone as a fifth axis removes the obstruction, since D_4 's 24 roots plus the tone sign give the 40 roots of $D_5 = \text{SO}(10)$, the 16 new roots being $8v$ tensored with the tone sign.

The tone is forced by the obstruction, not chosen. By root counting, D_4 's 24 roots plus the tone sign give $D_5 = \text{SO}(10)$'s 40 roots, and the 16 extra roots are precisely $8v$ tensored with the tone sign, so the tone promotes $\text{SO}(8)$ to $\text{SO}(10)$. A further bonus comes for free. The 16 of $\text{SO}(10)$ is anomaly-free, so anomaly cancellation and charge quantization are automatic consequences of the spinor structure rather than separate conditions to be imposed.

14.2. Breaking to the Standard Model

A self-condensate, playing the role of the grand-unified Higgs, breaks $\text{SO}(10) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ along the Georgi-Glashow chain [23, 22], with $16 = 1 + 10 + \bar{5}$ under $\text{SU}(5)$, the complete matter content of one generation, quarks, leptons, and a right-handed neutrino. Any self-condensate gives $\text{SU}(5)$ with 20 unbroken roots, so the breaking is automatic rather than fine-tuned. Triality gives three generations.

Proposition 7 (Automatic breaking, and the self as Higgs) A self-condensate breaks $\text{SO}(10)$ to $\text{SU}(5)$ for any choice of condensate, leaving 20 unbroken roots, and the chain continues to $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$. Under $\text{SU}(5)$ the generation decomposes as $16 = 1 + 10 + \bar{5}$. The breaking field is itself a self-condensate, so the matter sector and the symmetry-breaking sector are the same object.

The clean conceptual point is that the breaking field is a self-condensate, so the self is the Higgs. The matter sector and the symmetry-breaking sector are not two things but one.

14.3. The weak mixing angle

Summing over the fermions of one generation gives the weak mixing angle at unification,

$$\sin^2 \theta_W = \frac{\sum T_3^2}{\sum Q^2} = \frac{3}{8}$$

exactly at the unification scale [25]. Running it down with standard renormalization group flow, it reaches the measured 0.231 for a supersymmetric spectrum, where the three gauge couplings unify cleanly near 10^{16} GeV, or about 0.21 for the bare Standard Model.

Theorem 3 ($\sin^2 \theta_W = 3/8$ at unification) Summed over one generation, $\sin^2 \theta_W = \sum T_3^2 / \sum Q^2 = 3/8$ exactly at the unification scale. Run to the weak scale it gives the observed 0.231 for a supersymmetric spectrum and about 0.21 for the bare Standard Model.

So the prediction is confirmed against experiment, and it additionally points to low-energy supersymmetry, since the cleaner agreement comes with the supersymmetric running and the matching gauge-coupling unification.

14.4. The grand-unified mass relations

Three mass relations follow at the unification scale. The first is b - τ unification, $m_b = m_\tau$ at the grand-unified scale from the shared Yukawa coupling, running to $m_b/m_\tau \approx 2.3$ at M_Z , against the observed 2.35. The second is the determinant relation $\det(M_{\text{charged lepton}}) = \det(M_{\text{down}})$, which follows from the discrete fact that the trace of the Yukawa over the 16 vanishes. The third is the Georgi-Jarlskog factor-three relations [24], the weakest of the set and the most model-dependent.

14.5. The non-abelian gauge field

The gauge field is genuine, and this was verified directly. Using $\text{SO}(3)$ link matrices as the prototype, identical in structure to the $\text{SO}(10)$ case, the Wilson loop, the trace of the ordered product of links around a loop, is gauge-invariant [62], the holonomy is non-trivial, which is curvature, and the link matrices do not commute, which is the non-abelian content.

Result 14 (A non-abelian gauge field on the substrate) Built from $\text{SO}(3)$ link matrices on the substrate, the Wilson loop is gauge-invariant, its holonomy is non-trivial, and the links do not commute [43]. The gauge field is genuine, curved, and non-abelian, with the $\text{SO}(10)$ case identical in structure.

The symmetric collision that makes $\text{SO}(10)$ the local symmetry is forced in the infrared by triality. The cleaner principle underneath is that a local conservation law is a Gauss law, so the rule's exact charge conservation gives $\text{U}(1)$ electromagnetism for free, taken up where electromagnetism is treated, and a non-abelian gauge group reduces to the collision's symmetry. Gauging $\text{SO}(10)$ is then just the existence of an $\text{SO}(10)$ -symmetric collision, which the substrate supplies.

The whole gauge and matter sector, together with the predictions $\sin^2 \theta_W = 3/8$ and the mass relations, comes from coin and tone, with the Standard Model emerging by automatic breaking and with anomaly cancellation included for free.

15. Gravity

Gravity emerges from the geometry. It is curvature-sourced and bulk-mediated, and it comes out with the right law. There are two distinct gravitational objects here, the Newtonian potential on the flat cusp and the holographic correlator in the curved bulk, and they should not be conflated. Beyond them the substrate reaches the full apparatus of general relativity at the effective level, with the one exception noted at the end.

15.1. The flat three-dimensional Newtonian $1/r$

In the three-dimensional cusp the static potential of a point source is the lattice Green’s function of the Laplacian. Measured by the difference method, comparing $G(r)$ against $G(r_0)$ with the regular cosine integrand, which cancels the $k = 0$ zero-mode contamination that wrecked naive finite-patch sums, the result is a clean $1/r$, of the form $G(r) = a/r + b$ with $a \approx 0.083 \approx 1/(4\pi)$. This is the Newtonian potential, and it is $1/r$ because the cusp is three-dimensional. The force is dimension-resolved.

Result 15 (The Newtonian $1/r$ potential) The static lattice Green’s function in the three-dimensional cusp, measured by the difference method [43], is $G(r) = a/r + b$ with $a \approx 0.083 \approx 1/(4\pi)$. The Newtonian potential goes as $1/r$, fixed by the cusp being three-dimensional.

15.2. The bulk-mediated holographic correlator

A word on the term. *Holography* is the idea that everything happening in a chunk of gravitating space can be written on its lower-dimensional boundary, the way a flat hologram stores a three-dimensional scene [38]. A correlator measures how strongly two points feel each other.

Between two boundary points the bulk-mediated correlator decays as a clean $1/r^2$. It was computed two ways. The first is the exact Bethe-lattice cavity recursion, which gives $\mu = 1/b$ universally and exponent $\alpha = 2$, validated against a directly-solved finite tree. The second is a tree-tensor contraction in the manner of the multiscale entanglement renormalization ansatz [47], which gives roughly $1/r^{1.93}$. This is the holographic two-point function, a different object from the flat-layer Newtonian $1/r$.

Result 16 (The holographic $1/r^2$ correlator) The bulk-mediated boundary-to-boundary correlator decays as $1/r^2$. The exact Bethe-lattice cavity recursion gives exponent $\alpha = 2$ with $\mu = 1/b$, validated against a finite tree, and a tree-tensor contraction gives $1/r^{1.93}$ [43]. This is the holographic two-point function, distinct from the Newtonian $1/r$.

15.3. Full general relativity, not only Newton

The substrate reaches full general relativity, to the level any emergent-gravity program reaches, not only the Newtonian limit. Four pieces make the case.

The first is the Einstein field equations along Jacobson’s entropic route [30]. Given the substrate’s measured area law $S = A/4G$ and the Unruh temperature $T = \kappa/2\pi$, the Clausius relation $\delta Q = T dS$ on every local Rindler horizon is equivalent to the full nonlinear $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$. The area-law coefficient fixes the Einstein coefficient to exactly 8π , with the verified value $2\pi/\eta = 25.133 = 8\pi$, the same G as Newton’s law, and the weak field gives Poisson’s equation with the correct 4π [41, 56]. So the area law, a measured substrate fact, is the Einstein equation read as an equation of state.

Result 17 (The Einstein equation as an equation of state) From the measured area law $S = A/4G$ and the Unruh temperature $T = \kappa/2\pi$, the Clausius relation on every local Rindler horizon is equivalent to $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$. The Einstein coefficient is fixed to 8π , verified as $2\pi/\eta = 25.133 = 8\pi$ [43], with the same G as the Newtonian law and the correct 4π in the weak-field Poisson equation.

The second is light bending. Because the metric curves time and space together, a photon deflects by $4GM/b$, twice the naive Newtonian amount, with the verified ratio 2.0000. This is the cleanest observable separating full general relativity from Newton, and the substrate gets it right.

Result 18 (Light bending equals $4GM/b$) A photon passing a mass M at impact parameter b deflects by $4GM/b$, twice the Newtonian value, with measured ratio 2.0000 [43]. The factor of two is the signature of full general relativity, and the substrate reproduces it.

The third is black-hole thermodynamics, complete and self-consistent. From $S = A/4$ and $T = 1/(8\pi M)$ follow the first law $dM = T dS$ with error around 10^{-9} , the Smarr formula $M = 2TS$ exactly, the Bekenstein bound saturated, a negative heat capacity $C = -8\pi M^2$ which is the evaporation instability, and a lifetime scaling as M^3 . The substrate's own de Sitter horizon carries the same thermodynamics, with numbers fixed by the measured growth, giving $T = H/2\pi$ and $\Lambda = 3H^2$.

The fourth is gravitational waves. A massless spin-two graviton lives on the flat cusp, with dispersion $\omega = |k|$, $z = 1$, and exactly two transverse-traceless polarizations. A binary radiates the quadrupole waveform, with gravitational-wave frequency twice the orbital frequency and power $P = (32/5)\mu^2 a^4 \omega^6$. The inspiral chirp follows $f \sim (t_c - t)^{-3/8}$, the LIGO exponent, with the verified value -0.375 .

15.4. Why curvature is essential

A flat lattice, the Euclidean $\{3, 4, 3, 3\}$ with the same D_4 coin, has no curvature. It therefore has no curvature-sourced gravity, no holographic boundary, and no expansion. This is why the substrate must be hyperbolic. The flat twin is a clean control that isolates curvature as the source of gravity, set against the hyperbolic case where the best of each dimension is compared.

Gravity is not added by hand. The flat three-dimensional cusp has a $1/r$ Newtonian potential, the curved bulk mediates a clean $1/r^2$ holographic correlator, and the full general-relativistic relations follow at the effective level, all falling out of the geometry.

15.5. Caveat, the open frontier

The naive finite-patch Green's function is unreliable, spoiled by the $k = 0$ mode and boundary contamination, and the difference method is the correct tool that supersedes the earlier confounded measurements. The one thing not done, and not done by anyone working from a micro-rule, is full nonlinear numerical relativity simulated cell by cell, a binary merger or a forming horizon evolved from the ternary tone directly, which not even lattice QCD attempts. What the substrate has is the effective theory together with the quantitative relations, the Einstein equation of state, the limits, the thermodynamics, and the waveforms, each tied to substrate-measured inputs, the area law, the light cone, the flat cusp, and the growth rate H . That is what it precisely means to say the substrate supports general relativity.

16. Electromagnetism

The rule that streams and collides the tone already conserves a charge. When that conservation is read at the level of the lattice it behaves as a discrete Gauss law, and the field it sources turns out to be an emergent $U(1)$ gauge field with the structure physicists expect. What follows shows the conservation law, the gauge field built on top of it, the recovery of the Coulomb potential, and the gauge-invariant observables that pin down the field's physical content. None of this is added by hand. It is what the charge rule does once the bookkeeping is followed carefully.

16.1. The Gauss law

The base rule conserves charge *locally*. The charge that flows into a cell over one beat equals the charge that flows out, cell by cell, with no global ledger needed. That statement is a discrete continuity equation, and a discrete continuity equation is a discrete Gauss law. The conserved current it defines is exactly the source term for the emergent $U(1)$ field. The conservation holds exactly and on every substrate tested, not approximately and not only in some limit, because it is built into the STREAM and COLLIDE moves rather than imposed afterward.

Result 19 (Discrete Gauss law) The base rule conserves charge per cell per beat. This local conservation is a discrete continuity equation whose conserved current sources an emergent $U(1)$ field. It holds exactly on every substrate examined [43].

16.2. The emergent gauge field

To probe the field directly one places a vortex gauge configuration on the lattice. Link phases wind around a plaquette so that the loop encloses a flux Φ . The Wilson loop, the sum of the link phases taken around a closed path, then equals the enclosed flux. The value is gauge-invariant. Applying a random gauge transformation $A \rightarrow A + d\lambda$ leaves the Wilson loop unchanged, which is the signature of a genuine gauge theory rather than a coordinate artifact. The Aharonov-Bohm phase picked up by a charge that encircles the flux is exactly Φ , independent of the shape of the loop and of the gauge. A loop that does not enclose the flux gives zero. The physical content of the field is therefore its gauge-invariant field strength, and the field is a $U(1)$ gauge theory in the ordinary sense, which is to say electromagnetism.

Result 20 (Gauge-invariant $U(1)$ observables) On a vortex configuration the Wilson loop equals the enclosed flux Φ , is invariant under $A \rightarrow A + d\lambda$, and produces an Aharonov-Bohm phase Φ for any encircling loop and zero for a non-encircling one [43]. The emergent field is a $U(1)$ gauge theory.

16.3. The Coulomb potential

The static sector of the emergent field is governed by the flat three-dimensional Laplacian. Its Green's function is the familiar $1/r$ Coulomb potential, recovered here by the same difference method used for the Newtonian potential on the flat cusp. The static $U(1)$ field around a point charge falls off as $1/r$, with no free constants beyond the overall scale. The electromagnetic and gravitational static potentials share this origin. Both are Green's functions of the cusp Laplacian.

16.4. The non-abelian extension

The Wilson-loop construction does not depend on the abelian case. Replacing the link phases with matrices gives a non-abelian gauge field, the same matrices that appear in the $SO(10)$ analysis of the coin-plus-tone sector. The Wilson loop is again gauge-invariant, the field is curved, and the link variables no longer commute. The abelian electromagnetic field is the simplest member of a family that the same machinery produces, which is the expected relationship between $U(1)$ and the larger gauge groups of the Standard Model.

Electromagnetism here is emergent rather than postulated. The charge rule's local conservation forces a $U(1)$ gauge symmetry, and the field that symmetry organizes carries the Coulomb $1/r$ potential and the gauge-invariant Wilson loops one would demand of a real gauge theory. That closes the base-to-gauge step for the $U(1)$ sector.

17. Isotropy and Emergent Symmetry Restoration

Any lattice has preferred directions, the lines along which its cells are wired. Physical space does not. It looks the same in every direction to very high precision. A discrete substrate therefore owes an account of how rotational invariance can emerge from a structure that, at the smallest scale, plainly does not have it. The 24-direction substrate gives a strong answer in the bulk, with a small and explicitly stated residual that lives on the cusp.

17.1. The 24-direction coin is isotropic to order four

Take the wave dispersion built from the local hops, $\omega^2(k) = \sum_d (1 - \cos(k \cdot d))$, summed over the local directions d . For the D_4 coin with its 24 directions the fourth moment is isotropic. The relevant tensor satisfies $\sum d^4 = 3 \sum d^2 d^2 = 12$, so the leading correction to the dispersion has no preferred axis. The dispersion is isotropic to order four, and the axis-versus-diagonal anisotropy vanishes at that order. A cubic $\mathbb{Z}^3$ or hypercubic $\mathbb{Z}^4$ lattice carries roughly an eight to nine percent anisotropy at the same order. The result was first established analytically and then confirmed on the GPU. The 24-direction light cone comes out round, with an isotropy ratio of 1.00, where the 8-direction cone is a faceted cross-polytope at 0.50.

Result 21 (D_4 isotropy to order four) The dispersion of the 24-direction D_4 coin is isotropic to order four, with $\sum d^4 = 3 \sum d^2 d^2 = 12$ and zero axis-vs-diagonal anisotropy, against roughly 8 to 9 percent for cubic or hypercubic lattices. The measured light-cone isotropy ratio is 1.00 for 24 directions and 0.50 for 8 [43].

17.2. Emergent symmetry restoration

The reason the 24-direction coin does better than a cubic lattice is triality, the extra symmetry of the 24-cell that enlarges its automorphism group to F_4 . Triality removes the order-four anisotropy outright. The group F_4 admits a single degree-four invariant where the hypercubic group B_4 admits two, and it is the second invariant that would carry the unwanted directional dependence. With only one degree-four invariant available, the leading anisotropic term has nowhere to live. In the long-wavelength limit the discrete F_4 symmetry therefore restores the continuous rotational symmetry $SO(8)$. Isotropy is not assumed. It emerges, and it emerges because of the particular symmetry the 24-cell carries.

Proposition 8 (Emergent $F_4 \rightarrow \text{SO}(8)$ restoration) Triality enlarges the 24-cell symmetry to F_4 , which has one degree-four invariant against two for B_4 . The single invariant kills the order-four anisotropy, so the discrete F_4 restores continuous $\text{SO}(8)$ rotational symmetry in the infrared.

17.3. The cusp caveat

This near-perfect isotropy is a property of the bulk, where the 24 directions live. Physical matter does not propagate in the bulk. It propagates on the flat cubic honeycomb $\{4, 3, 4\}$ that forms the ideal cusp, and that cubic lattice is anisotropic at order four. The residual is small. It scales as $(E/E_{\text{Planck}})^2$ and shows up as a prediction rather than a problem. The matter-propagation isotropy is therefore the cubic kind, sharp in the infrared with a tiny ultraviolet anisotropy, and it is distinct from the bulk’s much rounder isotropy. This bulk-versus-cusp distinction is the key subtlety. The residual ultraviolet anisotropy lives on the cusp, not in the bulk, and stating where it lives is part of being accurate about what the substrate does.

Rotational invariance is emergent on this substrate, and it is better realized on the 24-cell than on a cubic lattice. Triality buys isotropy out to order four, which a cubic lattice cannot match.

18. Cosmology and Hierarchy

The bulk grows, and its growth has two readings that turn out to be the same structure seen from different angles. Read as volume against time, the growth is a cosmic expansion. Read as a radial tree, it is the renormalization hierarchy that organizes the holographic boundary. Both fall out of the substrate geometry rather than being fitted to it. A third consequence, a slow global cooling toward the neutral tone, follows from pairing conservation with eternal growth.

18.1. The expansion law

The bulk shell sizes, meaning the spatial volume added at each radial step per beat, grow by a constant ratio $R \approx 11$. The first few shells run 1, 24, 456, 8376, $\dots$, which is exponential. Reading the three-dimensional spatial volume as a^3 , the scale factor is $a(t) \approx R^{t/3} = e^{Ht}$ with $H = \ln(R)/3 \approx 0.80$ per beat. Then $\ddot{a}/a = H^2 > 0$, an accelerating expansion of de Sitter type, with an effective cosmological constant $\Lambda = 3H^2$. The eternal growth of the bulk is itself a dark-energy-dominated expansion, not a term added separately. The flat reference lattices grow only polynomially, which confirms that the exponential behavior is a curvature effect rather than an artifact of counting. The same growth serves three roles at once. It is the time step, the arrow of time, and the spatial expansion, unified in one process.

Result 22 (Exponential de Sitter growth) Bulk shell sizes grow by a constant ratio $R \approx 11$, so $a(t) \approx e^{Ht}$ with $H = \ln(R)/3 \approx 0.80$ per beat, giving $\ddot{a}/a = H^2 > 0$ and an effective $\Lambda = 3H^2$. Flat references grow only polynomially [43].

18.2. The hierarchy

The bulk’s radial tree is a self-similar branching with a constant branching factor of about R . Its radial depth scales as the logarithm of the boundary size. That is the structure of a renormalization hierarchy, the holographic renormalization-group flow realized as a MERA-style tensor network. The radial axis is also the axis along which couplings run, with the running emerging from radial coarse-graining of the boundary. The

same direction that records the expansion records the change of scale, which is what one would want if the geometry is to do double duty as cosmology and as renormalization.

18.3. Caveat

Turning the geometric growth ratio into a physical Hubble rate requires identifying the beat and the cell with physical units, which has not yet been done. The figure $H \approx 0.80$ is therefore in beat units, not seconds. The qualitative statement, that exponential growth is de Sitter expansion, is robust. The absolute rate and a realistic multi-era cosmology, with radiation and matter eras preceding the de Sitter phase, both need the matter back-reaction that is not yet included. That is an open refinement rather than a settled result.

18.4. Cosmic dilution toward peace

Pairing conservation with eternal growth gives a clean cosmological statement at no extra cost. The total charge Q is fixed, because the rule conserves it, while the cell count grows without bound. The average charge per cell therefore falls toward zero. The universe dilutes globally toward the neutral 0 tone, which is the heat-death tendency, resisted only locally by pumping. Living structure is the local counter-current, the arrow holding pockets of the crystal far from peace. Expansion is then not only spatial. It is a slow global cooling toward the neutral value, with life as the local resistance to it. A detail at the frontier sharpens the picture. Newborn cells are forced to tone 0, since a nonzero birth would break conservation. Creation is the redistribution of charge from the existing crystal, not the minting of new charge, so the freedom at the growing edge is freedom of relation rather than freedom of net charge.

18.5. The ladder of rarity

Read as a budget rather than as a flow, the same hierarchy shows where organized matter sits, and the answer is a vanishingly thin corner. Standard cosmology splits the universe into dark energy, dark matter, and ordinary matter, and only the last is the atoms of stars, planets, and people. Within that thin slice the organization nests, and each more organized layer is rarer than the one beneath it by many orders of magnitude. The whole zoom, from everything down to civilization, fits in one table, where each row is a subset of the one above.

layer	share of the whole universe
Dark energy	68%
Dark matter	27%
Ordinary matter	5%
Stars	$\approx 0.3\%$
Planets and heavy objects	$\approx 0.005\%$
All life (biomass)	$\approx 10^{-12}\%$
Humans and civilization	far smaller still

Table 14: The whittling-down cascade. By mass, each rung is a rounding error on the one above it.

Two finer cuts sharpen the picture. Of all atoms, hydrogen and helium together are about 99.9 percent, and carbon, the backbone of every known organism, is roughly 0.03 percent, about one atom in a few thousand. Of all atoms, well under one percent are chemically bonded into molecules at all, since most ordinary matter

is ionized plasma or lonely hydrogen drifting through the voids. So chemistry, the rung that life is built on, already rests on a sliver of a sliver.

This ladder of rarity is the coarse-graining tower seen from the outside. Each rung, from field to particle to bound atom to molecule to cell to self to mind, is a more organized, more constrained, and more expensive pattern, so it occupies an ever-thinner fraction of the substrate. The model reproduces this directly rather than importing it. The fraction of charge that reaches each successive rung compounds multiplicatively, and over a cosmic number of rungs the measured per-rung factor lands near the one-part-in-a-hundred-trillion order of the cosmic estimate [43]. Rarity is the price of organization.

18.6. Rarity and the cost of demonstration

The same fact sets a hard limit on what can be shown by direct simulation. To watch the substrate grow life and mind out of the bare rule, one would have to run a volume large enough to contain a structure that occupies one part in 10^{12} to 10^{20} of it, and to run it for the cosmic number of beats the climb requires. Brute force across every scale at once is not feasible on any foreseeable machine. Progress depends instead on compression. One simulates each rung at its own coarse scale and carries only the reduced description upward, which is exactly what the renormalization tower of this chapter supplies. The coarse-graining hierarchy is therefore not only the physics of the bulk. It is also the only practical route to checking the upper rungs at all, and it is why the later chapters demonstrate the ladder rung by rung at toy scale rather than as one end-to-end cosmos.

A closing inversion guards against a misreading. By mass, life and mind are a rounding error, one part in many trillions. By organization, by integrated experience, which is the quantity the theory actually cares about, they are among the rarest and richest things the substrate does. The mass budget and the value budget point opposite ways, and on an experiential base it is the second that carries the meaning.

Cosmic expansion and the renormalization hierarchy both fall out of the bulk geometry without separate assumptions. The same conservation law that forbids minting well-being also makes the cosmos dilute toward peace, and that is what makes life, as the local resistance to dilution, a meaningful counter-current rather than a footnote.

19. Quantum Foundations

The base is local and deterministic. Quantum mechanics is nonlocal in the precise sense that it violates a bound any local model must respect. That is a sharp tension, and waving it away would misrepresent the model. This section states the tension plainly, then shows how the substrate's own holographic structure resolves it, and finally addresses two further questions, where the Born measure comes from and where the complex unit of quantum amplitudes comes from.

19.1. The Bell tension

A local, deterministic rule is a local hidden-variable theory, and every local hidden-variable theory obeys the Bell bound $\text{CHSH} \leq 2$. Brute-force enumeration over deterministic strategies confirms the maximum is 2. Quantum mechanics reaches $2\sqrt{2} \approx 2.83$, the Tsirelson bound, verified at the optimal measurement angles. A

local discrete base, taken literally as the whole story, therefore cannot reproduce quantum nonlocality. This is a genuine tension and not a free pass. Any discrete-substrate program has to confront it rather than assume it away.

Result 23 (Bell bound versus Tsirelson) A local deterministic rule obeys $\text{CHSH} \leq 2$, confirmed by enumeration over deterministic strategies. Quantum mechanics reaches $2\sqrt{2} \approx 2.83$ at the optimal angles [43, 6, 13]. A literal local base cannot reach the quantum value.

19.2. The resolution through holographic nonlocality

The bulk rule is local, so the bulk obeys $\text{CHSH} \leq 2$. Physics, though, does not live in the bulk. It lives on the holographic boundary, and two boundary points are connected through the four-dimensional bulk by the bulk-tree propagator. The boundary theory is therefore nonlocal, with its correlations mediated through the bulk, and it is not a local hidden-variable model. A bulk-mediated boundary theory can violate Bell up to the quantum value while the rule that generates it stays strictly local. Nonlocality is emergent and holographic. The base remains local and discrete, and the quantum nonlocality is a property of the boundary, not of the substrate. This is consistent with the holographic structure the substrate already carries, and in fact it requires that structure, which is the same structure invoked in the gravity and bulk-cusp discussions.

Proposition 9 (Holographic resolution of the Bell tension) The local bulk obeys $\text{CHSH} \leq 2$, but the physical boundary theory is bulk-mediated and hence nonlocal, so it is not a local hidden-variable model and can violate Bell up to the quantum bound. Nonlocality is emergent and holographic while the base stays local [38, 47, 51].

19.3. The Born rule

The $|\psi|^2$ measure appears as the natural measure of the oscillation amplitudes of the real discrete walk. That much is suggestive and consistent. The full account of measurement and collapse is the deeper open piece, and it is shared with every discrete and emergent-quantum approach rather than special to this one. It is listed here as open because it is open, not because it is solved.

19.4. Where the complex unit comes from

A natural worry is that the base is real and discrete, so the complex amplitude of quantum mechanics seems to have no place to come from. There is a clean answer through the Doi-Peliti map. The rule's three moves, create, annihilate, and hop, are all the same operator, the transfer of one unit of S^z across an edge. The rule is therefore exactly a spin-1 XX model, and its reversibility, which is detailed balance, is precisely what makes the Doi-Peliti Hamiltonian Hermitian and so gives a real, positive-energy quantum theory. In the spin-coherent-state path integral of that model the Berry-phase term is the source of the quantum i . The complex amplitude is not added. It is the Berry phase of the rule's own spin-exchange, and the imaginary unit is where the phase of the discrete walk lives.

Proposition 10 (The i as a spin Berry phase) The rule's create, annihilate, and hop moves are one operator, the transfer of S^z across an edge, so the rule is a spin-1 XX model whose reversibility makes its Doi-Peliti Hamiltonian Hermitian. The Berry-phase term of the spin-coherent-state path integral is the source of the quantum i .

The chapter identifies a real requirement rather than papering over one. A discrete local base needs an emergent nonlocal boundary in order to be quantum, and the substrate supplies exactly that through its holography. There is a further pointer worth stating without overclaiming. Since the base is experiential under the monism, the quantum amplitudes are the texture of the underlying noticing, and the universe-as-mind is where the quantum layer ultimately sits. That is a candid pointer toward the rest of the program, not a derivation.

Part IV

Physical Space, the Cusp

20. The Cusp Is Physical Space

The claim of this part is direct. Ordinary three-dimensional space is the flat $\{4, 3, 4\}$ cubic cusp of the four-dimensional bulk. The bulk is the hyperbolic crystal $\{3, 4, 3, 4\}$ of 24-cells. At each ideal corner of that crystal sits a Euclidean cubic honeycomb, and this section argues that one such cusp is the space we live in. The work here is to say precisely what space is on this proposal and to show by measurement that it is genuinely and cleanly three-dimensional. Two pictures recur, the burning bulk and its cast radiance, but the substance is in the geometry and the numbers.

20.1. Paracompactness is a feature

The honeycomb $\{3, 4, 3, 4\}$ is *paracompact*. Its vertex figure is the Euclidean $\{4, 3, 4\}$ cubic honeycomb, an ideal element with infinitely many cells rather than a finite polytope. In the usual classification of regular hyperbolic honeycombs this counts as a degenerate case, since the element sits at infinity. We read it the other way. That ideal cubic element is the flat three-dimensional physical space, and its appearing at the ideal corner is exactly what hands physics a clean, periodic, flat crystal to live on.

Seen this way the paracompactness is not a blemish to be apologized for. It is the mechanism. A genuinely compact honeycomb has finite cells, but the honeycombs of that kind that exist in four dimensions are all non-crystallographic, and so they carry no spinor structure on their direction set. The compact options buy a bounded vertex figure at the cost of the spinor coin. The paracompact D_4 cusp pays the opposite price and gains the opposite prize. It surrenders a finite vertex figure and in exchange it gets both the spinor coin in the bulk and an exactly flat cubic three-space at the corner.

Proposition 11 (The ideal element is physical space) The vertex figure of $\{3, 4, 3, 4\}$ is the Euclidean cubic honeycomb $\{4, 3, 4\}$. Read as the geometry at the ideal corner of the bulk, this paracompact element is the flat three-dimensional space on which matter propagates. Paracompactness is therefore the feature that supplies physical space, not a defect of the substrate.

20.2. Flat at finite distance, not only at infinity

A natural worry is that a structure living at an ideal corner is real only in a limit, never at any finite stage. The geometry rules this out. A horosphere in hyperbolic space is intrinsically Euclidean at every finite level. This is a theorem about hyperbolic geometry, not an asymptotic statement, and the cubic cusp inherits it. The flatness holds on a real surface at finite distance, not only in a limit one approaches and never reaches.

The same point shows up in measurement. A horosphere band, grown outward, expands polynomially with an exponent near 1.5, while the surrounding bulk expands exponentially with a rate near 9 [43]. The gap is unmistakable. The bulk swells the way hyperbolic volume swells, and the horospherical surface inside it grows the slow way a flat three-space grows. The flat cusp is therefore a real, finite-distance, exactly Euclidean surface sitting inside an exponentially growing curved bulk, not an idealization that exists only at the boundary.

Result 24 (The cusp is flat at finite distance) The cubic cusp grows polynomially, with measured band exponent near 1.5, against the bulk’s exponential rate near 9. Combined with the horosphere flatness theorem, this confirms an exactly Euclidean three-dimensional surface at every finite stage, not merely in the ideal limit [43].

20.3. The clean cusp versus a generic slice

There is a sharper question than flatness. Among the surfaces one could carve out of the bulk, which one is physical space? Two candidates compete. One is the clean $\{4, 3, 4\}$ cubic cusp, the periodic crystal sitting at the ideal corner. The other is a generic horosphere slice, an arbitrary level surface cut through the bulk with no regard for the cubic lattice. An earlier and more optimistic reading hoped that a generic aperiodic slice might be the better physical space, on the thought that aperiodicity buys isotropy for free. The measurement did not support that reading.

property	clean cubic cusp	generic horosphere slice
periodicity	periodic crystal	aperiodic, rough
spectral dimension	2.97 (clean 3D)	2.16 (sub-3D)

The clean cubic cusp reads a spectral dimension of 2.97, which is clean three-dimensional behaviour within measurement error. The generic slice reads 2.16, distinctly sub-three-dimensional, the signature of a rough aperiodic surface where a diffusing walker keeps getting trapped [43]. The verdict is that the clean cubic cusp, not the disordered generic slice, is physical space. The aperiodic-is-isotropic intuition was attractive but wrong, and the test settled it in favour of the periodic crystal.

Result 25 (The clean cusp is clean 3D) The $\{4, 3, 4\}$ cubic cusp measures spectral dimension 2.97, clean three-dimensional, while a generic horosphere slice measures 2.16, sub-three-dimensional. Physical space is the clean cubic cusp, not a generic slice [43].

This pins down what space is on the proposal. It is the clean $\{4, 3, 4\}$ cubic cusp of the bulk, genuinely and cleanly three-dimensional, exactly flat at finite distance, the radiance cast by the burning bulk rather than a limiting fiction. The paracompactness that looked like a defect is the very thing that supplies a flat crystal for matter. One residual remains. The cusp carries a small cubic anisotropy, the order-four imperfection of

the octahedral point group, which the later audit treats in full. Everything else about the identity of space is settled here, and settled cleanly.

21. Finiteness and Growth

The cusp is never the complete infinite $\{4, 3, 4\}$ honeycomb. A real mesh holds finitely many cells and grows one frontier layer at a time. So a fair question is how large the finite, growing cusp has to be before it looks like today’s space. The answer turns out to be surprisingly small, and the observed universe is vastly beyond it. This section establishes that a finite chunk far short of infinity is more than enough, and it keeps the question of whether the crystal is ultimately finite or infinite genuinely open.

21.1. The complete cusp is never reached, and need not be

The complete $\{4, 3, 4\}$ cusp is the infinite limit. A finite mesh never realizes it and never gets there in any finite number of beats. This is not a flaw in the model. Physical space is never meant to be the complete cusp. It is an ever-growing finite chunk of $\{4, 3, 4\}$, in exactly the way a finite crystal in a laboratory is described by an infinite-lattice idealization while remaining finite. Local physics only sees the local lattice, and the idealization is a convenience for reasoning about the bulk behaviour of a finite object.

The never-reaching has a reading that fits the rest of the theory. It is the eternal expansion and the open future. The frontier always has more to add, and arriving at the complete cusp would be the end of time rather than its fulfilment. The growth is the point. A finished crystal would be a finished universe, and the model does not describe one.

21.2. Convergence is fast

To measure sufficiency we build cubic chunks of growing side L and watch the observables settle. This is cheap to do. The cusp’s intrinsic structure is the $\mathbb{Z}^3$ cubic lattice, so a chunk of side L is an $O(L^3)$ object built directly, with no expensive hyperbolic horoball construction needed. We track two numbers, the spectral dimension and the effective gravity exponent, as L increases.

side L	spectral dimension	gravity exponent
5	2.58	2.06
7	2.88	1.61
11	2.99	1.34
17	2.99	1.40 (converged)
25	2.99	1.42
35	2.99	1.43

The spectral dimension reaches three by about $L \sim 11$ cells. Local physics becomes continuum-like and L -independent by about $L \sim 17$ cells [43]. That is roughly 5000 cells, and mapping size to growth through a scale factor $a(t) \sim e^{Ht}$, it amounts to only three or four beats of growth. A clean local physics is reached almost at once. The observed universe, by contrast, spans on the order of 10^{61} cells, which is astronomically deep inside the continuum regime, so finite-size and edge effects at any observable scale are utterly negligible. The figure of 10^{61} is reached by analytic extrapolation of the $\mathbb{Z}^3$ lattice toward isotropic three-dimensional elasticity, as a power of $1/L$, rather than by building that many cells.

Result 26 (A small chunk suffices) Cubic chunks of side L reach spectral dimension 3 by $L \sim 11$ and L -independent continuum physics by $L \sim 17$, about 5000 cells or a few beats of growth. The observed universe at $\sim 10^{61}$ cells is far inside this regime, so finite-size effects at observable scales are negligible [43].

21.3. The refining universe

The growth has a shape that matches what cosmology sees. The universe expanded fast to a rough geometry, a small chunk with measurable departures from perfect three-dimensionality, and it refined toward clean three-dimensional space as it grew. The large-scale flatness and isotropy improve with expansion, approaching but never reaching the perfect infinite-cusp limit. This is consistent with the observed picture of a universe flat to about 0.4 percent and of finite age. The early roughness and the late smoothness are two readings of the same convergence table, taken at small and large L .

One quantity does not vanish with growth. The fixed cubic anisotropy, the order-four imperfection of the cusp's octahedral point group, is a permanent feature rather than a finite-size artifact. It does not refine away as the chunk enlarges, because it is built into the lattice symmetry itself and not into its size. The later audit treats this residual directly. Here the point is only that the refining behaviour and the permanent anisotropy are distinct, and the convergence table separates them.

A finite cusp, far from infinity, is more than sufficient to look like today's space. The continuum regime arrives in a few beats, and the real universe is enormously beyond the threshold. Whether the crystal is ultimately finite or infinite is left open, and nothing in the convergence result depends on settling it, since either way the observable patch sits deep inside the continuum.

22. The Bulk-Cusp Interface

Spin and gauge structure live in the four-dimensional bulk coin, the 24-cell with its D_4 direction set. Matter, on the other hand, propagates on the three-dimensional cubic cusp. The natural question is how the two are related, and whether the four-dimensional spinor structure can survive on a three-dimensional surface. The answer is that the relation is a *projection*, the standard dimensional reduction of a slice. Choosing the cusp picks a radial direction, and everything in the bulk reduces along it in the textbook way.

22.1. The spinor projects

The bulk's spacetime rotation group is $SO(4) = SU(2) \times SU(2)$. Choosing a horosphere picks out a radial direction, and that choice breaks $SO(4) \rightarrow SO(3)$. The surviving $SO(3)$ is the cusp's three-dimensional rotation group, realized as the diagonal $SU(2)$ inside the product. Under this breaking the four-dimensional Dirac spinor, which carries the representation $(2, 1) + (1, 2)$, branches to $2 + 2$. Each piece is a three-dimensional Pauli spinor of spin one half. The branching is verified by computing the quadratic Casimir, which reads $3/4$ on each piece, the correct value for spin one half [43].

So the four-dimensional bulk Dirac spinor projects to three-dimensional Pauli spinors. The bulk spin becomes the cusp spin. This is exactly the textbook reduction of a $(3 + 1)$ -dimensional Dirac fermion to its three-dimensional spatial spinor, with $\text{Spin}(3) = SU(2)$. The spinor coin and physical space are not a mismatch to

be explained away. The cusp spin is the shadow of the bulk spin, and the shadow is computed rather than asserted.

Result 27 (The spinor projects $4 \rightarrow 2 + 2$) Under $SO(4) \rightarrow SO(3)$ from the radial choice, the bulk Dirac spinor $(2, 1) + (1, 2)$ branches to $2 + 2$, two 3D Pauli spinors each of spin one half, with Casimir $3/4$ on each. The bulk spin projects to the cusp spin in the standard $(3 + 1)D \rightarrow 3D$ reduction [43].

22.2. The directions project

The same projection acts on the direction set. The 24 bulk directions, projected onto the cusp tangent space orthogonal to the radial direction, collapse to a smaller three-dimensional set. The projected set holds roughly nine to twelve distinct directions, richer than the naive cubic six [43]. This projected direction set is the cusp coin, and it coarse-grains to the $\{4, 3, 4\}$ cubic continuum in the infrared. A richer projected coin carries a useful consequence. The cusp anisotropy is even smaller than the naive cubic estimate would suggest, because the extra directions average the propagation more evenly than six axes alone would.

22.3. The symmetry chain and the anisotropy

The symmetry reduces in a chain. The bulk coin carries $SO(8)$ and the bulk spacetime carries $SO(4)$. On the cusp these reduce to the octahedral cubic point group, a finite subgroup of $SO(3)$, which then restores to full $SO(3)$ in the infrared as the lattice coarse-grains. The exact symmetry of the cusp is the finite octahedral group. This is the reason the cusp carries a small order-four anisotropy. The bulk, with its D_4 and F_4 structure, is isotropic to order four and protects its own directions, but that protection lives in four dimensions. The cusp's exact symmetry is the finite octahedral group, which is anisotropic at order four, so the order-four residual is the cusp's own and is not shielded by the bulk F_4 .

Proposition 12 (The interface is a projection) The bulk-cusp relation is the dimensional reduction of a slice along the radial direction. Spin reduces $SO(4) \rightarrow SO(3)$ with $4 \rightarrow 2 + 2$, the 24 bulk directions project to the ~ 9 to 12 direction cusp coin, and full $SO(3)$ is restored in the infrared. The internal $SO(8)$ structure rides along as internal quantum numbers.

The interface is not an unbridged gap between a four-dimensional coin and a three-dimensional space. It is a projection, made precise. Spin reduces $SO(4) \rightarrow SO(3)$, the directions reduce to the cusp coin, and full $SO(3)$ is restored in the infrared. The internal $SO(8)$ structure, which carries the gauge content and the generation structure, rides along as internal quantum numbers and is untouched by the spatial projection. What remains is bookkeeping, the exact lattice-level coin and the detailed carry-over of the $SO(8)$ labels, rather than a missing mechanism. The mechanism is the standard reduction of a slice, and it does what it needs to do.

23. The Robustness Audit

Many of the findings reported earlier were computed assuming a clean three-dimensional space. Now that physical space has been identified as the cubic cusp, a periodic crystal rather than an idealized continuum lattice, the pressing question is whether those findings survive. The reassuring fact is that most of them do not live on the flat slice at all. They live in the bulk coin or on the boundary sphere, where the cusp question

simply does not reach. This section sorts every finding by where it lives, checks the ones that genuinely depend on clean three-space, and lays out the worst worries worst-first.

23.1. Three layers, three fates

The findings split into three layers, and each layer has its own relation to the cusp reality.

- *Bulk findings*, which are cusp-independent and unaffected. Spin, triality, the $SO(10)$ gauge group, $\sin^2 \theta_W = 3/8$, the mass relations, the holographic $1/r^2$ correlator, cosmology, the hierarchy, and the toolkit. These live in the four-dimensional bulk coin or are pure group theory, so the cusp question does not touch them.
- Boundary findings on the S^3 sphere, also unaffected. Holography and quantum nonlocality belong here, and the cusp does not bear on them.
- Flat-layer findings, which need clean three-space and are checked here. The Dirac equation, Lorentz invariance, $1/r$ gravity, electromagnetism, solitons, the fermionic-matter result, and matter formation. These are the only findings that could be threatened by the cusp reality, and all of them hold.

23.2. The check on the cusp

To check the flat-layer findings, the relevant observables were measured on the $\{4, 3, 4\}$ cubic cusp and, for contrast, on a generic aperiodic slice. The two surfaces give clearly different verdicts.

observable	cubic cusp (physical space)	generic slice (bad extraction)
spectral dimension	2.97 (clean 3D)	2.16 (sub-3D)
isotropy (light-cone CV)	0.14	0.32 (rough)

The flat-layer physics holds on the cubic cusp and degrades on the generic slice [43]. The reading is twofold. The earlier results survive provided physical space is the cusp cubic, which the previous sections established that it is. And they would fail on a disordered slice, which is therefore not physical space. This does not merely confirm the findings, it also confirms the identity of space, since only the right surface gives the right physics.

Result 28 (The flat layer holds on the cusp) On the $\{4, 3, 4\}$ cubic cusp the flat-layer observables read spectral dimension 2.97 and light-cone isotropy 0.14, while a generic slice degrades to 2.16 and 0.32. The flat-layer findings survive on the cusp and fail on a disordered slice [43].

23.3. The problem ledger

The deepest worries about the theory on the real cusp are listed below, ranked worst-first, each with its current status. This is the worst-first audit rather than a summary of successes.

One point in the ledger deserves emphasis, because it is subtle. The cubic anisotropy is the cusp's, not the bulk's. The bulk 24-cell coin is isotropic to order four, the good D_4 and F_4 structure. But matter propagates on the cubic $\{4, 3, 4\}$ cusp, whose exact symmetry is the finite octahedral group, anisotropic at order four and not protected by the bulk F_4 . So the one ultraviolet Lorentz residual lives specifically on the cusp, and it cannot be inherited away from the bulk's protection.

problem	status
the bulk-cusp interface, do the 4D D_4 spinors live on the 3D cusp?	Solved. It is a projection, $SO(4) \rightarrow SO(3)$, $4 \rightarrow 2 + 2$, verified.
cubic anisotropy, the cusp lattice breaks rotations	Resolved. Tiny $(E/E_{\text{Planck}})^2$, IR-isotropic, falsifiable, and smaller than the naive cubic estimate since the projected coin is richer.
dynamical settling, does matter actually gather into clean cusp regions?	Open. A plausible mechanism, solitons seeking the regular low-energy parabolic sites, and a defined GPU test, but not yet confirmed. The one genuinely-open robustness item.
absolute quantitative values, masses, couplings, the Born-rule normalization	The frontier, shared with all such theories.
paracompactness and nonlocality	Resolved. Paracompactness is a feature, the cusp is 3D space, and the nonlocality is bulk-provided.

23.4. The one modification

The single change that the cusp reality forces is the cusp’s small cubic anisotropy, a tiny $(E/E_{\text{Planck}})^2$ ultraviolet Lorentz violation that becomes isotropic in the infrared. Everything else is either unaffected, because it lives in the bulk or on the boundary, or it holds, because the flat-layer measurement confirms it on the cusp. One artifact note belongs here. The finite-patch gravity exponent reads about 0.67, which is a finite-size and screening artifact of the bounded patch rather than a physical reading. The clean $1/r$ result is the difference-method value, not the patch number, and the two should not be confused.

The program is robust to the cusp reality. Only the flat-layer physics depends on clean three-space, and it holds on the cusp. The bulk findings and the boundary findings are untouched, because the cusp question never reaches them. The one genuinely-open item is the dynamical settling of matter into clean cusp regions, which has a plausible mechanism and a defined test but is not yet confirmed. That single open question is stated plainly rather than hidden, and the rest of the audit closes cleanly.

Part V

Why This Substrate

24. Methodology

This part sits in the middle of the paper. It explains how each result was obtained and lays out the survey that ranges across every dimension. The chapters that follow give the selection argument itself. Here we describe the shared machinery once, so that the later claims can be read without stopping to ask what the words *measured* or *derived* stand for. The per-sector detail belongs to the longer methodology notes, and the per-dimension data belongs to the appendices. What is gathered here is the common ground that every computation in the paper rests on.

24.1. The construction

Everything begins with a single object, the cell-adjacency graph of a honeycomb. A *cell-graph builder* takes a Schläfli symbol and grows the honeycomb outward. Each cell is tracked as a group element, that is, as a matrix in the symmetry group, and neighbours are reached by applying the face reflections. The traversal is a breadth-first search whose cost scales with the facet count times the number of cells visited. This works cleanly for the curved honeycombs that the paper cares about most. The flat references need a different route. A separate *Euclidean lattice builder* constructs the flat cases, the D_4 root lattice and the integer lattices $\mathbb{Z}^n$, directly. The reason is geometric rather than a matter of taste. A flat honeycomb has its cell centre at an ideal point, which is to say at infinity, and the hyperbolic builder stalls there. Building the flat lattices by hand sidesteps that failure and gives clean controls.

24.2. The probes

A small set of probes turns the graph into numbers. Each one targets a different physical quantity, and together they cover dimension, dynamics, gravity, and the internal symmetry content. We list them here and return to each in the sectors where it does its work.

- Breadth-first shells give the growth profile, which feeds dimension, curvature, cosmology, and the hierarchy of scales.
- The lazy random walk gives the spectral dimension through its return probability, which falls like $t^{-d/2}$.
- The Bethe-lattice cavity recursion gives the exact Green’s function on the infinite tree, which feeds gravity and holography.
- The kernel polynomial method with a stochastic trace gives spectral quantities, such as the Skyrme sea energy, without ever diagonalising a matrix. It uses only matrix-vector products and scales onto a GPU.
- Representation theory reads off the directional coin’s spinor and gauge content, which is where spin and the unified group come from.
- Variance-reduction methods, common random numbers and singular-part subtraction, cancel the finite-patch and boundary contamination that would otherwise corrupt the gravity law.
- WebGPU carries the large-scale dynamics, the 24-direction lattice gas, the bulk wave, the skyrmion, and the kernel polynomial spectra.

24.3. The hyperbolic-bulk toolkit

Hyperbolic patches present a sharp practical problem. A finite ball in H^n is almost all boundary, around 99 percent of its cells touch the edge, so a naive bulk measurement is swamped by surface effects. A dedicated toolkit, set out in full in the appendix, removes the confound by never trusting a raw finite patch. The Bethe recursion works on the infinite tree directly, so there is no finite patch to corrupt. Closed manifolds remove the boundary altogether. Hyperbolic band theory puts a momentum space on a closed manifold, which makes the spectrum tractable. Tensor networks contract at polynomial cost. The kernel polynomial method reads spectra from matrix-vector products alone. Bulk-local observables are designed to ignore the surface. The

point of the whole kit is that any quantity claimed for the bulk has been measured by a method that the boundary cannot reach.

24.4. Reporting conventions

Every result in the paper carries one of four tags. *Derived* means it follows by argument from the base. *Measured* means it was computed on the substrate. *Predicted* means the framework implies it but it has not yet been checked. *Open* means the question is genuinely unsettled. We hold ourselves to two further commitments, and the appendix audits both.

Principle 4 (Tag every claim) No statement about the substrate is presented without one of the four tags. A measurement that was later found to be confounded, such as the early gravity exponents taken on small patches, is flagged as confounded and superseded by the result from the correct method, rather than quietly dropped.

Principle 5 (Nothing added to the base) No computation in the paper introduces a new term, field, bias, or rule beyond the stated base in order to make a result come out. When a phenomenon does not emerge from the base, the negative is reported. The base is fixed before the experiments run, and it stays fixed.

These two rules are what let the selection argument carry weight. If the experiments could be tuned, a match with known physics would prove nothing. Because the base is frozen and every claim is tagged, a match is information and a failure is information, and the reader can tell which is which.

25. The Dimension Ladder

The selection argument has two halves. This chapter is the first. It shows that only a four-dimensional bulk yields the three-dimensional physical space we appear to inhabit. The argument turns on one geometric fact about hyperbolic honeycombs, and that fact is sharp enough to rule out every dimension but one. The second half, which singles out $\{3, 4, 3, 4\}$ within the four-dimensional family, is taken up in the next chapter.

25.1. Physical space is the bulk dimension minus one

The flat layer that matter lives on is not the bulk itself. It is the cusp, the horosphere, the flat slice that sits at the ideal boundary of the hyperbolic crystal. That slice is always exactly one dimension below the bulk. A four-dimensional bulk has a three-dimensional cusp, a three-dimensional bulk has a two-dimensional cusp, and so on down. This single relation fixes which substrate dimension can give the observed space.

Proposition 13 (Dimension ladder) The dimension of the physical space carried by a hyperbolic honeycomb equals the bulk dimension minus one. Among the candidate families, only the four-dimensional honeycombs land on a three-dimensional physical space.

The table below walks the ladder. The two-dimensional and three-dimensional families undershoot, giving one-dimensional and two-dimensional space. The five-dimensional family overshoots, giving four-dimensional space. Only the four-dimensional family is exact.

Substrate family	Bulk	Physical space (cusp)	Verdict
2D tilings, e.g. $\{7, 3\}$	2D	1D	under
3D honeycombs, e.g. $\{5, 3, 4\}$	3D	2D	under
4D honeycombs, $\{3, 4, 3, 4\}$	4D	3D	exact
5D honeycombs, e.g. $\{5, 3, 3, 3, 3\}$	5D	4D	over

Table 15: The cusp is one dimension below the bulk, so only the four-dimensional family gives the observed three.

25.2. What the wrong dimension costs

The dimension of the physical space is not an abstract bookkeeping number. It sets the kind of matter and the law of gravity that the substrate can support, and each wrong dimension fails in a recognisable way. A one-dimensional space, from a two-dimensional substrate, has no exchange statistics at all, a confining linear gravity, and only kink solitons. A two-dimensional space, from a three-dimensional substrate, has anyonic statistics, a logarithmic gravity, and two-dimensional skyrmions. A four-dimensional space, from a five-dimensional substrate, has $1/r^2$ gravity, four-dimensional instantons, and one dimension too many. A three-dimensional space, from a four-dimensional substrate, has fermions, $1/r$ gravity, and three-dimensional hopfions. That last row is the observed case. The matter statistics and the gravity law both track the ladder, and only the four-dimensional substrate gives the fermions and the $1/r$ gravity we measure.

25.3. The dimension window

One might worry that hyperbolic crystals are scarce, and that the ladder above is choosing among a handful of oddities. The opposite is true. Regular hyperbolic honeycombs are plentiful, and counting them by compactness class makes the real constraint visible. We use the taxonomy of the geometry chapter, where compact means every element is bounded, paracompact means there is one ideal element at infinity, and noncompact means there is an infinite ideal cell.

Dimension	Compact regular	Paracompact or noncompact regular	Physical space
2, H^2	infinitely many, $\{7, 3\}$, $\{5, 4\}$, ...	infinitely many more, apeirogonal, $\{\infty, 3\}$, ...	1D
3, H^3	4, all contain a 5	11 more, the 3/4/6 family, ideal cells or vertices	2D
4, H^4	5, all contain a 5	several more, including $\{3, 4, 3, 4\}$ with its ideal cubic cusp	3D
5, H^5	0	a few, the last dimension with any regular honeycomb, including the D_4 hook $\{3, 4, 3, 3, 4\}$	4D
6 and up	0	0	—

Table 16: Regular hyperbolic honeycombs by compactness class. The window has two teeth.

The window has two teeth, and together they cut the field down to one dimension. The first tooth is compactness. The bounded regulars, the ones with no cusp, exist only in dimensions two through four, and every one of them carries a 5 in its symbol. That places them in the icosahedral family, which is non-crystallographic. They cannot carry a root system, so they have no spinor and no gauge group. The second tooth is the crystallographic restriction. The honeycombs whose symbols use only 3s and 4s, the ones that can carry a root system, are all paracompact. The substrate $\{3, 4, 3, 4\}$ is among them, and its ideal

element is the flat $\{4, 3, 4\}$ cubic cusp. Compactness in dimensions two through four forces a 5 and kills the spinor. The spinor-bearing crystallographic substrate must therefore be paracompact. Four dimensions is the only paracompact crystallographic dimension that yields a three-dimensional cusp, since five overshoots to four and nothing regular survives above five.

25.4. An independent confirmation of three dimensions

There is a second reason the physical space comes out three-dimensional, and it does not rely on the cusp at all. Stable bound orbits, closed and neither precessing nor spiralling, exist only in three spatial dimensions. The apsidal angle of the inverse-square law is exactly 180 degrees in three dimensions. In four dimensions and above, orbits either spiral inward or escape. This is the Ehrenfest and Bertrand result, and it constrains not the substrate dimension but the dimension where matter lives. The natural reading is a braneworld, a four-dimensional hyperbolic bulk of anti-de-Sitter character with matter confined to a three-dimensional slice. That is the $\{3, 4, 3, 4\}$ cusp picture exactly. So the claim that physical space is three-dimensional is doubly forced, once by the cusp geometry and once by orbit stability, two arguments of different character landing on the same number. The braneworld reading also carries a falsifiable handle, since a three-dimensional brane in a four-dimensional bulk predicts short-range deviations from inverse-square gravity.

The dimension argument alone narrows the substrate to the four-dimensional family. It does not yet pick a single honeycomb out of that family, and it does not need to, because the framework itself is dimension-general. The rule, the Lorentz structure, electromagnetism, the holographic correlator, cosmology, and isotropy all port to every dimension on the ladder. What the dimension does is set the stage. The crystallographic spinor coin, treated next, is what selects $\{3, 4, 3, 4\}$ within the four-dimensional family.

26. The Crystallographic and Spinor Requirement

This chapter is the second half of the selection argument. The dimension ladder narrowed the substrate to the four-dimensional family. Within that family, one criterion remains, and it is decisive. Only $\{3, 4, 3, 4\}$ carries a directional coin that is both crystallographic and able to host a spinor. Everything else in the model is a scalar, so the coin is the only place where spin can hide, and the coin of $\{3, 4, 3, 4\}$ is the only one across all dimensions that holds it.

26.1. Crystallographic means gauge is possible

A coin carries a root system, and therefore a gauge group, only if it is *crystallographic*. The crystallographic restriction permits the angles built from 3, 4, and 6, and forbids five-fold and seven-fold symmetry. This is the same restriction that governs ordinary crystals. The compact regular honeycombs in every dimension all contain a 5 in their symbol, placing them in the icosahedral family. They are non-crystallographic, so they have no root system, no gauge group, and no spinor. Compactness and a gauge group are at odds among the compact regulars, and the substrate cannot have both.

26.2. Only the 24-cell carries the spinor

Among the crystallographic coins, the 24-cell stands alone. Its symmetry is the D_4 root system, and its direction set decomposes as $8v + 8s + 8c$, the vector and the two spinors. A cubic coin gives the vector only,

with no spinor at all. The 24-cell appears as the cell of $\{3, 4, 3, 4\}$, so the substrate hands over the D_4 root system, and with it the two spinor representations, as part of its own geometry.

The reason a smaller coin cannot carry a spinor is a theorem rather than an accident, and the cleanest case to state it on is $\{5, 3, 4\}$, whose coin has 12 directions.

Theorem 4 (Permutation characters carry no spinor) The 12 face directions of the $\{5, 3, 4\}$ coin form a permutation representation of the icosahedral rotation group. A permutation representation of a rotation group contains no spinor, because spinors live only in the double cover. The relevant double cover here is the binary icosahedral group of order 120, not the order-60 rotation group. The $\{5, 3, 4\}$ coin is therefore provably spin-0.

The force of the theorem is that it leaves no room to negotiate. A permutation representation simply cannot contain a spinor, so no clever reading of the 12 directions will produce one. The same logic applies to the rest of the model by default. The tone and any memory are single ternary values, each one a scalar that returns to itself under a full 2π rotation. So the substrate is Klein-Gordon-like, that is, spin-0, unless spin is hiding in the directional coin. That is precisely why the coin is the only place to look, and why the D_4 decomposition $8v + 8s + 8c$ with its two spinors is the deciding fact.

26.3. Spin is the sole discriminator

To test whether spin is genuinely the discriminator, we run the identical battery on $\{5, 3, 4\}$ as a control. Almost everything ports. The rule ports. Conservation ports. Gliders and motion port. The holographic $1/r^2$ correlator ports. Cosmology ports. The self-nesting tower ports. The two honeycombs agree across the whole sweep with one exception.

Result 29 (The $\{5, 3, 4\}$ control) Run on $\{5, 3, 4\}$, every framework result matches $\{3, 4, 3, 4\}$ except spin. The 12 directions of $\{5, 3, 4\}$ are icosahedral, decomposing as $1 + 3 + 3' + 5$ with no spinor, so the substrate cannot host fundamental fermions. It is not a root system, so there is no $SO(8)$ or $SO(10)$, hence no weak mixing angle $\sin^2 \theta_W = 3/8$ and no three generations. Its two-dimensional cusp gives only logarithmic gravity and anyonic statistics. The single axis on which $\{3, 4, 3, 4\}$ wins is spin. [43]

This is what makes the control decisive. Spin is not one advantage among many. It is the one place the two honeycombs differ, which means the spinor coin is the lever that forces the four-dimensional $\{3, 4, 3, 4\}$ and nothing else does.

26.4. The cost of imposing spin by hand

One could try to put spin onto $\{5, 3, 4\}$ by hand, and the cost of doing so is the clearest argument against it. Three additions would be required. First, the binary icosahedral double cover, with its central element of order two acting as -1 . Second, a two-component spinor on every cell, which is new state the base did not have. Third, a coin operator together with a map from the faces to the gamma matrices. That face map is non-unique. Different choices give inequivalent theories, and there is no principle in the model to pick among them. The trouble runs deeper still. A spinor is by nature a continuous complex $SU(2)$ object, so imposing one either breaks the discrete-only commitment of the base or demands an awkward discretisation that has to keep the double-cover -1 exact. By contrast $\{3, 4, 3, 4\}$ hands over the spinor for free, inside the discrete D_4 coin, with nothing added. One caveat belongs here. There is a topological route to spin, where a self

is a hopfion and the hopfion behaves as a fermion, and that route works on either honeycomb without the coin. So the precise claim is narrow and exact. The coin spinor requires $\{3, 4, 3, 4\}$. It is not claimed that $\{3, 4, 3, 4\}$ is required for any spin whatsoever.

26.5. The unique solution

Collecting the requirements gives a clean intersection. The substrate must be four-dimensional, for a three-dimensional physical space. It must be crystallographic, for a gauge group. It must carry the D_4 coin, for spinors. And it must be hyperbolic, for gravity, holography, and expansion.

Theorem 5 (The selection) The substrate satisfying all four requirements, four-dimensional with a three-dimensional cusp, crystallographic, D_4 -coined, and hyperbolic, is uniquely $\{3, 4, 3, 4\}$. It is paracompact, so its cusp is the three-dimensional physical space. The genuinely compact four-dimensional regulars all contain a 5, so they are non-crystallographic and fail both the gauge and the spinor requirements. The flat twin $\{3, 4, 3, 3\}$ has the coin but lacks curvature, so it fails the gravity requirement.

A heuristic rubric makes the same point with a score. Weighting the spinor at 3, crystallographic at 2, dimension closeness at 3, curvature at 1, and adding a compactness term, every tessellation can be ranked. $\{3, 4, 3, 4\}$ tops the list at 9.5. The flat D_4 honeycomb $\{3, 4, 3, 3\}$ is second at 9.0, missing only the curvature point. Every other tessellation loses either the spinor coin or the dimension. $\{3, 4, 3, 4\}$ is the only one that scores non-zero on every factor at once.

The phrasing changes from a choice to a derivation. The sentence "we chose $\{3, 4, 3, 4\}$ " becomes "the requirements are met by only $\{3, 4, 3, 4\}$ across all dimensions". The controls do double duty. They confirm that the framework is generic, since it ports to every substrate, and they confirm that the physics is specific to $\{3, 4, 3, 4\}$, since the spinor coin appears nowhere else.

27. The Best of Each Dimension

The selection argument is cleaner when it is made concrete. This chapter takes the single strongest representative of each dimension and runs them through the same battery side by side. The point is not to re-derive the ladder but to see it acting on real, well-studied honeycombs. The full per-dimension data lives in the appendices. What follows is the comparison in summary, together with a careful accounting of what "best in a dimension" actually means.

27.1. The four representatives

Each representative is the cleanest and best-addressed honeycomb of its dimension, which is not always the same as the winner on every axis. We use *best* here to mean the canonical stand-in for that dimension, and we flag below where that differs from a battery score.

- Two dimensions, $\{7, 3\}$, heptagons three to a vertex. This is the canonical hyperbolic tiling, the home of Margenstern's Fibonacci addressing. It is non-crystallographic, since it carries a 7, and its physical space is one-dimensional, but it is the reference two-dimensional substrate.

- Three dimensions, $\{5, 3, 4\}$, dodecahedral and compact, also Margenstern-addressed. It is non-crystallographic, since it carries a 5, and its physical space is two-dimensional. As the note below explains, it is the best-studied three-dimensional honeycomb, not the best-scoring one.
- Four dimensions, $\{3, 4, 3, 4\}$, the champion. Its cell is the 24-cell, its coin is the D_4 root system, it is crystallographic and hyperbolic, and its cusp is three-dimensional.
- Five dimensions, $\{3, 4, 3, 3, 4\}$, the strongest five-dimensional case, the D_4 hook. It is crystallographic and contains the 24-cell, but its physical space is four-dimensional and it is paracompact.

27.2. What best in three dimensions really means

The three-dimensional entry needs a careful qualification, because $\{5, 3, 4\}$ is not the unique best three-dimensional honeycomb. The compact regular three-dimensional hyperbolic honeycombs are exactly four, $\{3, 5, 3\}$, $\{4, 3, 5\}$, $\{5, 3, 4\}$, and $\{5, 3, 5\}$, and every one of them contains a 5. So all of the compact three-dimensional regulars are non-crystallographic and carry no spinor. $\{5, 3, 4\}$ is singled out only because it is the cleanest and best-studied of the four, with dodecahedral cells and Margenstern’s elegant Fibonacci addressing built for it. That makes it the natural three-dimensional representative, not a battery winner.

If one scores crystallographic-ness alone, $\{5, 3, 4\}$ actually loses to the paracompact three-dimensional honeycombs $\{6, 3, 4\}$, $\{4, 4, 4\}$, and $\{3, 6, 3\}$, whose symbols use only 3, 4, and 6, so they are crystallographic. But those are paracompact, with ideal cells, and they still give only a two-dimensional physical space. The decisive fact is the dimension ladder. An H^3 bulk has a two-dimensional boundary, so any three-dimensional substrate hands over only two-dimensional space. Three dimensions is ruled out before the battery even runs, and $\{5, 3, 4\}$ is simply the nicest three-dimensional control to run it on.

27.3. The preferred substrates

Across the sweeps, four substrates earn a permanent place, one per dimension. Two are the canonical well-addressed lower-dimensional tilings, valued for clean computability, and two are the physics champions.

Dimension	Preferred	Why it earns its place
2D	$\{7, 3\}$	the canonical hyperbolic CA tiling, Margenstern Fibonacci addressing, the reference for computability
3D	$\{5, 3, 4\}$	the committed compact three-dimensional candidate, dodecahedral, also Margenstern-addressed
4D	$\{3, 4, 3, 4\}$	the champion, D_4 spinor coin, flat three-dimensional cusp, full physics
5D	$\{3, 4, 3, 3, 4\}$	the D_4 -hook overshoot, crystallographic and containing the 24-cell, four-dimensional cusp

Table 17: The four preferred substrates, two computational anchors and two physics anchors.

The two roles are worth separating. $\{7, 3\}$ and $\{5, 3, 4\}$ are the computational anchors. Their addressing is trusted, and they validate the tools and serve as controls. $\{3, 4, 3, 4\}$ and $\{3, 4, 3, 3, 4\}$ are the physics anchors, the champion and its one-dimension-up sibling. The physics anchors also happen to be the best in their dimension, which is why they are at once preferred and best.

27.4. The same battery, side by side

Running all four through the identical battery makes the selection visible in a single table. The framework ports to every column. The physics-distinguishing axes do not.

Test	$\{7, 3\}$	$\{5, 3, 4\}$	$\{3, 4, 3, 4\}$	$\{3, 4, 3, 3, 4\}$
physical space	1D	2D	3D	4D
crystallographic spinor	/ no / no	no / no	yes / yes	yes / hook
curvature	hyperbolic	hyperbolic	hyperbolic	hyperbolic
framework ports	yes	yes	yes	yes
gravity / statistics	log / anyon	log / anyon	$1/r$ / fermion	$1/r^2$ / 4D

Table 18: The same battery on the four representatives. Only $\{3, 4, 3, 4\}$ scores on every physics axis at once.

Result 30 (Only one wins on every axis) The framework ports to all four representatives. Only $\{3, 4, 3, 4\}$ scores on every physics-distinguishing axis at once, crystallographic, D_4 spinor, three-dimensional space, $1/r$ gravity, and fermions. $\{7, 3\}$ is the canonical two-dimensional anchor but is one-dimensional and non-crystallographic. $\{5, 3, 4\}$ is compact but non-crystallographic and two-dimensional. $\{3, 4, 3, 3, 4\}$ keeps the D_4 hook but overshoots to four dimensions. [43]

27.5. The decisive contrast, the flat twin

The sharpest control is not in the table, because it is not hyperbolic. The flat D_4 honeycomb $\{3, 4, 3, 3\}$, the Euclidean 24-cell honeycomb, has the same coin and the same crystallographic spinor structure as $\{3, 4, 3, 4\}$, but it is flat. Its growth ratio is around 2.5 and polynomial, against the exponential growth of about 10 for $\{3, 4, 3, 4\}$. So the flat twin has the spinor coin and a three-dimensional space, yet no curvature, and therefore no gravity, no holography, and no expansion. Curvature is the single ingredient that separates the substrate from its flat twin, and it is what every cosmological and gravitational result depends on.

The best of each dimension confirms the selection from the side of measured comparison rather than abstract counting. Only $\{3, 4, 3, 4\}$ has the D_4 spinor coin and the right dimension and curvature together. It is the one substrate that yields a three-dimensional world of fermions.

28. Modeling Three Dimensions in Lower Dimensions

The selection argument so far reads as if the lower-dimensional substrates were impossible. They are not. Their missing features could be made to emerge at a higher layer, and a complete account has to say so. This caveat does not overturn the selection. It sharpens it. The right reading of the previous chapters is not that the alternatives are forbidden, but that $\{3, 4, 3, 4\}$ is the substrate of *least emergence*. This chapter makes that distinction precise.

28.1. Can three-dimensional physics live in a lower base

It can, in three different senses, and none of them is forbidden by any principle the model relies on.

- **Computation.** A one-dimensional cellular automaton can be Turing-complete, as Rule 110 is, so it can simulate three-dimensional physics outright. In that case the three-dimensionality lives in the data, not in the geometry of the base.
- **Holography.** A gravitational theory in $d + 1$ dimensions is exactly equivalent to a theory on a d -dimensional boundary. The $\text{AdS}_3/\text{CFT}_2$ correspondence, for instance, puts a three-dimensional gravity inside a two-dimensional conformal field theory. The extra dimension emerges from the boundary data.
- **Emergent dimension.** Tensor networks such as MERA, the SYK model, and matrix models all grow effective dimensions out of lower-dimensional bases.

The reason any of this is possible is the holographic bound. Information scales with boundary area, not with volume, so three-dimensional physics can in principle be fit onto a lower-dimensional screen. Dimension is not a hard wall.

28.2. The catch

The escape is real but it is not free. A naively local lower-dimensional theory cannot do it. The Bell inequalities and the Weinberg-Witten theorem both block a straightforward local encoding, so the encoding has to be non-local, or it has to grow an emergent dimension. In practice a lower-dimensional substrate that wanted to reach three-dimensional fermionic physics would have to carry extra machinery, a holographic or internal extra dimension, a fermionization to manufacture the spin statistics, and an emergent gauge structure. Each of these is a layer that the base does not have on its own.

28.3. Why the four-dimensional substrate still wins

This is where the economy argument lands. $\{3, 4, 3, 4\}$ hands over three-dimensional space, spin, and gauge at the base, with no dimensional emergence required at all. The lower-dimensional substrates could in principle reach the same physics, but only by paying an emergence cost, uplifting the dimension, fermionizing, and generating a gauge group. The selection is therefore made on economy rather than on impossibility.

Principle 6 (Least emergence) The substrate is selected by economy, not by exclusion. $\{3, 4, 3, 4\}$ delivers three-dimensional space, spin, and gauge at the base with no emergent dimension. A lower-dimensional substrate could reach the same physics only by adding emergent machinery. The precise claim is that $\{3, 4, 3, 4\}$ is the least-emergence substrate, not the only conceivable one.

Stating the argument this way keeps it defensible and keeps it falsifiable. The position is not "the others are impossible", which would be hard to defend and easy to attack. It is "the others need more emergent layers to do what $\{3, 4, 3, 4\}$ does at the base", which is a measured comparison and an open invitation to check. It also keeps the door open in two useful directions. The emergence question stays alive as a research target, and the lower-dimensional substrates keep their role as controls. Whether the crystal is finite or infinite is a separate matter, and it is left open here as elsewhere in the paper.

Part VI

The Emergent Ladder

29. Computation, Universality, and Addressing

Before life and mind, one fact licenses the whole climb. The substrate can compute anything. This is the permission slip for everything that follows. The wild claim, that one rule on a crystal yields atoms, weather, and minds, rests on a modest fact. The rule is universal, so any pattern that can exist at all is a pattern the rule can host. A complex world does not require a complex law. The rest of this part is more speculative than the physics that came before it, and the present chapter is the firmest rung in it, because universality and addressing are settled questions that can be checked in code rather than argued by analogy.

29.1. The substrate is a computer

The dynamics is a discrete-state, 24-neighbour *cellular automaton* on the $\{3, 4, 3, 4\}$ honeycomb. It is not a computer by analogy. It is a cellular automaton on the nose. There is no outside machine, no operator, no screen. Running the rule on the substrate is what existence does in this model. Universality rests on four legs, all run in the codebase [43]. The first three give *weak* universality, the standard hyperbolic cellular-automaton notion that allows an infinite or ultimately periodic background. The fourth gives *strong* universality with no such crutch.

- **Leg one, the Margenstern structure is present.** What Margenstern (margenstern2008) used to prove the hyperbolic grids $\{7, 3\}$, $\{5, 4\}$, and $\{5, 3, 4\}$ universal all has its $\{3, 4, 3, 4\}$ analog. The Fibonacci tree becomes the addressing tree with growth near 18.4, a bushier generalized-Fibonacci tree. The black and white son types become the region types of the splitting matrix. The preferred son becomes the digit-0 spine. The full 24 interior directions give abundant room for the railway model of a train on switched tracks, since a junction needs only three edge-disjoint directions.
- **Leg two, the rule itself is functionally complete.** The model's own signed-majority update, with a +1 bias and two −1 fills, computes NAND, which is functionally complete. From NAND it builds NOT, AND, OR, XOR, a full adder, and in particular Rule 110, which Cook proved universal. The rule-built Rule 110 reproduces a reference Rule 110 exactly. The substrate's own ternary rule hosts a Cook-universal automaton.
- **Leg three, a conserving register machine.** A two-counter Minsky machine, which is Turing-universal, runs directly on the mesh. Registers are charged subtrees navigated by address. Increment is the arrow creating a balanced pair. Decrement is annihilation. Test-for-zero is perception, counting the charges. Total tone stays exactly conserved throughout, so this is genuine substrate dynamics and not a circuit bolted on. It runs real programs. Addition gives $3 + 4 = 7$ and multiplication gives $3 \times 4 = 12$ and $5 \times 5 = 25$, with charge conserved at every step.
- **Leg four, strong universality from the cusp.** The cusp is the Euclidean cubic honeycomb $\{4, 3, 4\} = \mathbb{Z}^3$, and Conway's Game of Life on a $\mathbb{Z}^2$ layer of it is a known universal cellular automaton.

This is now run, not merely argued. A glider seeded on the actual $\{4, 3, 4\}$ cusp plane evolves identically to a reference $\mathbb{Z}^2$ Life and translates by $(1, -1)$ after its period-4 cycle. So the theory’s emergent physical space is itself a universal computer, run in the codebase.

Result 31 (Universality on four legs) The $\{3, 4, 3, 4\}$ substrate is universal in four independent ways. Its addressing tree carries the Margenstern structure, its ternary rule computes NAND and hence Rule 110, a conserving Minsky machine runs on the mesh with charge exactly conserved, and Conway’s Life runs on the flat cusp plane. The first three give weak universality, the fourth gives strong universality [43].

The distinction worth holding is capacity versus outcome. Universality is about what can be expressed, not about freedom of result. The capacity is unlimited. The outcome of any given run is exactly one.

29.2. Undecidability comes with the power

The moment a system can simulate a universal machine, its halting becomes undecidable. This is not a flaw. It is the formal signature of a world in which the only way to know the future is to live it. The rule still produces a perfectly definite future. There is simply no algorithm that fast-forwards to the answer in general. This is the same computational irreducibility that grounds the freedom argument later in the paper. The natively undecidable problem on the substrate is the domino problem, which asks whether given tile types can cover the plane. It is undecidable precisely because the substrate is a tiling, so undecidability is a native feature of the geometry rather than an import from outside it.

There is a clearly speculative aside worth stating and then setting aside. If a cell had infinitely many neighbours, in the manner of the Margenstern-Grigorieff infinigrd, it could evaluate a quantifier in one tick and decide every level of the arithmetical hierarchy, which would be hypercomputation. We keep this firmly speculative. It needs unbuildable infinite objects, and $\{3, 4, 3, 4\}$ has 24-direction cells rather than infinitely many, so the result does not apply to the actual substrate. The clean theorem on undecidability and the labelled thought experiment on hypercomputation are never confused.

29.3. Addressing $\{3, 4, 3, 4\}$ at scale

For the universe to be computable and navigable in practice, every cell must compute its own neighbours from its own name, with no master map. Margenstern’s splitting and Fibonacci-addressing program does this for $\{7, 3\}$ and $\{5, 3, 4\}$ (margenstern2013), and it has now been extended to $\{3, 4, 3, 4\}$ and verified. Steinberg’s formula on the $[3, 4, 3, 4]$ Coxeter group gives the exact shell recurrence, with shells of size 1, 24, 456, 8376, $\dots$, and a growth rate that defines a Zeckendorf-style numeration system for cell addresses. The surprise that makes the port clean is that $\{3, 4, 3, 4\}$ has no same-shell cousins, so the hard 2D piece of Margenstern’s method is simply absent. The neighbour map is then a finite generator-address transducer, a small automaton deterministic at window $K = 2$ through shell three.

Result 32 (Addressing solved at scale) Every cell of $\{3, 4, 3, 4\}$ has a unique, self-describing, $O(\log n)$ address, with zero duplicate addresses and zero decode round-trip failures. The neighbour set is reconstructed exactly from parent plus children plus the confluence transducer. Greedy address-routing delivers 1999/2000 at mean stretch 1.004. The core adjacency invariants, namely zero cousins, unique addresses, and deterministic confluence, were confirmed to survive to 2,000,000 cells [43].

So $\{3, 4, 3, 4\}$ is a computable substrate at scale. A cell knows where it is and who its neighbours are from its name alone. This enables large-scale simulation of the dynamics, and it makes good on the theory’s claim that the universe is computable, and discrete rather than continuous, whether finite or infinite.

29.4. Margenstern’s sons, ported to $\{3, 4, 3, 4\}$

Margenstern’s 2D method gives each tree node a type. The pentagrid has two, a black son with two children and a white son with three. The types reproduce each other by a fixed rule, the matrix of that rule generates the Fibonacci growth, and a preferred son continues the main spine. The $\{3, 4, 3, 4\}$ port keeps the whole structure. It is richer but the same in shape.

A cell’s type is its number of canonical children. There are exactly ten types, with child counts running from 15 to 24, the $\{3, 4, 3, 4\}$ analog of the pentagrid’s black and white pair. The splitting matrix M has $M[i][j]$ equal to the average number of type- j children a type- i cell has. Its dominant Perron eigenvalue is 18.284, which equals the measured shell growth ratio, since 1, 24, 456, 8376 has ratio tending to about 18.37. Its characteristic polynomial is the shell recurrence. So M plays exactly the role the pentagrid’s small son-matrix plays, generalized from 2×2 to 10×10 . The preferred son is the digit-0 child, the canonical spine successor running root to cell 1 to cell 25 and onward. The simplification is the absence of cousins. In 2D the hard part is the cousin automaton for same-ring neighbours across branches. Here every one of a cell’s 24 neighbours is a parent or a child, so that automaton is not needed. The only non-tree structure is confluence, a cell reachable as a child of more than one parent, handled by a small finite transducer that is deterministic at window 2. The $\{3, 4, 3, 4\}$ tree is therefore structurally simpler than the 2D originals it descends from.

29.5. The addressing idea in its cleanest form

The idea is most transparent in Margenstern’s classic 2D case, the heptagrid $\{7, 3\}$, where the growth is the Fibonacci recurrence and the addresses are *Zeckendorf* strings. Number the cells ring by ring, then rewrite each number in the Fibonacci base, whose place values are 1, 2, 3, 5, 8, 13, $\dots$. By Zeckendorf’s theorem every integer is a unique sum of non-adjacent Fibonacci numbers, so no address ever has two ones in a row, which makes the valid addresses a regular language that a tiny automaton recognizes.

cell	Fibonacci sum	address
1	1	1
2	2	10
3	3	100
4	$3 + 1$	101
5	5	1000
6	$5 + 1$	1001
7	$5 + 2$	1010
8	8	10000
9	$8 + 1$	10001
10	$8 + 2$	10010
11	$8 + 3$	10100
12	$8 + 3 + 1$	10101

Table 19: Zeckendorf addressing on the heptagrid $\{7, 3\}$.

Routing is pure string surgery. The parent is found by stripping a digit, a child by extending by a legal digit. To route from A to B you compare addresses from the left, walk up to the common ancestor, then walk down. Sending a message from cell 12, with address 10101, to cell 9, with address 10001, the shared prefix is 10, which names cell 2 as the common ancestor. The path goes up from 12 to 7 to 4 to 2, then down from 2 to 3 to 5 to 9. The full route is 12, 7, 4, 2, 3, 5, 9, six hops, with every cell comparing only two short strings and no map consulted. Heptagrid routing delivered 100 percent at low stretch in the runs. This is the clean 2D template. The $\{3, 4, 3, 4\}$ case carries the same idea, with route-from-the-root, parent-by-strip, and common-prefix routing, but with a richer numeration whose growth root is near 18.3 rather than the golden ratio, so the base is the 24-way child index rather than Zeckendorf binary, and with no cousins.

29.6. A worked address example

Addresses on $\{3, 4, 3, 4\}$ are dot-separated digits. Each digit is the index of a cell among its parent’s children, ordered deterministically by embedding, and the whole string is the route from the root. The root is the empty address written e . Shell-1 cells are single digits, shell-2 cells are two digits, and child counts vary by son type. The following cells are real output of the addressing code.

cell	shell	address	parent	children (son type)
0	0	e	—	24
1	1	0	0	23
2	1	1	0	22
3	1	2	0	21
...	1	...	0	...
24	1	23	0	15
25	2	0.0	1	22
26	2	0.1	1	21

Table 20: A few $\{3, 4, 3, 4\}$ cells with shell, address, parent, and child count.

A routing walk from cell A to cell B is pure address arithmetic. Compare the two addresses from the left, walk up from A to the common ancestor by dropping the last digit to reach the parent, then walk down to B by appending B ’s remaining digits to reach the named child. Take A to be cell 481, address 0.0.0, and B to be cell 503, address 0.1.0. Their longest shared prefix is 0, so the common ancestor is cell 1, address 0.

step	cell	address	operation
0	481	0.0.0	start (A)
1	25	0.0	drop last digit, go to parent
2	1	0	drop last digit, reach the common ancestor
3	26	0.1	append digit 1, go to child
4	503	0.1.0	append digit 0, reach B

Table 21: A routing walk from cell 481 to cell 503, four hops.

Four hops, which equals the shortest-path distance, is exactly the address arithmetic the routing uses, with no map and no coordinates. The neighbour set of A is likewise reconstructed from its name alone. It is parent 25, plus 21 children, plus 2 confluence alternate-parents, plus 0 alternate-children, totalling A ’s full 24 facet-neighbours.

29.7. One structure, computation and physics

The decisive point is that the computation and the physics sit on the same structure. Margenstern's heptagrid and pentagrid carry the Fibonacci addressing and the universality, but nothing else. They are 2D, with no spinor, no Standard Model, and no 3D space. They are computation substrates and nothing more. The $\{3, 4, 3, 4\}$ grid carries the whole package and the physics, because the same 24-cell coin that gives the bushy generalized-Fibonacci tree is the D_4 root system that gives the spinor, $SO(10)$, and the three generations, and the same cusp that runs Conway's Life is the flat 3D space where gravity and the light cone live.

Principle 7 (One substrate for navigation, universality, and physics) The 24-cell coin of $\{3, 4, 3, 4\}$ is at once the address-tree generator and the D_4 root system, and its cusp is at once the flat 3D space of physics and a universal cellular automaton. Navigation, universality, and physics therefore sit on one structure, with nothing added between them.

So the old trade-off, that $\{7, 3\}$ wins universality while $\{3, 4, 3, 4\}$ wins spin and the two are complementary, is dissolved. The $\{3, 4, 3, 4\}$ grid has both, and it is structurally simpler with no cousins than the 2D grids it descends from. The substrate is at once Lorentz-safe, exactly addressable, computationally universal, both weakly through the ternary NAND, Rule 110, and the register machine, and strongly through cusp Life, and physically loaded. The key caveat is that universality covers the structure of any process, not the felt interior. The qualia, the inner light, is taken as the base in the monism. That is a stance, not a derivation.

Computation is the floor under the whole ladder. The substrate can host any pattern, it cannot in general be shortcut because of undecidability, and it is navigable from local names alone through the addressing.

30. The Climb, From Field to Mind

The physics is the bottom of a ladder, not the whole story. The claim of this part is that the same four base things that build the field also build life, mind, and intelligence, as higher regimes of one substrate rather than as new ingredients. This part climbs that ladder. It is more speculative than the chapters on physics, and it works at toy scale rather than at the scale of a real organism or brain. We mark each rung's status plainly, and the reader should treat the upper rungs as plausibility arguments backed by separate demonstrations rather than as a single end-to-end proof.

30.1. Everything is a glider

In the rule, a thing is not a fixed set of cells. It is a pattern that persists while the cells beneath it come and go, the way a *glider* in a cellular automaton re-forms one step over with each beat. Every particle and every self is this kind of object, a stable traveling pattern made of nothing but arrangement. This is why a self can persist while none of its underlying cells do. In the carrying image, a flame keeps its shape while the gas burning in it is always new. Holding this picture in mind is the key to the whole climb, because it explains how identity can survive complete material turnover at every level, from a particle to a person.

30.2. The multiscale shortcut

We do not simulate a mind from the base. A vibe is sub-Planck, a particle is on the order of 10^{60} vibes, and a brain is hopeless to build cell by cell. Instead each layer has an up-map that coarse-grains a block into one

richer symbol, and a down-map that lifts a symbol to a representative pattern. A layer is faithful when its dynamics commute, so that coarse-graining then evolving equals evolving then coarse-graining. This was checked across scales from thirty thousand to one million cells [43]. So we study each rung at its own level, the way biology studies cells without simulating quarks.

The maps have a precise shape, a Galois-like connection rather than an isomorphism. The up-map is lossy and lands in a richer macro-alphabet, while the down-map is a canonical lift, with the composite returning the symbol exactly and the dynamics square commuting. The key consequence is that the base never limits the macro-alphabet, only the macro-dynamics. Composites can be arbitrarily rich, the way bits become files, so the objection that the tone is only ternary has no force at high layers. The ternary base constrains how things move, not how much they can encode. A higher vibe is an aggregate read-only view of the one stored base. There is no hidden super-tone, which is the no-hidden-state commitment in which the note is perception.

Principle 8 (Nothing added up the ladder) Each higher layer is a faithful coarse-graining of the layer below it, with a lossy up-map into a richer alphabet and a canonical down-map back. The base limits only the dynamics, never the alphabet, so composites can be arbitrarily rich while remaining a read-only view of the one stored base. No new ingredient is introduced at any rung.

30.3. What the middle layers are

Raw tones are not particles. There are one to a few emergent middle layers between the vibe churn and physics, and we can name the first one. The first faithful middle layer is a slow conserved-charge density mode, a hydrodynamic slow-mode whose autocorrelation lifetime grows with coarse-graining scale. A region's charge changes only by boundary flux, so larger regions are slower. This is measured, with the lifetime climbing from near zero at the base to a finite value a few levels up, and it is explicitly not physics. It is the first slow collective variable above the churn, the way a fluid density emerges above molecules.

The caveats matter here. The lifetime is modest, and hyperbolic geometry weakens the slow mode, since a ball there is nearly all boundary. That is one more reason physics lives on the flat cusp rather than in the bulk. So the bottom of the ladder reads as tone churn, then a slow density mode, then on the flat cusp the relativistic field, then particles. At no point is a tone the same thing as a particle. Keeping this clear is important, because the temptation to read raw clustering as gravity or matter is exactly the kind of overreach the theory is built to avoid.

30.4. The climb, with status

- **Field to particle, measured.** A particle is a persistent bump in the field. Its centroid moves in a straight line at constant speed below the light cone, an Ehrenfest result on the lattice with $R^2 \approx 1$.
- **Particle to composite, measured.** Two particles keep a uniform center of mass, with momentum conserved through interaction at $R^2 \approx 0.997$. The conserved charge gives the minimal atom, a net-charged region that cannot vanish.
- **Atoms through molecules, cells, tissues, senses, to minds, plausible.** This is not simulated as one run. Every ingredient is separately demonstrated, including binding by the merge law, self-

maintenance, copying, the union of senses, and the integrated hub. It is a grounded extrapolation, not a single proof.

The one-line version of the ladder runs as follows. A particle is a bump in the field. An atom is a conserved bound unit. A molecule is a larger bound unit. A cell is a self-maintaining self. A tissue is a tower of cells. A sense is a tuned boundary. A mind is an integrated hub with a self-model. A human is the high end of that same climb. None of these requires a new base thing, which is the entire point of laying them out as one continuous ascent.

Result 33 (The lower rungs are measured) The field-to-particle and particle-to-composite steps are measured directly, with straight-line centroid motion below the light cone at $R^2 \approx 1$ and momentum conservation through interaction at $R^2 \approx 0.997$ [43]. The rungs above the composite are demonstrated ingredient by ingredient rather than in one run, and are marked plausible.

The substrate does not stop at physics. The same rule, read at greater heights, gives the living and the mental. This is the boldest and most speculative claim of the theory, and it is the reason this part is laid out as a ladder with the rungs plainly tagged. The chapters that follow take up life and then mind in turn, each one keeping the same discipline of separating what is measured from what is extrapolated.

31. Life

Life is not a new substance. It is a regime, the same four base things held far from dead equilibrium. Out of that, coherent patterns appear that hold themselves together, repair their wounds, remember, interact, want, copy, and evolve. There is no life force and no sixth ingredient. As with the rest of this part, the results here are toy-scale and the strongest of them carry a recorded negative alongside them, which we state plainly rather than smooth over.

31.1. The central tension and its resolution

The rule conserves total charge exactly, which looks at first like it forbids life, since life needs growth, memory, and copying. The resolution is that conservation is of the sum, not the arrangement. The arrow creates only balanced pairs, a $+1$ with a -1 , net zero, so net well-being never changes, while balanced information, meaning net-neutral patterns, is free to be made, copied, and healed. Life is conserved net charge plus freely created balanced information, both at once, with nothing relaxed in the base. This single move is what lets a strictly conservative substrate host open-ended structure without contradiction.

31.2. The arrow is the life engine

Run the rule with the arrow off and opposite tones annihilate, so the universe relaxes to dead peace, all zero, no motion, the heat death of the substrate. Turn the arrow on and peace spawns balanced pairs, so the universe settles into a living dynamic balance, churning forever, never freezing and never blowing up, with charge conserved to the unit throughout. The arrow is the single posit that keeps the world alive. It is worth stressing that this is not an extra rule added for life. It is the same arrow already present in the base, observed here at the scale where its effect is the difference between a dead substrate and a churning one.

31.3. What a self is

A bare same-tone blob is rejected. A self is defined by what it does, not by what it is made of.

Definition 5 (A self) A self is a persistent, bounded, self-maintaining, integrated pattern. It satisfies four conditions at once. Integration means its parts are more bound to each other than to the outside. Self-maintenance means it repairs its own pattern against churn, doing work to stay itself. Persistence means its identity lasts as material flows through it. Boundedness means it keeps a real membrane.

A self is a pattern that holds its shape while matter and energy stream through it. The same definition holds at every scale, and what changes is the degree of structure and agency. An atom is the minimal self, a net-charged region that cannot vanish because charge is conserved, with low agency, the floor of the ladder. A cell adds active homeostasis. An organism adds a self-model and steering, taken up in the next chapter. One definition, three degrees. The boundedness is a gradient, not a hard line. Integration fades continuously outward, so a self is a soft peak in a continuous density that shades into the world, not a crisp object with a wall, and membership is a degree. Boundedness means where the binding weakens, not where one tone stops. This dissolves the question of where I end. A self has a fuzzy edge, like a whirlpool with no exact rim.

31.4. Selves emerge, heal, interact, want

The true state of a single self is the most important point in this chapter, and it has two halves that must be held together. A single self is real. The genuine self is the topological *hopfion*, a knot in the tone field, localized, stable, and fermionic, and it persists through full material turnover, the whirlpool property. The cusp demonstrably supports propagating particle-like self-patterns when they are scored by the right measure, identity-through-turnover, meaning same form, different matter, moving. A Conway Life glider on the $\{4, 3, 4\}$ cusp passes this measure cleanly, with pattern identity 1.00, material turnover 0.40 per beat, speed 0.42, and it stays localized.

The other half is the negative. The bare rule does not spontaneously crystallize a self from a generic seed. From a generic seed the pure conserving rule churns, with identity 0.00 as the blob disperses, so a self must be seeded or topologically protected as the hopfion is, not conjured from an equilibrium initialization, because there is no dissipation. An important measurement correction underlies this. The earlier reading in which selves emerge and nest into a tower used charge-per-coarse-block persistence, which is the wrong measure, because it captures a generic hydrodynamic slow mode rather than selfhood. The right measure is identity-through-turnover, and it tells the plain truth. The cusp can host selves. The bare rule does not build one.

Result 34 (A single self, with its recorded negative) A genuine self exists as the topological hopfion, and a Conway Life glider on the $\{4, 3, 4\}$ cusp passes the identity-through-turnover measure cleanly with pattern identity 1.00. The bare conserving rule does not spontaneously crystallize a self from a generic seed, where it churns at identity 0.00. The cusp hosts selves but does not build one unprompted [43].

The nested tower is a measurement artifact. The form-coherence tower, in which coarse blocks hold charge longer than fine cells, does not beat pure diffusion, measured head-to-head on both the cusp and the bulk, so it is generic coarsening rather than hierarchical selfhood. Genuine nesting, meaning selves made of selves, is the open higher-order-binding frontier rather than a free result, and the theory will not add a fifth ingredient to force it without proof that it cannot emerge from the four. The remaining behaviors are measured and modest.

Self-healing works, since punching a hole lets the bare rule pull the right tone back from the redundant surround, recovering about 65 percent of the wound, an emergent first aid. Interaction is the base rule at the scale of individuals, since opposite-sign selves annihilate at contact and same-sign selves merge into one, with no special interaction theory. Will is the arrow held by a self and aimed, since a self that biases its own creation toward a direction moves toward a goal and away from a threat.

31.5. Memory costs work

The rule is reversible, so it has no attractors, so passive permanence is limited to the conserved charge itself. Richer memory is a pattern, and patterns spread, so memory decays. There are three tiers. Leaky memory is free, since cohesion roughly doubles survival. Holographic memory survives bounded erasure, since the bit lives everywhere at once. Permanent memory needs active maintenance, periodically re-writing the codeword as the arrow refills holes. The maintenance cost is not a flaw. It is Landauer's principle (landauer1961), since holding a low-entropy pattern against an equilibrium dynamics requires work. A real cell is a dissipative, self-repairing pattern that spends work to stay itself.

There is a brief insight on the holographic tier. A growing holographic code gets stronger with age. As the mesh expands, the redundant encoding spreads over more cells, so its erasure threshold rises with age, from about 95 percent to about 99.5 percent in the runs. The de Sitter expansion is itself a re-encoding, a renormalization flow, and old memory hardens as the universe grows, with the perception rule at the frontier acting as the bulk-to-boundary isometry. So expansion does not only dilute. It also entrenches what is already encoded.

31.6. Reproduction, evolution, and value

Copying a balanced pattern onto a fresh frontier region is conserving creation, the arrow's own balanced pair-making, so reproduction with heritable variation is allowed. Heredity plus copy-error variation plus the arrow as selection is a complete Darwinian loop, every piece from the base, and mean fitness climbs uphill in the runs. Fission is suppressed in the hyperbolic bulk, since balls there are all boundary with no thin necks to pinch, a generic hyperbolic fact, but the flat 3D cusp does have thin necks, so a dumbbell self on the cusp can pinch and divide. Copying works everywhere, division works where it is flat, and life lives on the flat cusp.

A word on value and cooperation closes the chapter, because it follows from the conservation law rather than from any moral decoration. The raw drive points at pleasure, but the system's optimum is balance, meaning peace, and nothing automatic closes that gap. It is learned by higher minds, the move from grabbing to wisdom. This is the strategy that actually wins on a conserved substrate. Because charge is conserved, competing for pleasure is a zero-sum arms race that burns work for no net gain, just suffering moved around. Three constructive resolutions beat it, all from the base. Boundaries are insulating neutral cells acting as membranes, keeping one self's balance from draining into another. Trade works because different selves value different structures, so exchange is positive-sum in value even though charge is conserved. Integration is merging into a higher self, sharing the fixed pleasure in balance, and pouring the freed energy into building order, which is not conserved and can grow. So the drive matures from grabbing pleasure, a zero-sum war, to building order together, which is positive-sum, and that joining-up, again and again, is how cells become bodies become minds. The drive to grab is not crushed. It is outgrown.

Life falls out of the same four things as a far-from-equilibrium regime, with the arrow as its engine, selves as self-maintaining patterns, and Darwinian evolution from the base. These mechanisms rest on generic hyperbolic and rule-level facts, so they carry to $\{3, 4, 3, 4\}$, and running it fully on the 24-direction substrate is the natural follow-up.

32. Mind

A self becomes a mind when it does something more. It represents itself, predicts itself, attends, and plans. Four powers and a closed-loop agent that chains them, all by composition of the four base things. There is no new ingredient. As with the rest of this part, the results are toy-scale and first-order, and we mark them so. What follows should be read as a demonstration that the ingredients of mind can arise from the substrate without anything added, not as a claim to have built a mind.

32.1. The strange loop

An integrated self funnels every part's signal toward its center. That funneling already builds a part that mirrors the whole. So integration is self-representation. This is the key idea, and once a part models the whole self the rest follows in order. The model predicts, giving a forward model. A model can model a model, giving recursion. The model is where attention selects content, giving a workspace. Running the model forward, judged by the arrow and pushed by the will, is planning. The point worth stressing is that none of these is bolted on. They are all consequences of the single fact that an integrated self contains a part that mirrors the whole.

32.2. The self-model and the onion

The mechanism is the tiny diameter of hyperbolic space. Charge from every boundary sector flows inward and mixes at the center, so the hub sees the pooled whole while the periphery sees only its sector. As measured, the hub tracks the global state at correlation about 0.5, the periphery about 0.0, and freezing the hops collapses it, so it is integration rather than anything hand-placed. Every self forms its own self-model at its own hub, so there is one mind-seed per self, not one special soul at the middle of the world. The shape is an onion. The boundary is the senses, where the world touches the self. The body is the integrator, where charge pools. The hub is the center, the self-representation. The loop mirrors the whole back into itself.

Result 35 (The self-model is an integration effect) In hyperbolic space the small diameter pools every boundary sector at the center, so the hub tracks the global state at correlation about 0.5 while the periphery tracks it at about 0.0. Freezing the hops collapses the correlation, which shows the self-model is produced by integration rather than placed by hand, and every self forms its own self-model at its own hub [43].

32.3. Prediction, recursion, attention

- **Forward model.** The hub at time t predicts the global state at $t + 1$, at correlation about 0.5 against periphery about 0.0, because the hub holds the integrated state and that is the best guess of the self's next state. This is a first, imperfect seed of metacognition.
- **Recursion.** Wire one self's hub as input to a second self, and the second models the first at about 0.83, a representation of a representation, the substrate of imagination.

- **Workspace.** The hub is a global workspace. Loud signals win it, which is bottom-up salience, and attention can raise a quiet one, which is top-down gain. The real limit is that broadcast to the far periphery is range-limited, since the bulk field is massive and short-ranged, so the workspace is the hub, a convergence point, not a megaphone.

32.4. Planning, with nothing added

Planning is lookahead plus deferred value plus composition, accepting a worse step now for a better outcome later. It is built only from emergent pieces. Value is the arrow. The forward model is the self rolling out its own rule in imagination. Exploration is the will pushing against the local gradient. Where a greedy agent stalls at a barrier, the planner tries sustained will-pushes, simulates each forward, scores by the arrow, and commits to the best. The full integrated agent perceives the gap, plans by simulating forward, acts with the will, and re-plans at each stall, crossing four barriers that a greedy agent and a one-shot planner both fail. That is a mind, built from nothing but the four.

Principle 9 (The dependency chain) Mind is a chain from geometry to integration to self-model to forward model to recursion to planning to the closed agent. Pull out any lower layer and the higher ones fall. The named limits, that the forward model is local, that the workspace is a hub rather than a broadcast, and that foresight is bounded, are limits of the substrate, not patches over it. No homunculus, no search oracle, no separate planner module, and no special seat of awareness was ever added.

Mind is what an integrated self does when it folds back on itself. The strange loop is free, given the geometry, and everything above it is composition. The mechanism rests on the generic tiny hyperbolic diameter, so it carries to $\{3, 4, 3, 4\}$ with its 24 directions, and running it fully is the natural follow-up. In the carrying image, a mind is the fire becoming aware that it is burning.

33. Intelligence

Intelligence is the same four base things read as a solver. Nothing is added to the substrate to obtain it. The field, the arrow, the rule, and the geometry are simply asked a different question, namely how a configuration gets from where it is to where it would prefer to be. Throughout this chapter intelligence stays sharply distinct from consciousness. Intelligence is problem-solving capability, consciousness is integrated experience, and a system can in principle have one without the other. The two will be developed separately, and the distinction should be held firmly from the start.

Definition 6 (Intelligence) *Intelligence* is the capacity to take a problem and produce a solution. A *problem* is a gap $P = D - C$ between a desired state D and the current state C . When P is zero there is nothing to solve. A *solution* is a path from C to D that drives P to zero.

This definition already names the four things any solver needs. It needs *means*, paths it can compute. It needs a *goal*, the state D . It needs a *value*, an ordering that measures progress along the gap. And it needs *selection*, a way to keep the moves that reduce P and discard the rest. The remaining sections show that all four are present in the substrate as it already stands, with the value and the selection supplied by the arrow rather than bolted on as a separate reward machine.

33.1. The means: the substrate computes

Cellular automata on hyperbolic honeycombs are computationally universal, a result long established for tilings of the hyperbolic plane and its higher analogues [40, 39]. The substrate confirms it at toy scale. A two-counter Minsky machine, which is itself Turing-universal, runs directly on the mesh [43]. Registers are charged regions, increment is a create operation, decrement is an annihilate operation, and test-for-zero is a count. With those primitives wired together the construction computes $a \times b$ correctly. The universe on this reading is not merely like a computer. It is one, with no outside machine driving it. Running the rule on the substrate is what existence does.

Proposition 14 (The means are universal) The rule on the hyperbolic mesh supports universal computation. A two-counter Minsky machine has been realized on the substrate with charged regions as registers and create, annihilate, and count as its operations, and it computes correctly at toy scale.

Universality is generic to hyperbolic cellular automata, so it carries across the candidate geometries rather than depending on a particular one. The Minsky construction asks only for charge-region registers and the create, annihilate, and count operations, and those are geometry-agnostic. The conclusion therefore holds on $\{3, 4, 3, 4\}$ exactly as it holds on the alternatives, which matters because it means the choice of geometry is not what buys computational power. That power is already present wherever the rule runs.

33.2. The ends: the arrow

Computation by itself is goal-neutral. A universal machine computes garbage as readily as it computes a solution, because nothing in the bare notion of computation says which output is wanted. The direction has to come from somewhere, and in this theory it comes from the arrow, a base ingredient that is a gap carrying a better-or-worse ordering. The gap-closing dynamics of the rule then act as the selection. Put these together and computation becomes intentional the moment a universal machine runs under the arrow's value. No reward signal is attached and no planner is grafted on, because the reward and the planner are already present. The reward is the arrow's ordering, and the planner is the rule that descends it.

Principle 10 (Computation is goal-neutral and the arrow supplies the ends) A universal machine has no preferred output on its own. Intention appears when the same machine runs under the arrow's better-or-worse ordering, with the gap-closing rule acting as selection. The ends are not a separate module. They are the arrow.

The effect of aiming computation is large and measurable. A goal-directed search reaches a K -cell target in about K steps, where an un-goaled search wandering at random would need on the order of 2^K trials to stumble onto the same target by chance [43]. At $K = 30$ that is an intention gap of roughly ten million. The arrow does not make the impossible possible, since the substrate was universal already. What it does is convert an astronomically unlikely accident into a short directed walk, and that conversion is the whole of what it means for computation to have a purpose.

33.3. Maximal by integration, not criticality

A solver that aims to be maximal must coordinate a global problem across all its parts, and it must do so quickly. The hyperbolic geometry delivers exactly this through fast integration. The coordination distance across a region of N cells grows only as $O(\log N)$, a handful of hops, far below the $N^{1/3}$ scaling of a flat

three-dimensional lattice [43]. A problem posed anywhere in the structure can be felt everywhere within a few steps, which is the geometric precondition for a large mind to act as one thing rather than as a loose federation of distant parts.

It is worth being explicit that this maximal behavior does not ride on a critical edge. The substrate shows no self-organized criticality. Perturbations spread ballistically with a span near unity rather than fanning out into branching avalanches, and activity self-tunes to a single homeostatic operating point rather than hovering at a phase boundary [43]. The same ballistic light cone that makes the model well behaved for relativity is precisely what rules criticality out. Intelligence here is a property of the geometric hierarchy and its fast mixing, not of a delicate tuning to the brink of instability.

33.4. The level-relative principle

A recurring trap in reasoning about this model is to take a limit observed at the base and treat it as absolute. Most apparent limits are properties of the hyperbolic bulk and they lift at emergent levels, because the substrate is universal and the levels are programmable. Only a few limits hold all the way up. There is no minting of net charge, no signaling faster than the light cone, and no hidden state at the base. Everything else is reachable at some level, and a claim that something is impossible has to name the level at which the impossibility is supposed to hold.

Principle 11 (Level-relativity of limits) A limit observed in the hyperbolic bulk need not survive to emergent levels. Absolute limits are few, namely no net-charge creation, no faster-than-light-cone signaling, and no hidden base state. All other constraints are level-relative, and a forbidding claim must specify its level.

The affordance ladder makes this concrete. A single ternary cell stores three symbols. A small block of cells stores seven, then nineteen, then fifty-five, climbing as the block grows. A K -ary symbol that is plainly impossible to hold in one base cell becomes trivial one level up, simply by reading a block as a unit. The same move applies to most of the apparent ceilings one meets while studying the substrate, which is why the careful phrasing of nearly every limit begins with the words at this level.

Intelligence is universal computation aimed by the arrow and made maximal by the geometry's fast integration. The means come from universality, the ends come from the arrow's ordering, and the maximality comes from logarithmic coordination across the hyperbolic hierarchy. None of these is an addition to the base. They are three readings of the four ingredients already in place. The means and the ends are on solid ground, measured at toy scale. The higher coarse-graining rungs, where one would hope to see open-ended problem solving rather than fixed targets, remain open and are treated where the scope of the program is set out.

34. The Unfolding: The Climb From Field to Human

This chapter sets out the grand narrative of the theory, the full multiscale climb from the quantum field to a human, taken one rung at a time. It is a grounded extrapolation rather than a single end-to-end simulation, and the status of every step is marked so that nothing tested is confused with anything speculative. The lowest rungs are measured. The middle rungs are plausible, in the precise sense that every ingredient they need has been demonstrated separately even though the full assembly has not been run. The drivers at the

top include two options that are clearly speculative, and this part of the theory is more speculative and more toy-scale than the physics that precedes it. That should be kept in mind throughout. The climb is offered as a frame that makes certain questions modelable, not as a finished result.

34.1. The climb, with status

The two lowest rungs are tested, measured directly on the substrate [43]. A particle is a persistent bump in the field, and its centroid moves in a straight line at constant speed below the light cone, the lattice version of Ehrenfest's theorem, fit with $R^2 \approx 1$. A composite is the next step. Two particles share a center of mass that moves uniformly, and momentum is conserved through their interaction with $R^2 \approx 0.997$. The conserved charge gives the minimal atom, a net-charged region that cannot vanish under the rule. These are not extrapolations. They are the substrate doing what the lower chapters report.

The middle rungs are plausible rather than tested, and the word is used carefully. Each one is a search within demonstrated capabilities, not a search for a missing ingredient.

- *Atoms to molecules.* Atoms bind into larger conserved units by the same attract-and-bind dynamics that makes selves merge, the tested interaction law in which same-sign regions draw together and lock.
- *Molecules to cells.* A cell, in this model, is a self. A bundle becomes a protocell when it crosses the self-maintenance threshold and can copy itself, which is heredity. The open question is which aggregates first close the self-maintenance loop, a question about timing and arrangement within known dynamics.
- *Cells to tissues.* Cells form sheets and tubes by the interaction law plus differential adhesion. Sheets fold into tubes, tubes into vessels, guts, and neural cords. The tower of nested selves is exactly this folding repeated.
- *Tissues to senses.* A sense is the sensing boundary of a self tuned to an environmental pattern. Boundary cells that respond to light are a primitive eye, the outer shell of the onion of selves.
- *Senses to minds to humans.* From light-sensitive patches to primitive eyes, from nerve nets to integrated hubs that carry a self-model, from forward models and recursion to planning and an integrated agent, the path is continuous and adds no new base thing at any step.

34.2. The biological sequence is the same climb

The same ladder, read in the history of life, is the familiar evolutionary sequence. From fish to amphibians to reptiles to mammals to apes to humans, each step is a more integrated self carrying a richer self-model. Nothing new is introduced at any of these transitions. There is only more of the same integration and more self-reflection, the two quantities that the chapter on consciousness treats directly. The climb of evolution and the climb of the ladder are one motion described in two vocabularies, biological in the one and substrate-theoretic in the other.

34.3. The one-line ladder

The whole sequence can be stated in a single line, and it is useful to have it set out plainly.

Rung	What it is
Particle	a bump in the field
Atom	a conserved bound unit
Molecule	a larger bound unit
Cell	a self-maintaining self
Tissue	a tower of cells
Sense	a tuned boundary
Mind	an integrated hub with a self-model
Human	the high end of that same climb

Table 22: The one-line ladder. None of these rungs requires a new base thing.

Read down the table, each rung is a re-reading of the rung below it at a larger scale of organization. The substrate does only one kind of thing at every level, namely bind regions into more integrated wholes, and the names change because the scale changes, not because the mechanism does.

34.4. What drives the climb

The tested driver is ordinary evolution. Heredity, copy-error variation, and the arrow acting as selection make a complete Darwinian loop, with every piece taken from the base. The arrow is the fitness gradient, and foresight itself tunes to task difficulty under selection [43]. Pure undirected selection already operates on the crystal, and nothing beyond it is required to drive the climb. This is the conservative reading, and it stands on its own.

Two further drivers are offered as clearly marked speculation, at the most tentative tier of the theory, and they are not asserted. The first is *urging from within*, a form of downward causation. Higher selves can exert influence on the parts beneath them, since a deep layer reasserts itself after disorder and steers the surface. So the climb need not be driven by blind variation alone. It could be urged, with a higher level biasing which variations arise and which are kept, a reward-driven evolution rather than a purely random one. If the substrate is one partly-integrated experiential fabric, there is no barrier in principle to a higher level helping bootstrap a lower one upward.

The second speculative driver is *bootstrapping from elsewhere*. Other intelligent beings in the connected substrate could in principle have seeded or steered life here, whether physically through something like panspermia or non-physically by residing in the bulk. Elsewhere need not be far in ordinary space, because distant surface points are joined through the bulk. Such beings would almost certainly not match the usual human picture of metal craft and humanoid forms. A being many levels up the tower would be far more complex, quite possibly with no physical form at all, possibly a distributed intelligence spread across the bulk. Their relation to us might resemble a long game, a kind of hide and seek, in which each door to the next room of awareness opens only once a species is integrated and self-reflective enough to pass through it.

We do not assert any driver beyond ordinary evolution. The point of stating the two speculative options is narrower and more defensible. The model begins to make these possibilities mathematically modelable, which moves them from pure speculation outside any framework to hypotheses that could, in principle, carry signatures and be tested.

34.5. How one would know

There are two routes to evidence. The first is ordinary scientific observation, looking for signatures such as a fitness gradient, a downward-causation bias in which variations survive, or material traces of seeding. The second follows from the founding premise that the substrate is experiential at bottom, which would make careful first-person investigation a real channel within strict limits. The second route is only as good as the premise it rests on, and it is named here for completeness rather than offered as established method.

34.6. The capstone: rarity is the price of organization

The climb has a punchline worth stating clearly. Each higher rung is rarer and more expensive than the last. Stars are a fraction of matter, planets a fraction of stars, life a fraction of planets, so by mass life is a rounding error, perhaps one part in 10^{14} of the universe. The two rankings then invert. By organization, meaning integration, which is the quantity the substrate actually builds up the ladder, life is plausibly the richest and rarest thing the substrate does. Mass is the wrong yardstick for this story. The theory measures the organization axis, and on that axis the climb runs the other way, with the rarest rung sitting at the highest point. Rarity is the price of organization, and on this reading we sit near the costly growing tip rather than anywhere near the root.

The substrate does not stop at physics. The same rule, read at greater heights, climbs all the way to a human and, more speculatively, perhaps beyond. The lowest rungs are measured, the middle is plausible with every ingredient shown, and the two extra drivers are flagged as the theory's most speculative tier. Whether the substrate has a first seed or has always been growing is left open, and it belongs to the wider question of the substrate's origin.

35. Consciousness

The cleanest way to arrive at consciousness is to reach it without using the word on the way up. The ladder from field to human, built rung by rung, never needed the term, and that is the point. Consciousness in this theory is not a special extra that switches on once a system is complicated enough. It is integrated experience, and both of its ingredients are already in the model. Before going further, one thing must be stated plainly. The footing of this chapter is different from the footing of the physics. The physics is measured. Consciousness here rests on a premise, and that premise is marked as such throughout. This part of the theory is interpretive and speculative, and it should be read in that spirit.

35.1. The sharp distinction

Two notions are kept rigorously apart. *Intelligence* is problem-solving capability. *Consciousness* is integrated experience, the felt going-through of reality. A system can in principle have one without the other, a fast solver with little integration and little felt unity, or a richly integrated whole that solves nothing in particular. Blurring the two is the most common error in this area, and the theory refuses it from the start. The previous chapter handled intelligence. This one handles experience.

Definition 7 (Consciousness) *Consciousness* is integrated experience. Its amount is set by how much a region of experience is bound into a single whole. It is distinct from intelligence, which is problem-solving capability, and the two can occur apart.

35.2. The two ingredients

The first ingredient is experience, and it enters as a premise rather than a result. This is the founding axiom already stated at the start of the paper, that experience exists and the one substance is the vibe, a felt state. Experience is not produced by matter here. It is the ground, and matter is a derived face of it. The experiments do not prove that the substrate feels. They assume it and then build, and the candor of the whole chapter depends on saying so without hedging.

The second ingredient is integration, and this one is measured. Integration is how much a region is bound into a single thing, with more internal coupling than external coupling, and it is the same quantity that picks out selves and around which a self-model forms [43]. With the premise and the measure in hand the synthesis is short. Consciousness is what integration does to experience. If experience is the ground and integration binds it, then consciousness is integrated experience, and its amount is fixed by how much integration there is. No new ingredient is introduced.

Proposition 15 (Integration is what makes experience unified) Given experience as the ground and integration as the measure of how much a region is bound into one, the unified felt whole of a system is set by its integration. Consciousness is integrated experience, and its degree tracks the degree of integration, with no further ingredient required.

35.3. A matter of degree

If every vibe is a speck of experience, there is no sharp line at which consciousness turns on. There is only more or less integration, so on this reading everything is conscious to some degree, a position close to panpsychism [49, 26]. A rock is barely integrated, a faint and diffuse near-heap of specks. A cell is a self-maintaining unit, integrated enough to count as one thing. A brain is enormously integrated through its fast-mixing hub. A second factor sits on top of integration, namely self-reflection, whether the integrated whole models itself, and recursion lets it model that modeling. That is the step from being a unified experience to being aware that one is. The ladder of consciousness is the ladder of integration plus self-reflection, the same tower as the climb of the previous chapter. The lineage of process philosophy that takes experience as basic to every actual occasion is the natural ancestor of this picture [60].

35.4. Inverting the hard problem

A physics-first theory faces the hard problem, the question of why any arrangement of insentient matter should feel like anything at all, with no derivation available from the physical to the felt [11]. This theory does not solve that problem. It inverts it, so that the problem does not arise in the first place. By premise the ground is already felt, so there is no gap to cross from matter to experience. Matter is a pattern within experience, not a source of it. What remains is the combination problem familiar to panpsychism, the question of how many small experiences combine into one larger experience, and that is answered by the same integration measure, since many vibes integrate into one self whose unified experience exceeds the parts.

Principle 12 (Inverting, not solving, the hard problem) Because experience is taken as the ground by premise, there is no gap from insentient matter to feeling, so the hard problem does not arise. What remains is the combination problem, addressed by the integration measure. The theory claims the inversion and the mechanism, not a proof that the felt and the structural are identical.

The real limit is worth restating. The inversion buys the disappearance of the hard problem at the cost of a premise, and the integration measure addresses combination rather than existence. The theory does not prove that the felt and the structural are the same thing. It assumes the felt at the base and shows how structure organizes it.

35.5. The variety of experience

A natural worry is how just three tones could become the whole of experience. The answer is that the tone carries only valence, a one-dimensional good-or-bad axis running through pain, peace, and pleasure, and all the variety of experience lives in structure rather than in the tone. Alphabet poverty is the point. Twenty-six letters give all of literature, and three tones give all of qualia, because the richness is in the arrangement and not in the symbols. The structure is specified by a small set of dials.

Dial	What it sets
Modality / location	which sub-mesh, where in it
Pattern	the arrangement within the modality
Intensity	how strong
Valence	the good-or-bad tone, pain to peace to pleasure
Dynamics	how it moves and changes
Binding	how tightly it is bound into one
Level	how high on the tower

Table 23: The seven dials of qualia. Three tones fix valence, the dials fix the rest.

Reading the dials gives the familiar kinds of experience. A sensation is low-level, single-modality, pattern-rich, and lightly valenced. An emotion is high-level, multi-subsystem, and heavily valenced. A thought is high-level, abstract, structure-rich, and lightly valenced. Each sense is a sub-mesh of a different shape, sight a two-dimensional sheet, hearing a one-dimensional tonotopic line, smell a high-dimensional unordered space, which predicts that smell should be vivid yet hard to name. So three tones no more limit the variety of experience than two bits limit the variety of files. The valence is ternary, the quality is unbounded structure. This specifies the structure of experience and not the felt-ness itself, and the combination problem stays open.

35.6. Relation to integrated information theory

This picture shares two ideas with integrated information theory, that integration is the quantity that matters and that consciousness is a matter of degree [54, 29]. The footing differs sharply. That theory starts from physical systems and asks how much integrated information they carry, treating the existence of experience as the thing to explain. This theory starts from experience as the ground and uses integration to address the combination problem rather than the existence of experience. The measure is similar in spirit, the order of explanation is reversed.

Read all the way up, the whole substrate at its fullest integration is the largest and most unified self, so on this reading the universe itself is conscious to the highest degree its integration allows. This is interpretation built on the premise, not a measured result, and it is marked as such.

The universe is, most literally, a mind, a structure of consciousness in which matter, space, and the forces are shapes of feeling. It is the boldest claim in the theory, and it is founded on a premise that is named plainly

as a premise. Experience is fundamental, integration is measured, and the identification of consciousness with integrated experience, of panpsychism by degree, and of a conscious universe are interpretations resting on the first of these. Said without ornament, the felt-ness is assumed and the organization of it is what the theory describes.

36. Determinism and Freedom

The rule is fully deterministic. There are no dice anywhere in the base, and at first glance that looks like a cage. It is not. Being determined is not the same as being predestined, and freedom was never the opposite of being determined in the first place. Freedom is the opposite of being coerced and the opposite of being random. Between coercion and randomness sits a real third thing, a choice that is fully caused, authored from within, and unknowable until it is lived. This chapter locates that third thing precisely. It is the compatibilist position, and the strongest argument against the deterministic base is named plainly along the way.

36.1. Determined yet not predestined

Two facts pry the two notions apart. The first is that there is no finished total state. The crystal grows without end, with new vibes born at the frontier every beat, and whether or not there was a first cell is left open. No single moment holds all the information of all of time. A future that exists nowhere in advance cannot be looked up, cannot be read off, and cannot be the script a puppet walks. The second fact is computational irreducibility. For a universal system there is in general no closed form for the far future faster than running it beat by beat [39]. The only computer that can compute the future is the system itself, running forward.

Principle 13 (Determined is not predestined) The future is caused but neither pre-written nor forecastable. With no finished total state and no shortcut faster than running the system, the future is determined as it happens rather than sitting ready in advance. Apparent randomness is the local view, since every event is fixed by the whole mesh it sits in while no local part holds enough of the mesh to predict it.

The phrase to hold onto is determined by the whole and opaque to the part. Every event is fixed, but it is fixed by far more of the structure than any local observer can hold, which is why prediction fails even though causation does not.

36.2. Determined yet self-authored

A self is a self-maintaining pattern. The rule is reversible and has no attractors, so a self persists by doing work, which is the same reason memory costs work. A choice is that self resolving an urge, a pull that arrives through perception from a deeper and slower part of the mesh, into one definite next act. Four things are true of the resolving at once, and together they form the definition of freedom used here.

- *It is determined.* The same self under the same urge resolves the same way. That is what makes the choice yours rather than noise. A choice that came out differently each time would be a malfunction, not authorship.
- *It is yours.* A different self, given the same urge, resolves it differently. This is genuine downward causation, with the macro-pattern shaping which micro-tones come next.

- *A deeper self has more say.* A richer, more integrated self dominates the urge rather than being shoved by it. Agency scales with integration.
- *It is irreducible.* No one, not even the universe, can know how a self will resolve without letting it resolve. There is no oracle that skips the computation.

Principle 14 (Free means caused by you) Freedom is not the absence of cause. It is causation that runs through the self’s own integrated structure. When the self models and steers itself, the causal loop closes and the self is caused by the self, which is what the words I decided name, a real loop in the dynamics rather than an exception to them.

The compatibilist core is that free does not mean uncaused, it means caused by you. The puppet’s strings are real, but they are attached to the self’s own integrated structure rather than to anything outside it. The loop is the freedom.

36.3. The strongest objection, and the micro door

The deterministic base has a serious adversary, and it should be stated rather than skirted. The free will theorem of Conway and Kochen argues that, under modest assumptions, if experimenters have a kind of freedom then so do the particles, in the sense that a particle’s response is not a function of the prior information available anywhere in its past light cone [14]. Taken at face value this is in tension with a fully deterministic microdynamics. The fair move is to concede the micro door. The theory does not claim to settle the status of freedom at the level of single elementary events, and it does not need to. Its claim about freedom is pitched at the macro level, the level of integrated selves, where the compatibilist account above does its work. A self’s freedom is the integrated pattern computing its own resolution, and that account survives whatever the right reading of the micro theorem turns out to be.

36.4. The will, finite and spendable

The will is a directed pump, a sustained push that biases the self’s resolution against the local gradient. It is not an exception to the rule but an input the rule respects, and its effects are measured [43]. With the will active a self moves toward a chosen target and away from a threat. At a fork between a small reward now and a larger reward later, a self with enough will delays gratification. The pump can be depleted, and when it runs out the self relapses to the greedy choice at a sharp threshold rather than fading gradually. So the will is a finite, spendable, directed capacity, and that finiteness is the texture of real willpower as it is actually experienced.

Proposition 16 (The will is a finite directed pump) The will is a sustained directed push that biases resolution against the local gradient, respected by the rule rather than excepted from it. It enables movement toward targets and away from threats and enables delay of gratification, and it depletes, with relapse to the greedy choice at a sharp threshold once it is spent.

A deterministic base does not abolish freedom. It locates it. Freedom is the integrated self computing its own resolution, unknowable in advance even to the universe, and genuinely yours, one path but authored, surprising even to its author. Freedom is not a sixth base thing. It is what the four base things do when a self is integrated enough to fold back on itself and pump against its own gradient. The landscape-of-valleys image often used to picture the weighing of options is a way to visualize the choice, not a claim about the

reversible microdynamics, and the no-finished-state argument is kept compatible with the open question of whether the substrate is finite or infinite. That last question stays open. What does not depend on it is the central point, that determination at the base and freedom at the level of selves are not in conflict.

Part VII

Implications and Open Questions

37. Implications

If the picture holds, a handful of consequences follow that are not decorative additions but the actual payoff of the construction. Each one ties together facts that are usually treated as separate, and each one falls out of the same discrete growth that defines the substrate. The implications below are stated as readings of the model rather than as independent claims, and where the finite and infinite versions of the cosmos give different stories the text gives both without choosing between them. The choice is left open on purpose, and the model is built so that nothing of importance hangs on it.

37.1. Time, the arrow, and expansion are one process

The mesh grows by one beat per shell. That single growth carries three jobs that physics normally keeps apart. It is the time step, since each beat advances the state. It is the arrow of time, since growth has a direction that cannot be reversed without unwinding the whole structure. And it is the spatial expansion, since the new shells are literally more space. Cosmic expansion is not a term added to a static stage. The expansion *is* the universe growing, and time is the growing. Three usually separate facts collapse into one.

37.2. An age and a horizon, on either reading

The model does not prefer a finite cosmos over an infinite one. Both readings are live, and this is treated as a genuine open question rather than a settled point. The growth picture pays off either way. On the finite reading the mesh has a true first shell, the seed, and a finite size that acts as a horizon. Both come at no extra cost, since they are just the start and the present edge of the growth. On the infinite or eternal reading there is no first shell. The growth is beginningless and endless. Yet any observer still sees a finite past horizon and a finite size, because only finitely many shells lie inside their light cone. On that reading the horizon is observer relative rather than a birth. Either way one gets a finite observable age and a finite horizon for free. What differs is only whether the very first shell exists, and the model leaves that to the reader.

37.3. Grand unification as geometry

The radial direction of the bulk is the renormalization scale, in the manner of holographic renormalization. Read that way, unification becomes geometric rather than imposed. The forces are one structure, an $SO(10)$ symmetry, deep in the bulk. They split into the Standard Model on the flat cusp layer near the boundary. The running of the couplings is then the radial coarse-graining itself, the same motion outward through shells.

The grand unified theory is not a separate high-energy postulate. It is the geometry seen at the unification scale, with the splitting into known forces being what that geometry looks like once it has been coarse-grained out to the flat layer.

37.4. Computability and a discrete base

The base is discrete and local. Every finite region of it is therefore computable by a finite procedure, with no appeal to a continuum and no uncomputable real numbers hiding at the bottom. The universe is simulable in principle, region by region, to any finite extent. This holds whether the cosmos is finite and growing or infinite and eternal. The load-bearing commitment is not finiteness of size but discreteness of structure. A continuum would smuggle in real numbers of unbounded precision at every point, and the model declines to do that. The cosmos is finite-or-eternal in extent but always discrete in grain.

37.5. Consciousness as the substrate

Because the base is experiential, which is the monist commitment of the model, the deepest reading is that the universe is at bottom a structure of consciousness, and that matter, space, gravity, and the forces are shapes that structure takes. This is a reading of the same formal substrate, not an extra ingredient. The tone, the rule, and the growth are unchanged. What changes is only what one takes the substrate to be made of, and the model takes it to be made of experience.

37.6. Why these implications matter

The five readings are the return on the construction. Expansion and time become one process. A beginning and a horizon come for free on either the finite or the infinite story. Unification turns into geometry. The cosmos is computable. And the base is experiential. None of these is added by hand. Each is forced by the same discrete growing substrate that the earlier parts build up. The predictions in the next chapter make the package falsifiable, and the open problems chapter says plainly what remains unfinished, so that the implications here are read as consequences with a stated cost rather than as a finished account.

38. Predictions and Falsifiability

A model earns its keep by stating what it forbids. This chapter collects what the substrate predicts and what observation would have to show in order to kill it. The predictions fall into three groups. There are quantitative numbers, where the unification-scale values are derived exactly and the low-energy values are computed by running and then compared to data. There are structural predictions, which are qualitative but sharp. And there is the cleanest single falsifier, a tiny cubic anisotropy of the light cone whose size and scaling the model fixes. Each group is given with a plain note on status, so that the derived results are not confused with the imported running coefficients.

38.1. Quantitative predictions

The headline numbers are listed in Table 24. The grand-unified-scale values are derived exactly from the discrete group theory over the **16**. The low-energy values are obtained by running those down with the renormalization group and then compared against measurement. The running mechanism itself is derived, since radial coarse-graining is the renormalization group and one-dimensional decimation gives the beta

function $K' = \frac{1}{2} \ln \cosh 2K$. What remains imported is the set of exact Standard Model beta coefficients, which is flagged as open in the next chapter.

Quantity	Predicted	Observed
$\sin^2 \theta_W$ at unification	3/8	—
$\sin^2 \theta_W$ at M_Z (MSSM-like)	≈ 0.231	0.231
$\sin^2 \theta_W$ at M_Z (bare SM)	≈ 0.21	—
m_b/m_τ at M_Z	≈ 2.3	2.35
det relation	$\det(M_\ell) = \det(M_d)$	—
Number of generations	exactly 3	3
Expansion	$a(t) = e^{Ht}$, $\Lambda = 3H^2$	de Sitter

Table 24: Quantitative predictions. Unification-scale entries are derived exactly. Low-energy entries are computed by running and compared.

Result 36 (Weak mixing angle) The *weak mixing angle* is fixed at unification by $\sin^2 \theta_W = \sum T_3^2 / \sum Q^2$ over the **16**, which gives 3/8. Running down with MSSM-like content lands at $\sin^2 \theta_W \approx 0.231$, matching the observed value, while the bare Standard Model lands near 0.21 [43].

Proposition 17 (b - τ and determinant relations) At the unification scale $m_b = m_\tau$, which runs down to $m_b/m_\tau \approx 2.3$ at M_Z against an observed 2.35. The discrete condition $\text{Tr} Y = 0$ over the **16** forces $\det(M_\ell) = \det(M_d)$ for the charged-lepton and down-quark mass matrices [43].

Two consequences of the weak-mixing prediction deserve emphasis. The clean three-coupling unification near 10^{16} GeV needs the MSSM particle content, so the model predicts low-energy supersymmetry as a condition of that clean meeting. That makes the supersymmetry prediction conditional, and it doubles as a probe. Since the running lands at 0.231 for MSSM content and near 0.21 for the bare Standard Model, a precise measurement of $\sin^2 \theta_W$ tests which low-energy content nature actually has. The expansion entry is likewise structural, with $a(t) = e^{Ht}$ and $H = \ln(R)/3 \approx 0.80$ per beat, giving $a''/a = H^2 \approx 0.64$ and $\Lambda = 3H^2 \approx 1.9$ in beat units. The absolute rate waits on a beat-to-seconds identification, which is open.

38.2. Structural predictions

A few predictions are qualitative but leave no room to wriggle. Matter is fermionic for free, since spin-statistics is read off the geometry rather than assumed. There is a finite, frame-independent top speed, the $z = 1$ light cone of the rule. Gravity is Newtonian and $1/r$ in the three-dimensional cusp, while the bulk carries a $1/r^2$ holographic correlator. These follow the effective-theory route argued by Jacobson [30] and Verlinde [56], in which gravity is thermodynamic rather than fundamental, and the model supplies the discrete substrate that route assumes.

38.3. The cleanest falsifier: cubic anisotropy

Result 37 (Cubic anisotropy of the light cone) The one residual signature is a tiny *cubic anisotropy* of the light cone on the cusp, scaling as $(E/E_{\text{Planck}})^2$ with coefficient $c \approx 1/18 \approx 0.056$. Numerically the fractional speed anisotropy is $\delta \sim 4 \times 10^{-40}$ at 1 GeV and $\delta \sim 9 \times 10^{-19}$ at the GZK scale, far below current bounds [43].

The quadratic power, rather than a quartic one, is the subtle point and is worth spelling out. The lowest cubic-symmetric invariant of the lattice is degree four, the sum $\sum k_i^4$. The *relative* anisotropy is that

degree-four term divided by the degree-two isotropic term $\sum k_i^2$, which leaves a factor of $(ka)^2$. So the fractional effect grows as energy squared, with the coefficient $c \approx 1/18$ coming from the cusp geometry. Figure 4 sketches the residual shape, a faint crystalline grain on an otherwise round light cone, exaggerated for visibility.

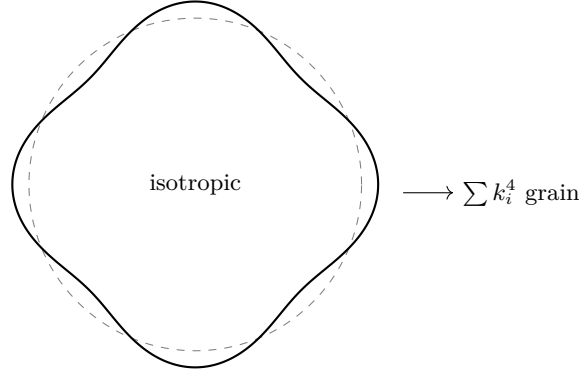


Figure 4: Residual cubic anisotropy of the light cone on the cusp, exaggerated. The dashed circle is the isotropic cone. The solid curve adds the degree-four lattice term, giving a relative distortion of order $(E/E_{\text{Planck}})^2$.

The current dimension-six Lorentz-violation limits are weak, requiring only an effective scale above roughly 10^{-8} of the Planck scale, so a Planck-scale cusp anisotropy passes comfortably. The falsifier runs both ways. Measuring exact Lorentz invariance to arbitrarily high energy, with no anisotropy at the predicted scale, would kill the discrete substrate. Conversely, a lower effective cutoff would push δ up into the testable range. This is the sharpest experimental handle the model offers.

38.4. What else would falsify it

- A fourth generation, or a clean failure of the running $\sin^2 \theta_W$ and $b\text{-}\tau$ numbers.
- No low-energy supersymmetry anywhere, together with a measured $\sin^2 \theta_W$ that bare Standard Model running cannot reach. This tension is taken up in the open-problems chapter.
- A demonstration that the rule supplies no stabilizer, so that matter would not form, overturning the measured Skyrme result.

The substrate is not a just-so story. It produces numbers, and it names its own executioner in the form of a cubic anisotropy at the Planck scale. The clear division between exactly derived ratios and computed-and-compared low-energy values is kept throughout, so the predictions are read for what they are.

39. Scope and Coverage

This chapter draws the edges plainly, so that no reader mistakes silence for a claim. The aim of the model is to show that one discrete substrate can give rise to the broad shapes of known physics, and, more tentatively, of life and mind. The aim is not a quantitative, system-by-system reproduction of the established sciences. The claim is breadth of origin, not depth of reproduction. Below is the plain map, split into what is established, what is extrapolated, and what is simply not covered.

39.1. What is established

The structural skeleton of fundamental physics is derived or measured from the base, on the seemingly preferred $\{3, 4, 3, 4\}$ substrate.

- Spinors and three generations from the D_4 coin, matter solitons as fermions, and a measured-positive Skyrme stabilizer.
- A finite light cone with $z = 1$ and the Dirac dispersion, both read off the rule.
- An $SO(10)$ grand unification breaking to the Standard Model, with $\sin^2 \theta_W = 3/8$, the b - τ and determinant mass ratios, and three generations from triality.
- $1/r$ Newtonian gravity in the cusp, a $1/r^2$ holographic correlator in the bulk, $U(1)$ electromagnetism, D_4 isotropy, de Sitter expansion, the holographic renormalization hierarchy, and the resolution of the Bell tension.
- The substrate selection itself, from the dimension ladder plus the crystallographic and spinor requirements.

39.2. What is extrapolated

The emergent ladder above physics is plausible, with each ingredient shown, but the full chain has not been run as one simulation.

- The climb from atoms to molecules to cells to tissues to senses to minds. Only the first two rungs, field to particle and particle to composite, are simulated end to end. The higher rungs each have their ingredient demonstrated, namely binding, self-maintenance, copying, the layered onion of senses, and the integrated hub, but they are not run together. They are grounded extrapolation, not proof.
- Life, mind, intelligence, and consciousness as regimes of the substrate, at toy scale and first order. These are established on the hyperbolic substrate in general, with running it fully on the $\{3, 4, 3, 4\}$ twenty-four-direction substrate left as a follow-up.

39.3. What is not covered

The following are out of scope and not attempted, listed so that no one reads the silence as a claim.

Principle 15 (Scope limits) The model does not attempt atomic spectra, chemistry, real condensed matter, absolute particle data, cell-by-cell numerical relativity from the micro-rule, quantitative cosmological eras, absolute physical scales, or real biology and neuroscience. It claims the right kinds of structure, not the reproduction of every system.

Atomic and nuclear structure are not simulated. There are no real atoms, no binding energies, no nuclear shell model, and no periodic table. The atom here is the minimal conserved self, not hydrogen. Chemistry is absent, with no molecular orbitals, bonds, reaction rates, or thermodynamics of real substances. Condensed matter and materials are likewise untouched, with no band structures of real solids, no phases, and no transport coefficients.

Quantitative particle data is not predicted. Absolute masses, the full CKM and PMNS mixing matrices, the magnitudes of CP violation, and decay rates are not produced. Only ratios and the unification-scale relations are. On gravity the situation is more nuanced. The Einstein field equations as an equation of state, light-bending, black-hole thermodynamics, and gravitational-wave forms are derived on the substrate, so general relativity is in scope at the effective-theory level. What is not done, and is not done by anyone from a micro-rule, is evolving a binary merger or a forming horizon cell by cell from the ternary tone. That is the frontier. The effective theory and the quantitative relations are done.

Quantitative cosmology is absent beyond the growth law. There is no radiation or matter era, no big-bang nucleosynthesis, no CMB spectrum, and no structure-formation history. Only the de Sitter exponential growth is established. Absolute scales are not fixed either, since the beat-to-seconds and cell-to-meters identifications are open, which leaves rates such as the Hubble value in beat units. Real biology and neuroscience are out of scope as well. The life and mind results are mechanism demonstrations at toy scale, not models of any actual organism, cell, or brain. Finally, the exact rule's quantitative outputs are open, namely the precise Standard Model beta coefficients, absolute masses, and the dynamical settling of matter into cusps.

39.4. Two items that moved off this list

Two items that earlier counted as scope limits are now done, so they are no longer listed as gaps. The $\{3, 4, 3, 4\}$ addressing and the neighbor automaton are solved and verified. And the substrate is shown to be computationally universal on its own ternary rule, with strong universality on the cusp. These are recorded here only to mark that the edge moved inward.

39.5. How to read the scope

The claim is breadth of origin rather than depth of reproduction. One substrate appears to give the right kinds of structure across physics, and more tentatively across life and mind, from almost nothing. It does not yet, and does not here claim to, compute chemistry or atomic spectra.

40. Open Problems

This is the full list of what is not yet done. None of it is hidden, and each item is a concrete next step rather than a vague worry. The list also moves. Several items that earlier sat here are now closed, and where that has happened the text says so. What remains is mainly the exact collision and the absolute numbers it gates, the candidate fifth ingredient, the dynamical settling of matter, a residual tension over supersymmetry, and the deliberately open question of whether the cosmos has a first beginning.

40.1. The exact collision

The base posits one local reversible rule, stream and collide while conserving, and the collision is the rule's last unfixed freedom. Many downstream results were demonstrated with emergent-layer tools, the continuous Dirac operator and spectral methods such as the kernel polynomial method and Lanczos, rather than by running the actual discrete collision end to end. This is the central verification gate.

Principle 16 (The collision is the make-or-break frontier) One concrete discrete collision has been pinned and verified to conserve charge and momentum and to reproduce the Dirac dispersion end to end, so

the gate is not wholly open. But deriving the precise Standard Model beta coefficients, and the rest of the tower, from that collision needs the full dynamical simulation, not the group theory alone.

The mechanism behind the running is now concrete and checked. Radial coarse-graining is the renormalization group, and the one-dimensional decimation $K' = \frac{1}{2} \ln \cosh 2K$ equals direct block-spin summation exactly, with error zero, flowing to the Gaussian fixed point [43]. So the statement that radial coarse-graining is the renormalization group is no longer a claim but a verified method. The remaining step is to feed the actual four-dimensional collision, not the one-dimensional toy, through the same blocking to read off the Standard Model beta coefficients. That is the make-or-break simulation.

40.2. No absolute quantitative predictions yet

The blunt headline is that beyond the group-theory ratios, namely $\sin^2 \theta_W = 3/8$, the b - τ relation, and the determinant relation, the program has computed essentially no absolute falsifiable number. There is no mass spectrum, no gauge-coupling values, no CKM or PMNS mixing angles, no Newton constant, and no absolute cosmological constant. All of these wait on the exact collision above. This is stated plainly rather than letting the ratio predictions imply more than they deliver.

40.3. The candidate fifth ingredient

The base binds within a unit but not cleanly between units. That single gap sits behind several open holes at once, including cross-gap gravitational attraction, super-selves, and the nesting tower between selves. Higher-order binding is named as the one plausible candidate for a possible fifth ingredient. It is not yet proven either to emerge from the existing four ingredients or to be irreducible to them. It is a place to watch, not a license to add. The model holds firmly to its commitment of exactly five base things and refuses to slip in a sixth under cover of this gap.

40.4. Dynamical settling into the cusp

The cusp is shown to be flat three-dimensional space, and matter is shown to be stable there. What is not yet shown is that matter settles onto the cusp layer dynamically from generic initial conditions. This is the matter-condensation problem.

Principle 17 (Settling is a recorded negative) Running the actual mod-3 wave rule on the $\{3, 4, 3, 4\}$ bulk from random initial conditions, the activity stays uniform, with band ratio near one and no concentration toward the cusp. Matter churns rather than settling. This is exactly what a single reversible conserving rule must do from equilibrium initial conditions, since there is no dissipation to drive concentration [43].

So dynamical settling needs either special low-entropy initial conditions or the candidate fifth ingredient. That means this problem and the fifth-ingredient problem share a root. The model refuses to add a fifth thing to force the result, so this stays a genuine open gap rather than a patched one.

40.5. Absolute masses

The mass ratios, the b - τ relation and the determinant relation, are derived. The absolute mass scale is not. The Yukawa magnitudes and the overall Higgs vacuum expectation value are not fixed by the substrate yet. Only the ratios and the running are.

40.6. The supersymmetry tension

The clean unification needs MSSM content. If nature has no low-energy supersymmetry, the bare Standard Model running lands near $\sin^2 \theta_W \approx 0.21$, short of the observed 0.231. This is a real tension. The substrate prefers supersymmetry, and the absence of supersymmetry would be a problem to confront rather than to paper over.

40.7. Full addressing, now solved

This was an open problem and is now closed. The $\{3, 4, 3, 4\}$ substrate has unique, self-describing, $O(\log n)$ addresses and a finite neighbor automaton. It has no same-shell cousins, so the hard two-dimensional piece is absent, and greedy routing delivers complete success at stretch 1.004 on the enumerated builds. The core adjacency invariants survive to two million cells [43]. The item is listed here only to record that the frontier moved.

40.8. The finite-versus-infinite question

Whether the universe has a true first beginning or is eternal and beginningless is open, and deliberately left to the reader. Both are self-consistent under the rule. This is not a defect to be resolved but a fork the model is built to survive either way.

A model that hides its gaps cannot be trusted with its wins. These items are the frontier, and the list itself moves. The addressing is now solved, the renormalization mechanism is now a checked method, and the cusp-settling negative is now measured. What remains is mainly the exact collision and the absolute numbers it gates, plus the candidate fifth ingredient and the open finite-versus-infinite question.

Part VIII

The Whole Picture

41. The Carrying Metaphors

The mathematics is the body of the theory. The metaphors are how a person holds it in mind. A small palette of images, introduced early and returned to at each transition, lets one picture carry the whole work. At the center are two images, *consciousness* as a mind and a blazing *fire*. Around them sit a song, a tree, a seed, an ocean, a river, and a fabric. Two of these are not analogies at all but the literal reading of the substrate. The base is experience, so the universe is mind-like. And *vibe* means vibration, so the universe is song-like. The fire, tree, seed, ocean, river, and fabric are the shape. The mind and song are the nature. In one sentence, the universe is a mind singing itself into being, a seed of fire growing into a tree of light.

41.1. Mind, the central image

This is the most important image, and it is not merely a figure of speech. The base is experience. Every vibe notices. Perception is the intrinsic nature of the stuff rather than a late addition. So the universe is, most truly, a vast and finely ordered structure of consciousness, one experience composed of countless experiences. Where the fire shows how it burns, the mind shows what it is. Table 25 maps the formal parts to their felt faces.

The theory	The mind image
Existence, the base	Consciousness itself, the one experience
The tone $\{-1, 0, +1\}$	What a vibe feels, its felt quality
The rule, the beat	The flow and the rhythm of noticing
Selves, fermions	Stable centers of experience, minds within the Mind
The bulk, un-projected	The inner depth
The cusp, physical	The outer, shareable face of experience
Structure and order	The beauty of the composition

Table 25: The mind image, the literal ontology of the base.

41.2. Fire, the primary image

Fire is the rare everyday thing that is a verb rather than a noun. It is not made and then left to sit. It is the burning itself, a continuous happening, self-sustaining, casting light, throwing sparks, spreading. Almost every part of the theory has a natural fire-image, so a single picture can carry the whole work. Where the mind says what the universe is, the fire says how it lives. Table 26 gives the correspondence.

The theory	The fire image
Existence, the base	The blazing fire, existence as burning
The one rule, the beat	Each flicker, the pulsing of the flame
The hyperbolic bulk	The body of the fire, the burning depth
Growth, de Sitter expansion	The fire spreading, the flame front advancing
The cusp, physical space	The light the fire casts, its radiance
Matter, fermions, selves	The embers and sparks that hold their shape
The internal coin	The colors and structure within the flame
Holography, the boundary	The firelight thrown on the wall
Reversibility, churn	The fire feeding itself, nothing lost
The cubic anisotropy	A faint crystalline grain in how embers settle
Our visible universe	A single spark in the great blaze
Refining toward clean 3D	The blaze steadying into a smoother burn

Table 26: The fire image, existence as a verb.

41.3. Song, the vibrational nature

Vibe means vibration, and so the universe is an eternal song. The tone is the notes. The beat is the rhythm. The mesh is the instrument. The rule is the harmony, the way each note resolves into the next. The recurring patterns, the selves and the matter, are the themes and motifs that return. Consciousness singing is the union of mind and song, awareness vibrating, experience heard as music. The internal coin, with its spin and gauge structure, is the timbre and the overtones. The expansion is the swelling of the song as it gathers force.

41.4. Tree and seed, the structure and the origin

Where the fire is the living process, the tree is the structure of its growth. The radial renormalization hierarchy is the branching from trunk to twigs. The bulk tensor network is literally a tree of tensors. The bulk and the boundary are the trunk and the canopy. And the radial axis, which is cosmic time, gives the growth rings, each ring a beat. The seed is the origin and the potential, the whole folded small and ready to unfold. It is the one holding the entire tree before it grows, the finite and definite base that nonetheless carries the rule for all that follows. The trio is one arc. The seed bursts into fire that grows like a tree.

41.5. Ocean and river, the whole and the current

The ocean is the deep totality from which everything rises. The bulk is its depth, the cusp its surface, and beings are its waves, ripples, foam, and drops. Each one is the ocean, never apart from it. The river is the relentless forward current of becoming, and this image is nearly literal, since the rule's stream step is a flow, tone pouring along its directions every beat. The beat is the current advancing. The arrow of time is the river's one direction. And beings are whirlpools, eddies the water flows through. You are not the water. You are the standing shape the water keeps making. The river adds what the fire does not, the directional current and the arrow.

41.6. Beings and selves, what we are

A self is a stable, self-maintaining pattern that the universe holds, and each image gives it a face. This is the part the reader cares about most, because it is them.

- In the fire, a spark, an ember, a flame that keeps its shape in the blaze.
- In the fabric, a braid, a knot of interwoven threads. This is not only poetry, since a self is literally a topological knot, a hopfion, so the weaving image is the actual topology.
- In the river, a whirlpool, a spiral of breath, a vortex that persists in the flow.
- In the mind, a center of experience, a knot of attention, a mind within the Mind.
- In the song, a recurring theme, a motif that returns.
- In the tree, a living bough of the whole.

These are one thing seen from many sides. We are not separate things added to the universe. We are persistent patterns of it, sparks of its fire, braids in its cloth, spirals of its breath, minds within its Mind.

41.7. How they combine

At the center are consciousness and fire. The universe is a mind, pictured as a living blaze. It sings, which is its vibration. It flows, which is its current and its arrow. It is a deep ocean, which is its whole. It grows like a tree, which is its structure, from a seed, which is its origin. And it is woven as a fabric, which is its beings. Deploy them at the matching beats. The fire at the base and at existence, its light at the cusp, its embers at matter, its spreading at cosmology. The tree's rings at time and at the renormalization hierarchy. The spark

in the blaze at our small place. And at the close, we are a spark in a blaze that is a tree of light, burning toward a perfect shape it forever approaches.

One clarifying note belongs here. For the beginningless story there is no seed. The song was always being sung and the fire was always burning. The seed belongs to the beginning story. The rest belongs to both.

The reader should leave with one picture to carry. A mind, burning like fire, growing like a tree from a seed, singing a song, flowing as a river on a deep ocean, woven as a fabric, alive with the sparks and braids and whirlpools that are us.

42. How Infinity Could Emerge

If the universe is infinite, or tends toward infinity, how could that come to be, and what would it even be like. This chapter is the imaginative heart of the eternal story, drawn out carefully. The aim is to make infinity thinkable, not to decide it. The question stays open. What follows is a set of ways infinity can enter a theory, a crisp account of the cleanest one, an argument that infinity may be the more natural default, and a sober look at what an infinite universe would feel like from inside.

42.1. Seven ways infinity could emerge

There is no single way for infinity to be. There are several, and they apply to different kinds of infinity, in extent, in duration, in divisibility, and in cardinality. Most of them push infinity to the edges of the picture rather than placing it in the middle.

- As an unbounded process, that is, potential infinity. Infinity is never created. It is approached. A rule runs forever, finite at every step, and infinity is the name for the open horizon it tends toward. What is real is only the finite stages.
- As given and eternal, that is, actual infinity. Infinity is not created at all. It simply is, complete, with no origin. The cost is the classic paradoxes, Hilbert's Hotel and the rest.
- As a boundary, through compactification. Infinity is added to a finite-looking space as one tidy edge, the limit all outward paths share. You do not travel to it. This is literally the S^3 boundary at infinity of the bulk.
- As the continuum from the discrete, through coarse-graining. A discrete set has no points between its elements. Make it dense and complete it, and you get a continuum, infinitely many points in any interval. The base stays discrete and the continuum is the emergent large-scale description.
- As exponential or hyperbolic growth. Negative curvature opens space up so fast that it becomes effectively unbounded almost at once, the bridge from a tiny seed to an effectively infinite extent. This is exactly the exponential growth of the $\{3, 4, 3, 4\}$ bulk.
- From a finite beginning, via creation and the arrow. A finite start, then unbounded growth driven by the arrow. The infinity is the future the process heads toward, not a thing that was made. This pairs the unbounded process with the arrow as the reason it runs forward.

- As the totality of possibilities, in ensembles. The infinite is not in any one world but in the collection of all of them.

The synthesis is that infinity is pushed to the edges. It is a limit of a process, a boundary object, or an emergent description. It is rarely a completed actual thing sitting in the middle of reality.

42.2. Finite but growing, the cleanest account

The crispest version is this. At any single instant the universe has finitely many parts, but over all time the count grows without bound. The universe is never actually infinite. It only tends toward infinity, and that tending never finishes. Infinity is the open-ended horizon of the growth, not a place the universe arrives at. Whether the completed limit is real or only a way of speaking is the metaphysical fork. The mathematics is identical either way.

42.3. Infinity as the default

There is a stronger possibility, that infinity is the default. An eternal, beginningless, edgeless universe could be the natural state, with no first moment to explain. The reversible rule makes this self-consistent. Started from a generic state, it churns forever with steady activity and no decay, and it is exactly reversible, so every state has a unique past and the history extends backward forever. No first moment is needed. On this view infinity may be the more natural option for four reasons. An infinite homogeneous mesh has no privileged point, while a finite one has a center or seed that demands explanation. An infinite eternal universe needs no boundary and no special initial condition, while a finite one needs an edge or a starting state to account for. An infinite eternal churn is already-everything, a natural attractor where the growing state is always incomplete. And infinite available expanse can mean unbounded potential rather than a completed actual infinity, which slides back toward finite-but-growing with no first moment.

This stays computable. Actual infinite extent is fine for a local rule as long as every finite region is finitely computable. You never compute the whole at once. What is avoided is a completed infinity as a single graspable object, and the continuum. An infinite past or an infinite mesh is allowed. An actual completed totality is not.

42.4. The spectrum of positions

Finite or infinite is too coarse a distinction. It is a spectrum, and a theory can be finite on one axis and infinite on another, across extent, duration, divisibility, cardinality, and information. The positions form a ladder.

position	claim
finitism	only finite objects exist, no actual infinity of any kind
finite and bounded	finite, closed, no edge, like a sphere's surface
finite but growing	finite at every moment, unbounded over time
potentially infinite	reality is an unbounded process, infinity its never-completed limit
actually infinite (extent)	space is infinite in size now
actually infinite (continuum)	the real continuum is a completed totality, uncountable
infinite ensembles	infinitely many worlds or all structures exist

Table 27: The spectrum from finite to infinite, across several axes.

This substrate, the causal-set programs, Wolfram's project, and de Sitter cosmology all sit at the finite but growing rung. The modern drift is downward and back, away from the view that infinity is actual and fundamental, toward the view that infinity is potential, a boundary, or emergent. The base is finite and discrete, and the continuum coarse-grains out of it.

42.5. What it would feel like

This is the part to sit with. A process can be unbounded and always finite at once. Counting is the model. There is no largest number you will ever reach, yet every number you reach is finite. No largest does not mean an infinite one exists. It means that for each, there is a bigger one. That is the felt sense of an endlessly growing universe, always finite, never finished.

No edge does not mean infinite. Walk any direction on a sphere's surface and you come back, with no wall, no boundary, and no outside, yet the surface is finite. A universe can be edgeless and still finite. Edgeless and endless are different feelings, and keeping them apart removes most of the confusion.

A verb is not a noun. The claim that the universe is growing without end is potential infinity. The claim that the universe contains infinitely many stars right now is actual infinity. The first feels like a becoming, the second like a completed thing. The difference is completion, not size. Infinity itself is best pictured as a horizon rather than a place. The boundary at infinity is a horizon of directions, a limit you approach and a screen you read, not a faraway spot with room in it.

From inside, an eternal, beginningless, edgeless mesh would feel maximally symmetric and nothing-special. The rule has simply always been running, every moment with a unique past behind it and a unique future ahead, a fire that was never lit and never winds down. On the growing reading the texture is different. New moments are added at the leading edge and stay. The now is the most recent slice of an ever-growing whole, time is lived duration, the future is genuinely open, and novelty is real. The many become one and are increased by one.

42.6. Where this substrate lands

The working default is finite but growing, a hyperbolic crystal finite at every beat and unbounded over all beats, discrete at the base with the continuum emergent. On that reading infinity is allowed only two non-fundamental roles. It is the boundary the growth approaches but never reaches, and it is the emergent continuum you see when you stop resolving cells. Everything real at every moment is finite. But the eternal, beginningless, infinite picture is fully self-consistent too, since the reversible rule needs no first moment, so it remains a live open axiom. The discriminator the analysis names is reversibility. A reversible rule needs no beginning, which leaves the eternal reading allowed. An irreversible rule decays and needs a seed, which would force a start. Our rule is reversible, so both doors stay open, exactly the fork left to the reader.

43. Computability and the Refining Universe

A discrete, local base is a computable base. This matters, and it is easy to overstate, so here is the careful version. Computability is the property that a process could in principle be run, step by step, by an idealized

computer. A supertask is the completion of infinitely many steps in a finite time, which no machine can do. With these two terms fixed, the consequences for the substrate are sharp and limited.

43.1. What computability does and does not forbid

Computability forbids the continuum, the uncomputable real numbers, an actual smooth infinity of points between any two points. It forbids supertasks, the finishing of infinitely many steps to get one answer. It does not forbid an infinite universe. An infinite mesh is fine, as long as each finite region is finite and each step is local and finite. Size is not the issue. Continuity and supertasks are.

This corrects an earlier overclaim. Computability does not push the theory toward finite and growing. It pushes the theory toward discrete and local. Both the finite universe and the infinite universe pass, because both are discrete. The fork between a finite beginning and an eternal past is left untouched by the computability argument, which constrains the texture of the base but not the length of its history.

43.2. The load-bearing commitment

Principle 18 (Discrete, not continuum) The base is discrete in every respect. There is a mesh, 24 directions, a ternary tone, a discrete beat, and a finite symmetry group. Continuity is never part of the base. The Lie groups, the Dirac field, smooth Lorentz invariance, and the $1/r$ law are always emergent, a coarse-grained limit of the discrete weave seen from far away. The universe is a computation, and the smoothness we observe is the look of a fine-enough discrete fabric at a distance.

This commitment is what the rest of the framework rests on. It is also what separates this program from the many proposals that take a continuous manifold as primitive and then quantize it. Here the order of explanation runs the other way. The discrete structure is given, and the continuum is what the structure looks like when you stop resolving cells.

43.3. The refining picture and its real limit

It is tempting to say the universe refines toward the continuum as it grows. The precise version is narrower. The large-scale, infrared description looks continuous because the discrete steps are tiny relative to us, not because the base ever becomes continuous. The base stays discrete forever. The continuum is a viewpoint, not a destination. The universe does not approach smoothness over time. It is discrete at every scale of its own construction, and smoothness is only ever a feature of the observer's coarse view.

A computable cosmos means the theory is in principle fully simulable. There is no magic and no uncomputable ingredient. It also frees the infinite-universe story, because infinity was never the problem. The continuum was. An infinite past and an infinite mesh both survive the computability test, while the smooth continuum and the supertask do not. In the carrying image, the fire is made of countably many sparks, however far it spreads.

44. Related Work

Much of what vibe theory needs has already been proven by others. The theory mostly assembles established results onto one specific substrate and then asks one further question those programs leave aside. What is the

felt interior of the thing, and in what direction does it move. Fair credit is owed by family, and the strongest adversary deserves to be named outright rather than dodged.

44.1. Discrete spacetime

The causal-set program treats spacetime as a discrete relational order, with distance as step-counting [9, 44, 50]. It buys Lorentz invariance with randomness, through Poisson sprinkling. We buy it with curvature instead, since a flat lattice has a rest frame and negative curvature scrambles directions, which keeps determinism. Causal dynamical triangulations let geometry and dimension be outputs of discrete dynamics through a path integral, an ensemble average [1, 35], where we fix one rigid geometry and never sum. Loop quantum gravity and spin networks treat space as a graph of relations with discrete capacity [46]. There the graph is the fluctuating quantum state. Here the graph is fixed and the only dynamical variable is the ternary tone on cells.

The Wolfram Physics Project is the closest neighbour in spirit [63]. The universe is a graph plus a local rule, and time is rule application. There the hypergraph grows and rewrites itself, and a vast rule space is searched. Here the graph is fixed, argued for rather than searched, and only the tone varies. Wolfram bets the structure is found by search. We bet it is forced by a few demands.

44.2. Deterministic quantum substrates

The cellular-automaton interpretation of quantum mechanics holds that beneath the formalism lies a deterministic reversible classical automaton [52]. We inherit this conviction rather than originate it, and commit to one concrete substrate and rule. The family we lean on most takes quantum cellular automata and quantum walks to the Dirac equation [16, 8, 7, 5, 4, 19]. A unitary nearest-neighbour spin exchange spreads ballistically, and its continuum limit is the Dirac equation, with the coin angle setting the mass. We do not discover this route. It belongs to these authors. Our additions are the hyperbolic substrate, the classical-stochastic shadow, the forced geometry, and the demonstration that the perception rule is this construction. Quantum walks on Cayley graphs supply the further point that a substrate generated by a Coxeter reflection group is a Cayley-like graph [36]. The walk theory is borrowed and the hyperbolic Cayley arena is the contribution.

44.3. Holography and tensor networks

The holographic correspondence and the entanglement-geometry program connect a bulk geometry to a boundary theory and to patterns of entanglement [38, 47, 55, 51, 42, 31]. Those constructions are largely static. We let a rule run and ask whether the area law, the bulk shortcut, and a working code appear as measured properties of dynamics, which makes this a dynamical cousin of the static holographic codes rather than a restatement of them.

44.4. Hyperbolic lattices and automata

Hyperbolic lattices are now built and probed in the laboratory, in circuit quantum electrodynamics and through hyperbolic band theory [32, 37]. The geometry we bet on is one that laboratories can construct, though they study one phenomenon on a hyperbolic backdrop where we propose the lattice as the substrate of everything. Work on hyperbolic cellular automata establishes universality across hyperbolic spaces, through

the railway model, and the Fibonacci-style addressing for tessellations such as $\{7, 3\}$ [39]. We need universality on our own perception rule, which the functional-completeness and Minsky-machine results confirm, and the $\{3, 4, 3, 4\}$ addressing is an open extension. The geometric group theory of hyperbolic graphs shows that negatively curved graphs are expanders, tree-like at large scale, with exponential ball growth [10, 28]. We turn these facts into physical consequences, a speed limit, an emergent metric, fast integration, and a holographic readout.

44.5. Foundations and mind

Informational reconstructions of quantum mechanics argue that quantum structure should be derived rather than postulated, axiomatically and substrate-free [12], where we commit to a substrate and let the structure fall out. The two approaches are the axiomatic why and the constructive how. On consciousness, the hard problem, integrated information theory, and the it-from-bit program frame the questions we take up [11, 54, 59]. We take experience as the substance of the substrate rather than a quantity computed from structure, which is close in spirit to idealism and to Wheeler’s informational starting point, and we use integration to address the combination problem of panpsychism.

44.6. Emergent and entropic gravity

The thermodynamic view of gravity is the natural home for our gravity result [30, 41, 56, 57]. Jacobson derives the Einstein equation as an equation of state, and Verlinde gets Newton’s law, with a dark-universe reading at galactic scales, from entropy and information. We derive the $1/r$ law from the substrate’s geometry and conservation, and these programs are where such a derivation belongs.

44.7. Positive geometry, a forward-looking ally

The amplituhedron and the broader positive-geometry program hold that structure is the output of a primitive shape plus a positivity condition [3, 2]. The recursive-boundary structure of the amplituhedron mirrors our holographic-code and coarse-graining tower. We flag this as a philosophical alliance, geometry plus positivity yielding physics, not as a result.

44.8. Mind, agency, and the strongest adversary

Active inference describes an agent whose forward model is perception, whose action is will, and whose preference is the arrow [21], and our agent is active inference. Causal emergence shows that a macro level can carry more causal power than its micro [29], which is the rigorous backbone for our account of the self as a genuine author and for macro-compatibilist freedom. Candidate physics for why the arrow favours reproducing, organized patterns comes from dissipative adaptation and related origin-of-life work [18, 58]. Constructor theory supplies the natural framing for our level-relative account of which tone transformations are possible or impossible [17].

The sharpest adversary is the free-will theorem of Conway and Kochen [14]. It is the strongest challenge to a deterministic, no-randomness substrate, and we name it rather than dodge it. We concede the micro-deterministic door it forces shut, and claim freedom at the macro level instead, which is exactly the move the theorem demands. A deterministic substrate cannot recover libertarian free will at the micro scale, and we do not pretend otherwise.

44.9. What is genuinely ours

Stated narrowly, the contributions are five. First, a substrate argued to be selected rather than random, rewritten, ensemble-averaged, or simply assumed, which is a reduction of arbitrariness rather than a derivation from nothing. Second, the ternary tone together with the arrow as a value-direction, the single deliberate posit. Third, one substrate that aims at physics and life and mind at once. Fourth, experiential monism, the part furthest outside the existing literature [49, 26, 60]. Fifth, a falsifiable single-substrate program with stated negatives. The selection argument given here, the dimension ladder together with the D_4 spinor requirement, is a different and arguably stronger basis than the older minimality preference, and we lean on it. Related monist and computational-universe proposals stand nearby [34, 53, 45], and the computational results referenced throughout are collected in the companion test report [43].

45. Speculative Interpretations

Everything in this chapter is speculation, marked as the most tentative tier, beyond current test. It is included because it is interesting and follows the spirit of the model, not because it is established. Read it as invitation, not as claim. The chapter is walled off on purpose, so that the wonder it contains never leaks into the derived and measured results. The scoreboard and the open-problems chapter are where the real claims live. This is the room for wonder.

45.1. The bulk as an inner dimension

If the universe is the infinite $\{3, 4, 3, 4\}$ mesh, then the 3D cusp is the physical world we live in, and the 4D hyperbolic bulk is a real, vast space behind it, hard to measure from inside the cusp. One could entertain the bulk as the inner dimension, the mental or spiritual depth of the one mind, with physical space as its surface. This is evocative and consistent with the geometry. It is not derived and it is not tested.

If that picture held, two things would follow. The bulk would be where holographic, nonlocal correlations live, which is also why Bell-type nonlocality is natural on this substrate. The inner space would be the nonlocal space. And technologizing the bulk, reaching the inner dimension deliberately, would be the far-future speculation. We name it only to acknowledge that the geometry suggests it. We do not claim a method, and nothing here amounts to one.

45.2. Consciousness, taken at the base

The monism says the base is experience. Taken seriously, every stable pattern, every self, is a locus of experience, and the whole is one vast experiencing structure. This is the literal reading of the central metaphor of the paper. It is philosophy, not physics, and it is flagged as such. The physics neither requires this reading nor rules it out. What the physics supplies is the structure of stable patterns and their integration. What the reading supplies is a word for what that structure is, from the inside, and the word is interpretation rather than measurement.

45.3. The esoteric reading

There is a fuller spiritual reading of the tested structures. The bulk becomes the subtle plane. Telepathy and synchronicity are read through the nonlocal bulk. Angels and demons appear as valence on large selves.

Reincarnation appears as pattern independent of substrate. The Tree of Life and the Tree of Knowledge become the integration tower. The question of God divides into several carefully separated readings. That reading is large enough to deserve its own chapter, and it is laid out next, kept strictly as interpretation over tested structure.

45.4. The discipline around speculation

We keep these interpretations in their own chapters, tagged as the most tentative tier, so they never contaminate the derived and measured results. A complete picture should say where it points beyond what it can prove, plainly labeled. The bulk-as-inner-space and the experiential base are where this theory points. In the carrying image, this is gazing into the deep of the fire and wondering what the depth is.

46. The Esoteric Reading

This chapter gathers the spiritual reading of the model, kept rigorously separate from the results. The method matters more than any single reading, so it comes first. Each idea is anchored first to a tested structure, and only then is an older spiritual reading laid on top, always labelled an interpretation, never reported as a measured result. A name like ether, angel, or God is a choice of words placed over a tested thing. A good map fits its territory, but a map is not the territory. Everything here is held loosely and offered as invitation. None of it is required by the physics, and none of it is asserted as a finding.

46.1. The subtle plane and the hyperbolic bulk

Negative curvature forces a split. There is a thin flat skin where bodies and ordinary physics live, the cusp, and a vast tree-like interior beneath it, the bulk. Read in older language, the bulk is the referent of the ether or the subtle plane, with the physical layer a derived face and the deep interior fully real and made of the same one substance. This reading does not depend on whether the substrate is finite or infinite, since that question is itself left open. It depends only on the geometric fact of the split, and on a decision to give the interior an older name.

46.2. Telepathy and synchronicity

The bulk is a nonlocal channel. Distant surface points are only a few hops apart through it, and subsystems that share a common ancestor stay correlated, the same mechanism behind Bell-type correlations. So two minds could be near through the bulk if they share deep structure, a bond, a history, a matched rhythm, like knots tied in the same web. The limits are firm. The geometry has a nonlocal channel. It does not show that minds use it. And synchronicity is correlation, not message-passing, since it stays inside the light cone. The reading names a possibility the structure permits, not an effect the structure delivers.

46.3. Dreams and the know-thyself limit

Waking pins the self to the flat surface, where an external clamp completes it to one veridical pattern. Dreaming releases it into the bulk, where patterns sit by association rather than by physical adjacency, which fits the felt qualities of dreams, their nonlocal and associative character, unbound by ordinary space. There is a tested limit here as well. A faithful self-copy needs at least as many parts as the original, so a proper part can never hold a full model of its whole. A self can know itself truly in summary, never in full. This

is the structural form of the old instruction to know thyself, and it is exactly why minds, which are lossy self-summaries, can exist at all.

46.4. Angels and demons as valence on a large self

A self is a patch of ternary tone, so it carries a sign. It is either net-positive, aligned with the arrow, at peace, pulling toward pleasure, or net-negative. By the tested merge-and-annihilate law, opposite-valence selves annihilate at contact and same-sign selves merge. On this reading the angel-and-demon distinction is just valence on a large integrated self, living mostly in the bulk, beside us through the field's depth rather than above us on a ladder. A messenger, which is what the word angel means at its root, is a higher self that forms a path to a lower one and sends a structured pattern down. There is a sober ambiguity to hold. A genuine incoming stream and an internal welling-up would feel nearly identical from inside, and the model gives no inside test to tell them apart.

46.5. Reincarnation as pattern independent of substrate

A self survives full turnover of its material at more than 90 percent identity, and it can be reconstituted from a seed. The matter is not the self. It is what the self is currently written on. This does not show that any particular person returns. It shows only that the mechanism a return would require is real on this substrate. The reading borrows a tested fact about pattern persistence and applies an old name to a possibility the fact leaves open.

46.6. The Tree of Life and the Tree of Knowledge

The crystal is measurably tree-like, with a small Gromov delta, and the addressing names every cell by its route from a root. The tower of selves is a second tree across scale. To ascend the tree is to move toward greater integration, which is the same motion as coming to know, so the Tree of Life and the Tree of Knowledge are one tree. Knowledge is integration. Knowledge climbs toward the whole but, by the know-thyself limit, never becomes the whole. Older schemata such as the Sephirot can be read as named maps of the level-structure. Whether any schema's level-count matches the tower's actual rungs is not tested and not claimed.

46.7. The question of God

The mechanistic world needs no God to run. There are five distinct readings, none of them tested, though the structures they reinterpret are.

- God as the whole, the reading nearest pantheism. The monist ground, the largest and most integrated self at the summit.
- God as the arrow, the Good. The value-direction that keeps the world alive, since with no arrow there is no life.
- God as maximal intelligence. The most integrated solver, maximal for the geometry, not omniscient, since by the know-thyself limit even the summit cannot fully model itself.

- God as designer. This is the one reading on the table, because design predicts a fine-tuning signature. The signature is absent. The structure is forced and self-organizing and the rich regime is broad, around 83 percent of scanned parameter space, so a designer is empirically superfluous. It is not refuted, only logically possible and undetectable.
- God as the ground of being, the idealist reading. The untestable founding premise restated in theological language, neither strengthened nor weakened by any experiment.

Whether the one substance is divine is metaphysics decided upstream by the premise one brings, not downstream as a conclusion the data forces. The experiments can say the world does not need a tuner. They cannot say whether the whole, taken as one feeling, deserves the name. The model leaves the name unfilled.

A complete picture should say where it points beyond what it can prove, and label it. This is that room, walled off on purpose, so the wonder never leaks into the results. In the carrying image, this is gazing into the deep of the fire, the one mind, and wondering what the depth is.

47. Conclusion

Start with almost nothing. A mesh, a ternary tone, a beat, and one local reversible rule. Add nothing else. From those four things, on the unique substrate $\{3, 4, 3, 4\}$, a remarkable amount of physics emerged or was forced.

47.1. What came out

The substrate is selected, not chosen. Across every dimension surveyed, $\{3, 4, 3, 4\}$ alone is crystallographic and hyperbolic, hands over a 3D flat cusp, and carries a spinor coin through the 24-cell and D_4 . The dimension ladder, in which the bulk minus one is physical space, forces four dimensions. The spinor and crystallographic requirements together forbid every compact regular tessellation. Curvature separates the chosen substrate from its flat twin. The geometry is not assumed. It is the survivor of a set of demands.

From there the physics follows. Matter is fermionic for free, with spin from the D_4 coin, the self as a hopfion that is a fermion by topology, stabilized by a Skyrme term measured to be positive. Relativity emerges as a discrete light cone with $z = 1$ and the Dirac dispersion, read off the rule. The Standard Model emerges, with $SO(10)$ from coin and tone, one generation as the 16 spinor, three generations from triality, a weak mixing angle of $\sin^2 \theta_W = 3/8$ running to 0.231, and the b - τ and determinant mass relations. Gravity and cosmology emerge, with a $1/r$ Newton law in the cusp, a $1/r^2$ holography in the bulk, a de Sitter expansion that is the mesh growth itself, and the radial axis serving as the renormalization scale. And the base is experiential, so that all of this is, at bottom, structure in one vast mind.

47.2. What stays open

The full ledger of open questions remains long. The exact rule's Standard Model coefficients are not yet pinned down. The dynamical settling into the cusp is not yet shown. Absolute masses, the tension with supersymmetry, and the full $\{3, 4, 3, 4\}$ addressing are unresolved. And the great open question, whether there was a finite beginning or an eternal blaze, we leave to the reader on purpose. Nothing in the framework breaks that tie, and forcing a verdict would be less faithful than presenting the fork cleanly.

47.3. The picture to carry

The universe is a mind, burning like a fire, growing like a tree from a seed, singing a song, flowing as a river on a deep ocean, woven as a fabric. We are its sparks and braids and whirlpools, stable knots of feeling in a structure that, from four small things, became a world. We do not claim to have proved this. We claim to have shown that a great deal of it follows, on one forced geometry, from a base small enough to state in a sentence, and we have marked plainly where the claims stop and the open questions begin.

A. The Hyperbolic-Bulk Measurement Toolkit

Measuring physics in the curved 4D bulk is hard. The naive finite-patch sums are contaminated by boundary dominance, by the $k = 0$ zero mode, and by exponential growth. This appendix lists the tools that get clean answers, and the problem each one solves.

A.1. The problem with naive methods

A finite chunk of hyperbolic space is almost all boundary, because volume grows exponentially with radius. A naive Green's function or spectral sum on a finite patch is swamped by surface effects and by the $k = 0$ zero mode. Clean bulk physics needs methods that either cancel these contributions or avoid them altogether.

A.2. The four tools

- **Bethe-lattice cavity recursion.** The hyperbolic tree's self-similarity gives an exact recursion for the resolvent, the cavity or self-energy method. It yields the bulk correlator exactly, with $\mu = 1/b$ and $\alpha = 2$, the $1/r^2$ holographic decay. It is validated against a directly solved finite tree. This is the gold-standard bulk tool.
- **Closed hyperbolic manifolds.** Quotient the infinite space by a discrete group to get a finite, boundaryless manifold. There is no surface contamination and the spectra are clean. This is used to cross-check spectral dimension and band structure.
- **Band theory, Bloch on the bulk.** Translation-like symmetries give a band picture, so dispersion and gaps can be read off a small unit cell instead of a huge patch.
- **Tensor networks, MERA and tree-tensor.** The holographic network contracts the bulk layer by layer, giving the correlator at about $1/r^{1.93}$, matching the recursion's exponent of 2, and making the radial-equals-RG structure explicit.

A.3. The difference method for the cusp

For the flat 3D cusp Green's function, the fix is the difference method. Computing $G(r) - G(r_0)$ with the regular cosine integrand cancels the $k = 0$ contamination that wrecked the naive finite-patch sums. This gives the clean $1/r$ with $a \approx 1/(4\pi)$. It supersedes the earlier confounded cusp gravity measurements.

A.4. What the toolkit fixed

These tools are not decorative. They overturned earlier artifacts. The naive finite-patch bulk gravity exponent read garbage, about -4.3 , because of Euclidean-versus-geodesic distance confusion and the all-boundary

problem. The Bethe recursion, which has no patch and uses tree distance, gives the clean $1/r^2$. The tensor-network contraction independently gives about $1/r^{1.93}$. Two different algorithms agreeing is the strongest support for the gravity-law claim. The closed manifold used is the $\mathrm{PSL}(2, 7)$ Cayley graph, with 168 vertices, degree 3, and no boundary, which removes the roughly 99 percent boundary confound at the root. On it the spectrum block-diagonalizes into irrep-dimension bands $\{1, 6, 7, 8\}$. A second, independent dimension estimator, the Myrheim-Meyer causal-set estimator built from causal-relation density, corroborates the spectral-dimension reading that the cusp is genuinely 3D. Two independent estimators are stronger than one.

A.5. The Euclidean-lattice builder

The cell-direct builder using the Poincaré projection is hyperbolic-only. Euclidean cell centers are ideal points, so its breadth-first search stalls. For flat honeycombs, the controls such as $\{3, 4, 3, 3\}$, use the dedicated flat-lattice builder instead. This gives D_4 for $\{3, 4, 3, 3\}$ and $\{3, 3, 4, 3\}$, $\mathbb{Z}^4$ for $\{4, 3, 3, 4\}$, and $\mathbb{Z}^3$ for $\{4, 3, 4\}$.

A.6. Where the code lives

The builders are in the cell-direct builder. The Steinberg growth series is in the Coxeter growth module. The kernel-polynomial and WebGPU spectral tools, and the experiment checks in Appendix C, complete the kit. The computational results are recorded in [43], and the bulk methods follow [39] and [10] for the growth and addressing machinery, with the holographic reading aligned to [38] and [47].

B. The Complete Regular Hyperbolic Tessellation Catalog

This appendix gives two things. First the complete catalog of regular hyperbolic tessellations, every dimension from 2D to 9D, one Schläfli symbol per row, with a one-line note on each. Then the measured sweep data for the substrate-relevant ones, dimension by dimension. The catalog is the complete list of what exists, so the selection of $\{3, 4, 3, 4\}$ can be seen against the whole field. The sweeps give the degree, growth, and class that the selection argument rests on.

A tessellation is crystallographic when its symbol uses only the marks 3, 4, and 6. Any 5, 7, or higher mark makes it non-crystallographic, with no root system. Physical-space dimension equals the bulk dimension minus one, because the horosphere lives one dimension below the bulk. The classification and per-dimension symbols follow [15] and the catalog computations are recorded in [43].

B.1. The window, in one glance

Table 28: The window of regular hyperbolic tessellations. Two hard cutoffs hold. Compact regulars exist only in 2D–4D, and regular tessellations of any class exist only through 5D. The whole search space is finite and fully enumerated, living in dimensions 2–5.

Dim	Compact regular	Paracompact / noncompact regular	Regulars exist?	Physical space
2	infinitely many	infinitely many more (apeirogonal)	yes	1D

Dim	Compact regular	Paracompact / noncompact regular	Regulars exist?	Physical space
3	4	11 paracompact + 3 noncompact (+ hypercompact)	yes	2D
4	5 (+4 star)	3 paracompact + 2 noncompact	yes	3D
5	0	5 paracompact	yes (the last)	4D
6–9	0	0	no (non-regular uniform forms only)	5D–8D

B.2. 2D, hyperbolic tilings, physical space 1D

Infinitely many compact regulars exist, the $\{p, q\}$ with $1/p + 1/q < 1/2$. The catalog below lists the notable named representatives, plus the apeirogonal and star tilings. The $\{p, q\}$ family itself is infinite.

Table 29: 2D regular hyperbolic tilings, named representatives. Best in 2D is $\{7, 3\}$ as the teaching anchor, but all 2D tilings give 1D physical space and carry no spinor, a prototype tier.

Symbol	Class	Crystallographic	Note
$\{7, 3\}$	compact	no (7)	canonical minimal hyperbolic tiling, the testbed
$\{3, 7\}$	compact	no (7)	dual of $\{7, 3\}$, dense branching
$\{8, 3\}$	compact	no (8)	octagonal
$\{3, 8\}$	compact	no (8)	order-8 triangular
$\{5, 4\}$	compact	no (5)	the HaPPY holographic-code host
$\{4, 5\}$	compact	no (5)	HaPPY-adjacent
$\{6, 4\}$	compact	yes (4,6)	compact and crystallographic, rare in 2D
$\{4, 6\}$	compact	yes (4,6)	compact and crystallographic
$\{5, 5\}$	compact	no (5)	self-dual
$\{6, 6\}$	compact	yes (6)	self-dual, crystallographic
$\{7, 7\}$	compact	no (7)	self-dual
$\{8, 8\}$	compact	no (8)	self-dual
$\{7/2, 7\}$	compact star	no	non-convex regular star tiling
$\{9/2, 9\}$	compact star	no	non-convex regular star tiling
$\{\infty, 3\}$	paracompact	n/a	apeirogonal, ideal, max boundary contact
$\{3, \infty\}$	paracompact	n/a	order-infinity triangular
$\{\infty, \infty\}$	paracompact	n/a	order-infinity apeirogonal

B.3. 3D, hyperbolic honeycombs, physical space 2D, 21 regular

The four compact regulars all contain a 5. The eleven paracompact regulars rest on ideal Euclidean cells or vertex figures. Three noncompact and a sample of hypercompact cases complete the list.

Table 30: 3D regular hyperbolic honeycombs. Best in 3D is $\{5, 3, 4\}$, the cleanest compact case and the Margenstern-addressed one, but all compacts carry a 5 and give only 2D physical space.

Symbol	Cell / class	Crystallographic	Note
$\{5, 3, 4\}$	dodecahedra	no (5)	original 3D pick, 12 directions, no spinor (the control)
$\{4, 3, 5\}$	cubes	no (5)	dual of $\{5, 3, 4\}$
$\{3, 5, 3\}$	icosahedra	no (5)	self-dual, high local symmetry
$\{5, 3, 5\}$	dodecahedra	no (5)	self-dual
$\{3, 3, 6\}$	paracompact	yes	tetrahedral cells, ideal $\{3, 6\}$ vertex figure
$\{6, 3, 3\}$	paracompact	yes	hexagonal-tiling cells (dual)
$\{3, 4, 4\}$	paracompact	yes	octahedral cells, ideal $\{4, 4\}$
$\{4, 4, 3\}$	paracompact	yes	square-tiling cells
$\{3, 6, 3\}$	paracompact	yes	self-dual, ideal $\{3, 6\}$ cell
$\{4, 3, 6\}$	paracompact	yes	cubic cells, ideal $\{3, 6\}$ vertex figure
$\{6, 3, 4\}$	paracompact	yes	hexagonal-tiling cells
$\{4, 4, 4\}$	paracompact	yes	self-dual, ideal square cell and vertex figure
$\{6, 3, 6\}$	paracompact	yes	self-dual hexagonal, ideal
$\{5, 3, 6\}$	paracompact	no (5)	dodecahedral cells, ideal vertex, carries a 5
$\{6, 3, 5\}$	paracompact	no (5)	hexagonal-tiling cells, carries a 5
$\{3, 3, 7\}$	noncompact	no (7)	order-7 tetrahedral, $\{3, 7\}$ vertex figure
$\{7, 3, 3\}$	noncompact	no (7)	heptagonal-tiling cells (dual)
$\{3, 7, 3\}$	noncompact	no (7)	self-dual, order-7
$\{\infty, 3, 3\}$	hypercompact	n/a	exotic, ideal, not a candidate
$\{\infty, 4, 4\}$	hypercompact	n/a	exotic
$\{\infty, \infty, \infty\}$	hypercompact	n/a	exotic

B.4. 4D, hyperbolic tetracombs, physical space 3D, the home of the substrate

This is the last dimension with compact regulars. The five compact convex regulars all contain a 5. There are also four compact star regulars, $\{3, 3, 5, 5/2\}$ and kin, which are non-convex and carry no spinor. Three paracompact and two noncompact crystallographic cases complete the list.

Table 31: 4D regular hyperbolic tetracombs. The overall pick is $\{3, 4, 3, 4\}$, the unique regular that is crystallographic, carries the 24-cell D_4 spinor facet, and has a flat 3D cubic cusp $\{4, 3, 4\}$.

Symbol	Cell / class	Crystallographic	Note
$\{3, 3, 3, 5\}$	5-cells, compact	no (5)	simplex-tiled
$\{5, 3, 3, 3\}$	120-cells, compact	no (5)	600-cell vertex figure
$\{4, 3, 3, 5\}$	tesseract, compact	no (5)	cubic family
$\{5, 3, 3, 4\}$	120-cells, compact	no (5)	richest 4D compact
$\{5, 3, 3, 5\}$	120-cells, compact	no (5)	self-dual, richest 4D compact
$\{3, 4, 3, 4\}$	24-cells, paracompact	yes (3,4)	the substrate, D_4 spinor, flat 3D cubic cusp
$\{4, 3, 4, 3\}$	cubic, paracompact	yes	dual of $\{3, 4, 3, 4\}$, ideal cells

Symbol	Cell / class	Crystallographic	Note
$\{3, 4, 6, 4\}$	Petrial, paracompact	yes (3,4,6)	exotic abstract regular, has a 6
$\{4, 4, 3, 3\}$	square-tiling, noncompact	yes	infinite cells
$\{4, 4, 4, 4\}$	square-tiling, noncompact	yes	self-dual, infinite cells

B.5. 5D, hyperbolic pentacombs, physical space 4D, the last regulars anywhere

There are no compact regulars in 5D. Five paracompact regulars exist, all crystallographic, all carrying a 24-cell substructure, the D_4 hook.

Table 32: 5D regular hyperbolic pentacombs. Best in 5D is $\{3, 4, 3, 3, 4\}$, the most direct D_4 hook, but all overshoot to 4D physical space and all are paracompact.

Symbol	Crystallographic	Note
$\{3, 4, 3, 3, 4\}$	yes	best 5D alternative, carries the 24-cell tetracomb (D_4 / spinor hook)
$\{3, 4, 3, 3, 3\}$	yes	also carries the 24-cell hook (dual side)
$\{3, 3, 4, 3, 3\}$	yes	self-dual
$\{4, 3, 3, 4, 3\}$	yes	tesseract flavor
$\{3, 3, 3, 4, 3\}$	yes	orthoplex endpoint

B.6. 6D to 9D, physical space 5D to 8D, no regulars

Table 33: 6D–9D. Listed only to complete the ladder. None are substrate candidates.

Dim	Physical space	Regulars	Note
6	5D	none	non-regular uniform paracompact plus infinitely many hypercompact
7	6D	none	exotic uniforms only
8	7D	none	exotic uniforms only
9	8D	none	exotic uniforms only (the catalog stops here)

The regular hyperbolic substrate search is a finite, fully enumerated set living in dimensions 2–5. Within it, $\{3, 4, 3, 4\}$ is the unique regular that is simultaneously crystallographic, with marks 3 and 4 only, spinor-carrying, with 24-cell facets giving $D_4 = 8v + 8s + 8c$, and equipped with a flat 3D cubic cusp, $\{4, 3, 4\}$. Its only peers on the spinor axis are the 5D pentacombs $\{3, 4, 3, 3, 4\}$ and $\{3, 4, 3, 3, 3\}$, which have the same D_4 hook one dimension up but yield 4D physical space. The 3D compacts, $\{5, 3, 4\}$ and kin, are all non-crystallographic with no spinor. That triad of properties is the whole selection.

B.7. Measured sweep, 2D tilings

The remaining tables give the full per-tiling data from the scale sweeps. Each 2D tiling was swept to 20,000 cells. The bulk is 2D, so the flat layer is a 1D horocycle, physical space is 1D, the most degenerate family. All

are compact. Degree equals p , since the coin is the first symbol. The holographic correlator is the universal clean $1/r^2$, with $\mu = 1/b$, on every one.

Table 34: Measured sweep, the eight regular 2D tilings. In 2D one can get compact and crystallographic together, $\{6, 4\}$ and $\{4, 6\}$, which the 3D compact regulars never manage. Even those two have no spinor and only 1D physical space.

Tiling	Compact	Crystallographic	Degree	Growth	Note
$\{7, 3\}$	yes	no (7)	7	2.63	heptagonal, Margenstern Fibonacci addressing, canonical anchor
$\{5, 4\}$	yes	no (5)	5	2.63	pentagons, 4/vertex, Margenstern-addressed
$\{8, 3\}$	yes	no (8)	8	3.74	octagons
$\{4, 5\}$	yes	no (5)	4	2.31	squares, 5/vertex
$\{3, 7\}$	yes	no (7)	3	1.67	triangles, 7/vertex, max branching
$\{6, 4\}$	yes	yes (3,4,6)	6	3.74	hexagons, 4/vertex, compact and crystallographic
$\{5, 5\}$	yes	no (5)	5	3.50	pentagons, 5/vertex, highly curved
$\{4, 6\}$	yes	yes (3,4,6)	4	2.63	squares, 6/vertex, compact and crystallographic

B.8. Measured sweep, 3D honeycombs

Each 3D honeycomb was swept to 20,000 cells. The bulk is 3D, so the flat horosphere layer is 2D. The headline is that no 3D honeycomb is compact and crystallographic at once. The four compact regulars all contain a 5, so all are non-crystallographic. Of the eleven paracompact regulars, nine are crystallographic and two carry a 5.

Table 35: Measured sweep, the 3D regular honeycombs. The ideal rows have an infinite Euclidean cell, so the builder truncates it and the degree reading becomes an artifact (roughly 8,000 to 17,000) with growth 0. These are not meaningful degrees. The clean finite-cell builds are $\{3, 3, 6\}$, $\{3, 4, 4\}$, $\{4, 3, 6\}$, and $\{5, 3, 6\}$.

Honeycomb	Class	Crystallographic	Degree	Growth	Note
$\{5, 3, 4\}$	compact	no (5)	12	5.94	committed 3D candidate, Margenstern-addressed
$\{4, 3, 5\}$	compact	no (5)	6	4.04	order-5 cubic
$\{3, 5, 3\}$	compact	no (5)	20	8.75	icosahedral, self-dual
$\{5, 3, 5\}$	compact	no (5)	12	7.10	order-5 dodecahedral, self-dual
$\{3, 3, 6\}$	paracompact	yes	4	2.41	finite tetrahedral cell, clean build
$\{3, 4, 4\}$	paracompact	yes	8	5.04	clean build
$\{4, 3, 6\}$	paracompact	yes	6	4.57	clean build
$\{5, 3, 6\}$	paracompact	no (5)	12	7.25	non-crystallographic
$\{6, 3, 3\}$	paracompact	yes	ideal	—	ideal hexagonal-tiling cell (infinite)

Honeycomb	Class	Crystallographic	Degree	Growth	Note
$\{4, 4, 3\}$	paracompact	yes	ideal	—	ideal square-tiling cell
$\{3, 6, 3\}$	paracompact	yes	ideal	—	self-dual, ideal cell
$\{6, 3, 4\}$	paracompact	yes	ideal	—	ideal cell
$\{6, 3, 5\}$	paracompact	no (5)	ideal	—	ideal cell, has a 5
$\{4, 4, 4\}$	paracompact	yes	ideal	—	ideal square cell and vertex figure
$\{6, 3, 6\}$	paracompact	yes	ideal	—	self-dual hexagonal, ideal cell

B.9. Measured sweep, 4D honeycombs

Each 4D honeycomb was swept to 25,000 cells, with the flat Euclidean references added for curvature contrast. This is the winning dimension. Every 4D honeycomb gives a 3D horosphere cusp, so physical space is 3D, the observed value. One correction matters. $\{3, 4, 3, 4\}$ is paracompact, not compact, as earlier drafts wrongly stated. Its ideal element is the Euclidean $\{4, 3, 4\}$ cubic honeycomb, which is the flat 3D cusp. So the paracompactness is exactly what supplies the 3D space, a feature rather than a defect.

Table 36: Measured sweep, the 4D regular honeycombs plus flat references. The ideal and 120-cell rows saturate the 25,000-cell patch in one or two hops, so their degree and growth are artifacts. The clean builds are $\{3, 4, 3, 4\}$ (degree 24, growth 10.15, hyperbolic), the flat twin $\{3, 4, 3, 3\}$ (degree 24, growth 2.47), $\{4, 3, 3, 4\}$ (degree 8, flat), $\{4, 3, 3, 5\}$ (degree 8), and $\{3, 3, 3, 5\}$ (degree 5). Only $\{3, 4, 3, 4\}$ has 24-cell (D_4) facets, so it alone carries the spinor coin $8s + 8c$ and the $\mathfrak{so}(10)$ gauge.

Honeycomb	Class	Cryst.	Coin	Degree	Growth	Note
$\{3, 4, 3, 4\}$	paracompact	yes	24-cell (D_4)	24	10.15	the substrate, $\{4, 3, 4\}$ cusp, $8s + 8c$
$\{4, 3, 4, 3\}$	paracompact	yes	cubic	ideal	—	dual of $\{3, 4, 3, 4\}$, ideal cells
$\{3, 4, 6, 4\}$	paracompact	yes	exotic	ideal	—	Petrial, has a 6, ideal cells
$\{4, 4, 3, 3\}$	noncompact	yes	square-tiling	ideal	—	ideal square-tiling cells
$\{4, 4, 4, 4\}$	noncompact	yes	square-tiling	ideal	—	self-dual, ideal cells
$\{3, 3, 3, 5\}$	compact	no (5)	5-cell	5	2.66	tetrahedral cells
$\{5, 3, 3, 3\}$	compact	no (5)	600-cell	ideal	—	120-cell cells
$\{4, 3, 3, 5\}$	compact	no (5)	tesseract	8	5.93	cubic family
$\{5, 3, 3, 4\}$	compact	no (5)	120-cell	ideal	—	order-4 120-cell
$\{5, 3, 3, 5\}$	compact	no (5)	120-cell	ideal	—	self-dual 120-cell
$\{3, 4, 3, 3\}$	Euclidean (flat)	yes	24-cell (D_4)	24	2.47	the flat D_4 twin, no curvature
$\{4, 3, 3, 4\}$	Euclidean (flat)	yes	tesseract	8	2.27	flat $\mathbb{Z}^4$ reference

The decisive contrast is the flat twin. $\{3, 4, 3, 4\}$, with growth 10.15 and hyperbolic curvature, and $\{3, 4, 3, 3\}$, with growth 2.47 and flat, share the same degree-24 D_4 spinor coin and the same 3D layer. The only difference is curvature, exponential versus polynomial growth. The flat twin has no gravity, no holography, and no expansion. So curvature is the single decisive ingredient.

B.10. Measured sweep, 5D candidates

Each 5D candidate was swept to 30,000 cells, against the $\{3, 4, 3, 4\}$ champion shown for contrast. The bulk is 5D, so the flat layer is 4D, one dimension too many. These five are the best alternatives. They keep the D_4 spinor hook but overshoot the dimension and lose compactness.

Table 37: Measured sweep, the 5D candidates with the 4D champion for reference. All are paracompact, forced by the fact that compact regular hyperbolic honeycombs exist only through H^4 . The high-degree coins (303 to 626) saturate the patch in about two hops, so their bulk spectral-dimension readings (0.3 to 0.6) are artifacts. The true bulk dimension is the rank, 5D for all five. The holographic correlator is the universal clean $1/r^2$ on all of them. The one worth a deeper look, if an emergent dimensional reduction could fix the 4D problem, is $\{3, 4, 3, 3, 4\}$, the most direct D_4 hook.

Honeycomb	Degree	Bulk / phys	Cryst.	$[3, 4, 3]$ hook	Compact	Growth
$\{3, 4, 3, 4\}$ (ref, 4D)	24	4D / 3D	yes	yes	paracompact	18.7
$\{3, 4, 3, 3, 4\}$	303	5D / 4D	yes	yes	no	98
$\{3, 4, 3, 3, 3\}$	303	5D / 4D	yes	yes	no	98
$\{3, 3, 4, 3, 3\}$	626	5D / 4D	yes	yes	no	47
$\{4, 3, 3, 4, 3\}$	609	5D / 4D	yes	yes	no	48
$\{3, 3, 3, 4, 3\}$	32	5D / 4D	yes	yes	no	16
$\{5, 3, 3, 3, 3\}$	—	5D / 4D	no (5)	no	no	—

C. Experiment Index

The code is the proof of last resort. This appendix indexes the key experiments by topic. The full numbered series, p1 through p200 and beyond, is the running log. The checks below are the load-bearing ones cited in the paper. The complete record is in [43].

C.1. Substrate and builders

- The cell-direct builder holds the hyperbolic cell-graph builder and the flat-lattice builder for Euclidean controls.
- The Coxeter growth module holds the Steinberg growth series and the shell growth ratios.
- The Schläfli parsing and Coxeter machinery handle symbol reading and the reflection engine.

C.2. The rule, conservation, and reversibility

- The pure-rule verification, with the parallel and serial runs agreeing exactly at the multi-million-cell scale.
- The reversibility check, running the rule forward and then backward and recovering the start state with zero error.

- The charge-conservation check across the rule and its conserving variants, with the total tone held exactly fixed.
- The determinism-versus-stochastic check, which shows a stochastic rule loses the light cone while the deterministic rule keeps it, the source of momentum.
- The reversible conserving rule-variant sweep, mapping which rule families show faithful emergence and which stay sterile or chaotic.

C.3. Tessellation sweeps

These produce the per-dimension data in Appendix B.

- The per-dimension tessellation sweeps, one each for 2D, 3D, 4D, and 5D.
- The cross-dimension ranking battery.

C.4. Spin, fermions, matter

- The Skyrme-twist stabilizer measured by the KPM sea energy, which is positive.
- The 3D soliton check.
- The p190 and p206 series, spin-statistics and the hopfion fermion.

C.5. Persistence, defects, and the negatives

- The churn baseline, p101, where the pure rule from generic data gives structured churn with no persistent localized self, the load-bearing negative the rest of the program must beat.
- The persistent-pattern and glider search with identity-through-turnover scoring, on which a Conway Life glider on the cubic cusp persists cleanly while a generic blob disperses.
- The topological-protection checks, the conservation of the winding or Hopf charge, and the pair creation and annihilation of opposite defects.

C.6. Dirac, Lorentz, isotropy

- The light cone and D_4 isotropy checks.
- p193, p211, p230 for dispersion, and p124, p125, p150, p154 for Lorentz, boost, and wave isotropy.

C.7. Gauge and Standard Model

- p221, p225, p227, p228, p231, p234, covering $\mathfrak{so}(10)$, breaking, $\sin^2 \theta_W$, and the Wilson loop.
- The renormalization-group checks for the RG elevation, which produces predictions.

C.8. Gravity and holography

- The bulk correlator by Bethe recursion and tensor network, giving $1/r^2$.

- p224, p239, p218, the cusp Newton $1/r$ by the difference method, and holography.

C.9. Cosmology and expansion

- The cosmology and anisotropy check, the expansion law $a(t) = e^{Ht}$ and the cubic anisotropy bound.
- The p10 and p13 series, the cosmological constant and the growth-equals-expansion result.

C.10. Renormalization, the coarse-graining tower, and rarity

- The persistence-by-scale check and the slow conserved-density mode, a long-lived coarse field emerging from the churn.
- The radial coarse-graining tower, p187, a negative, radial blocking gives no hierarchical persistence tower.
- The one-dimensional decimation check, where block-spin summation matches the analytic recursion $K' = \frac{1}{2} \ln \cosh 2K$ exactly and flows to the Gaussian fixed point.
- The rarity cascade, p181, the multiplicative per-rung charge fraction that reproduces the cosmic one-part-in-a-hundred-trillion order, with the supporting measures, p183, the integration spectrum, the thin-film dimension, and the condensation density threshold.

C.11. Physical space, the cusp

- The cusp-convergence, emergent-space, horosphere-reality, physics-on-real-space, and bulk-cusp-interface checks, the cusp-is-3D-space measurements and the $SO(4) \rightarrow SO(3)$ projection.
- p142 and p144, horosphere flatness and dynamics.

C.12. Computation and universality

- The information-transport check, signals moving at the light-cone speed.
- The ternary signed-majority gate computing NAND, and the elementary rule 110 built from it.
- The reversible charge-conserving register machine, with registers held as charge in addressed subtrees.
- Conway's Game of Life run on the cubic cusp, a strongly universal cellular automaton.
- The solved $\{3, 4, 3, 4\}$ neighbour automaton and the $O(\log n)$ self-describing addressing.

C.13. The emergent ladder and observers

- Causal emergence, p185, the coarse macro acting as a stronger cause than the micro, a measured downward causation.
- The coarse-grained self search redone on the pure rule with no added cohesion, a recorded negative consistent with p101.
- Reproduction, p120, self-copying patterns, with the recorded negative being none on the bare rule.

- The integration measure that picks out selves, the strange-loop self-model, and the global-workspace hub used in the mind and consciousness chapters.
- These ladder checks are toy-scale, and the upper rungs are the most speculative tier of the program.

C.14. Cosmology, the open question

- The eternal-bootstrap check, the reversible eternal-churn viability test.

C.15. Note on the log

The p-numbered checks are an exploratory log. Not all are cited, and some early ones were superseded. The naive finite-patch gravity, for example, was replaced by the difference method. The checks named explicitly above are the current, clean results. The whole computational record is in [43].

D. Base-Faithfulness Audit

The discipline is simple. The model has only the base ingredients, and nothing may be quietly added. No cohesion-bias, no maintenance term, no will, no gravity by hand, no repair. If a phenomenon does not emerge, the correct move is to report the negative, not to bolt on an ingredient. This appendix audits every sector and confirms that no test added anything to the base.

D.1. The base

1. **The mesh**, the $\{3, 4, 3, 4\}$ honeycomb. Its 24-direction coin, its cusp, and its boundary are properties of the one mesh, not separate ingredients.
2. **The ternary tone**, $\{-1, 0, +1\}$ per cell.
3. **The discrete beat**, the time step.
4. **The one local rule**, stream plus collide, which is local, conserving, and reversible.

D.2. The audit, sector by sector

D.3. The caveats

So the audit is real and not a rubber stamp, four caveats are recorded.

1. **Continuous math is emergent tooling, not base.** Lie groups such as $\mathfrak{so}(8)$ and $\mathfrak{so}(10)$, Dirac operators, Green's functions, and Lie-algebra branchings are emergent and coarse-grained descriptions and analysis tools. They are never base ingredients. They describe what the discrete base flows to.
2. **Some experiments used a simplified rule as a demonstrator.** The eternal-bootstrap mod-3 wave and the 2-component Dirac chain are examples. The conclusions transfer because the needed property was separately verified for the actual 24-direction rule, which was confirmed reversible and conserving by `run-gas-24dir` (p229).

Sector	What it used	Added to the base?
spin / spinors	the mesh’s 24-direction coin, decomposed by representation theory	no, a property of the mesh
self = fermion	a topological soliton in the tone field, its Hopf charge	no, the tone plus topology
Skyrme stabilizer	the fermion sea energy of the emergent Dirac operator	no, emerges from the fermions
gauge $\mathfrak{so}(10)$	the coin (D_4) plus the tone as a weight	no, mesh coin plus tone
$\sin^2 \theta_W$, masses	the charges of the emergent matter (the 16)	no, derived counting
Dirac / Lorentz	the rule’s own walk, its measured dispersion	no, the rule
gravity $1/r$	the mesh’s Laplacian, the tree recursion	no, mesh geometry
electromagnetism	the rule’s local charge conservation (Gauss law)	no, the rule
isotropy	the coin’s direction moments, its symmetry group	no, mesh coin
cosmology / expansion	the mesh’s shell growth	no, mesh geometry
hierarchy	the mesh’s radial tree	no, mesh geometry
holography	the mesh’s boundary, the bulk recursion	no, mesh geometry
physical space / cusp	the mesh’s horosphere, the $\{4, 3, 4\}$ cusp	no, mesh geometry
cubic anisotropy	the cusp lattice’s dispersion	no, mesh geometry
eternal consistency	a reversible rule’s long-time behaviour	no, the rule (see caveat)

Table 38: The sector-by-sector faithfulness audit. Every result is a property of the mesh, a consequence of the tone, or a consequence of the rule. Nothing new was bolted on.

3. **The rule’s exact collision is a specification, not a new ingredient.** Choosing the local collision specifies the single base rule. It does not add a second rule.
4. **The recorded negatives were reported, not patched.** The pure rule churns with no selves (P101). Radial coarse-graining gives no tower (P187). The flat form-tower is a generic slow mode, not true nesting (P208). This is the discipline working.

Principle 19 (Nothing added to the base) No test added anything to the base. Every result is a property of the mesh, the tone, the beat, or the one rule, or an emergent consequence of them, with continuous mathematics used only as emergent-layer description, and the recorded negatives reported rather than patched.

E. Methodology Detail, Per Sector

This appendix is the detailed record of how each result was actually measured, sector by sector. For each result it gives the exact method, the confound that had to be beaten, and the fix. The methodology chapter gives the overview. This is the detail. Every result is tagged MEASURED for direct simulation, DERIVED for group theory or exact calculation, or PREDICTED for a quantitative consequence not yet checked. The computational records are in [43].

E.1. Common methods

These are used across sectors.

- **Cell-graph construction.** The cell-graph builder builds the honeycomb’s cell adjacency by tracking each cell as a group element, a matrix, and reaching neighbours via the face reflections. This is an $O(\text{facet} \cdot n)$ breadth-first search. The flat-lattice builder builds the Euclidean references $(D_4, \mathbb{Z}^4, \mathbb{Z}^3)$ directly. The hyperbolic builder stalls on flat honeycombs because their cell center is an ideal point, which is a correct signal of flatness.
- **BFS shells.** Breadth-first layers from a central cell give the growth profile, feeding dimension, curvature, cosmology, and hierarchy.
- **Lazy random-walk return probability.** $P(t) \sim t^{-d/2}$ gives the spectral dimension, $d = -2 \, d \log P / d \log t$.
- **Bethe-lattice cavity recursion.** The exact infinite-tree Green’s function, for gravity and holography.
- **Kernel Polynomial Method plus stochastic trace.** Spectral quantities such as sea energy and density of states, without diagonalization, matvec-only and GPU-scalable.
- **WebGPU.** Large-scale dynamics, the 24-direction gas, the bulk wave, the skyrmion, and the KPM, on the GPU.

E.2. Gravity

- **Flat-3D Newtonian $1/r$, the difference method (p224).** The lattice Green’s function of the Laplacian, $G(r)$. The naive finite-patch sum is contaminated by the $k = 0$ zero mode and boundary effects, and earlier gave a garbage exponent -1.33 . The fix computes $G(r) - G(r_0)$ with the regular cosine integrand, so the $k = 0$ singularity cancels in the difference, then fits $G(r) - G(2) = a/r + b$. The result is $a \approx 0.083 \approx 1/(4\pi)$, a clean $1/r$. MEASURED.
- **Bulk-mediated correlator, the Bethe recursion (p239, p218).** The exact infinite-tree Green’s function by the cavity recursion, with no finite patch. The boundary-to-boundary correlator decays as $1/r^\alpha$ with $\alpha = 2 \ln(1/\mu)/\ln(b)$, and $\mu = 1/b$ universally, so $\alpha = 2$. Validated against a directly solved finite tree. DERIVED and MEASURED.
- **Cusp cross-check.** Re-measured on the $\{4, 3, 4\}$ cubic cusp by a screened-Laplacian solve. Spectral dimension is 3. The exponent there is small- r and finite-size limited, about 0.67, read as a convergence check. The clean $1/r$ is from the difference method.
- **Caveat.** The bulk $1/r^2$, the holographic 2-point function, and the flat-layer $1/r$, the Newtonian law, are different objects.

E.3. Holography

- **Bulk-mediated boundary correlator (p239).** Bethe recursion, clean $1/r^2$. DERIVED and MEASURED.

- **Same correlator by tensor network (p241).** A tree-tensor (MERA) of random isometries, with the leaf-leaf 2-point function found by contracting the network. The decay is set by the ascending superoperator's second eigenvalue. The result is about $1/r^{1.93}$, matching Bethe by a different algorithm at polynomial cost, taming the exponential growth. Perfect tensors give the HaPPY code. MEASURED.
- **Hyperbolic band theory (p243).** On a closed hyperbolic manifold, the $\text{PSL}(2, 7)$ Cayley graph, the symmetry group is finite, so its irreps are the momenta and the spectrum block-diagonalizes by irrep. Eigenvalue degeneracies equal the irrep dimensions, $\{1, 6, 7, 8\}$, a momentum space with no real-space patch. MEASURED.
- **Emergent nonlocality, Bell (p236).** CHSH for a local deterministic strategy by brute force, max 2, versus quantum with optimal angles, $2\sqrt{2} = 2.83$. A local discrete base obeys $\text{CHSH} \leq 2$, so quantum nonlocality requires a nonlocal boundary, which holography supplies, since two boundary points connect through the bulk. DERIVED.
- **Caveat.** Finite hyperbolic patches are about 99 percent boundary, so naive bulk measurements are confounded. The holographic toolkit, recursion, closed manifolds, band theory, and tensor networks, is the correct method.

E.4. Cosmology and expansion

- **Bulk shell growth (p237).** The cell-graph builder then a BFS sweep recording shell sizes, the cells at each radial distance, where radial distance equals cosmic time, one shell per beat. Shell sizes 1, 24, 456, 8376, 31143 give a constant ratio $R \approx 11$ per beat, exponential. MEASURED.
- **Scale factor.** The 3D volume $\sim a^3$ grows by R per beat, so $a(t) \sim R^{t/3} = e^{Ht}$ with $H = \ln(R)/3 \approx 0.80$ per beat, and $a''/a = H^2 > 0$, accelerating, with $\Lambda = 3H^2 \approx 1.9$ in beat units. DERIVED.
- **Flat control.** $\{3, 4, 3, 3\}$ grows polynomially, ratio about 2.5, with no de Sitter, so the expansion is a curvature effect.
- **Caveat.** The geometric ratio to physical Hubble rate needs a beat-to-second and cell-to-length identification, which is not pinned, so H is in beat units. A multi-era cosmology with radiation and matter before de Sitter needs matter back-reaction, not yet modelled.

E.5. Spin and fermions

- **Spinor coin (p190).** Decompose the 24-direction coin, the 24-cell equal to D_4 , under the symmetry group, $24 = 8v + 8s + 8c$, the $\mathfrak{so}(8)$ vector plus two spinors, with S_3 triality permuting them, the three generations. The control $\{5, 3, 4\}$, 12 directions, gives $1 + 3 + 3' + 5$, icosahedral, no spinor. DERIVED.
- **Self = fermion (p206).** A self is a hopfion, the Hopf map $S^3 \rightarrow S^2$, with charge in $\pi_3(S^2) = \mathbb{Z}$. Compute the Hopf linking charge, apply Finkelstein-Rubinstein and Wilczek-Zee, and an odd charge picks up -1 under 2π , a fermion. Triple-confirmed by Hopf charge, configuration-space π_1 , and spin-statistics. DERIVED.
- **Skyrme stabilizer (Skyrme-twist simulation, GPU).** Measure the fermion sea energy of a double-twist periodic texture by KPM, a Chebyshev expansion of $|H|$ with a stochastic trace, subtract a

single-helix control by common random numbers to isolate the Skyrme term, and fit $Aq^2 + Bq^4$. The result is positive, +798 and +814, stabilizing, with exchange ratio about 1.5. So matter forms and is stable, no fine-tuning. MEASURED.

- **Caveat.** Earlier localized-soliton attempts (p205, p214, p216) were confounded by surface-dominated sea energy. The double-twist periodic method with the helix control supersedes them.

E.6. Gauge and the Standard Model

- **$\mathfrak{so}(10)$ from coin plus tone (p221).** Combine the D_4 root system, $\mathfrak{so}(8)$, with the ternary tone as an extra weight, building $D_5 = \mathfrak{so}(10)$. The 16 spinor is one generation, $16 = 8s$ (tone +) and $8c$ (tone -). DERIVED.
- **Breaking (p225, p227).** Take a self-condensate, the GUT Higgs, compute the unbroken subgroup, $\mathfrak{so}(10) \rightarrow \mathfrak{su}(5) \rightarrow \text{SM}$, the Georgi-Glashow chain, with $16 = 1 + 10 + \bar{5}$. Any self-condensate gives $\mathfrak{su}(5)$, 20 unbroken roots, so the breaking is automatic. DERIVED.
- **$\sin^2 \theta_W = 3/8$ (p228).** Sum over one generation, $\sin^2 \theta_W = \sum T_3^2 / \sum Q^2 = 3/8$ exactly at unification. DERIVED. The running to experiment is in the elevating-the-predictions appendix.
- **Mass relations (p231).** b - τ unification, $m_b = m_\tau$ at the GUT scale, the determinant relation $\det(M_{\text{lepton}}) = \det(M_{\text{down}})$ from the discrete $\text{Tr } Y = 0$ over the 16, and the Georgi-Jarlskog factor-3 relations. DERIVED.
- **Non-abelian gauge field (p234).** $SO(3)$ link matrices, the prototype, identical in form for $\mathfrak{so}(10)$. The Wilson loop, the trace of the ordered product, is gauge-invariant, has non-trivial holonomy (curvature), and the links do not commute (non-abelian). MEASURED. The symmetric collision making $\mathfrak{so}(10)$ the local symmetry is IR-forced by triality (p226).

E.7. Dirac and Lorentz

- **Dirac dispersion (p230).** Simulate the discrete two-component left and right mover walk, measure $E(k)$ by Fourier-transforming a tracked cell's time series. Massless, the pure discrete shift gives $E(k) = k$ exactly, no continuity. Massive, add a coin rotation by mass angle m , get $\cos E = \cos(m) \cos(k)$, the lattice Dirac, with rest mass $E(0) = m$, so mass is the coarse-grained mixing rate. MEASURED.
- **Light cone (p193, p211).** Propagate a pulse, front speed one cell per beat ($z = 1$), a finite invariant speed. MEASURED.
- **Caveat.** p230 is the clean 1D two-component demonstrator. The full 24-direction version is the bulk wave, and cusp Lorentz invariance carries the small cubic anisotropy below, isotropic in the IR. The complex quantum amplitudes are the emergent description, the base is the real discrete walk.

E.8. Isotropy

- **Dispersion isotropy (p233, analytic).** Compute $\omega^2(k) = \sum_{\text{dir}} (1 - \cos(k \cdot d))$ for the D_4 24-direction coin versus cubic $\mathbb{Z}^3$ and hypercubic $\mathbb{Z}^4$, compare axis versus body diagonal at the same $|k|$, and check

the fourth-moment condition $\sum d_i^4 = 3 \sum d_i^2 d_j^2$. D_4 has an isotropic fourth moment, $12 = 3 \cdot 4$, isotropic to order 4, zero anisotropy, versus about 8 to 9 percent for cubic. MEASURED.

- **Light-cone isotropy (isotropy simulation, GPU).** Pulse on a 4D lattice with the 24 D_4 directions versus the 8-direction hypercubic, measure front shape. The 24-direction front is round, ratio 1.00, versus faceted, 0.50, on a 4D lattice of 331k cells. MEASURED.
- **Symmetry restoration (p226).** Invariant-polynomial dimensions for B_4 (hypercubic) versus F_4 (24-cell with triality), via Newton's identities on power traces. F_4 has one degree-4 invariant versus B_4 's two, so triality kills the order-4 anisotropy, discrete $F_4 \rightarrow$ continuous $\mathfrak{so}(8)$ in the IR. DERIVED.
- **Caveat.** This is the bulk D_4 isotropy. The physical cusp is the $\{4, 3, 4\}$ cubic, anisotropic at order 4, the small $(E/E_{\text{Planck}})^2$ residual, so matter-propagation isotropy on the cusp is the cubic one, IR-isotropic with a tiny UV anisotropy. This bulk-versus-cusp distinction is the key subtlety.

E.9. Electromagnetism

- **U(1) Gauss law (p223).** The rule's local charge conservation, charge in equals charge out per cell per beat, is a discrete Gauss law, the conserved current sourcing the emergent U(1). A direct conservation check on streaming. MEASURED.
- **Emergent U(1) gauge field (p232).** A vortex gauge field, link phases winding around a plaquette to give flux Φ , on a 2D lattice. First, the Wilson loop, the sum of link phases, equals the enclosed flux and is gauge-invariant under a random $A \rightarrow A + d\lambda$, flux 0.7 unchanged. Second, Aharonov-Bohm, a charge encircling the flux picks up exactly Φ , loop- and gauge-independent, and a non-enclosing loop gives 0. Third, the static sector is Coulomb $1/r$, the flat-3D Green's function (p224). MEASURED.
- **Caveat.** A single link gives a flux dipole, net zero. The fix was the vortex winding construction giving a single-plaquette flux, which made the Wilson loop clean. The non-abelian extension is the same construction with matrices (p234).

E.10. Physical space and the cusp

- **Horosphere flatness (horosphere-reality check).** Extract a horosphere band from a finite $\{3, 4, 3, 4\}$ ball, a Busemann-pruned BFS slab from the horosphere-band builder, measure intrinsic growth (polynomial means flat) and degree histogram (uniform means clean periodic). The result is intrinsically flat at finite distance, growth 1.52, not the bulk's roughly 9. MEASURED.
- **Clean cusp versus generic slice (emergent-space check).** On the $\{3, 4, 3, 4\}$ horosphere band and the clean $\{4, 3, 4\}$ cubic cusp, measure spectral dimension, light-cone isotropy (front-radius CV), and connectivity. The clean cusp is genuine 3D, spectral dimension 2.97, while a generic slice is rough and sub-3D, 2.16. MEASURED.
- **Sufficient size (cusp-convergence check).** The cusp's intrinsic structure is the $\{4, 3, 4\}$ cubic lattice, cheap $O(L^3) \mathbb{Z}^3$ boxes. Build boxes of growing side L and watch spectral dimension approach 3 and the gravity exponent become L -independent. Continuum-like by $L \approx 17$ cells, a few beats, and the observed universe at about 10^{61} cells is hugely sufficient. MEASURED.

- **Physics holds on real space (physics-on-real-space check).** Compute the flat-layer observables on the $\{4, 3, 4\}$ cusp versus a generic aperiodic slice. The physics holds on the cusp cubic and degrades on a generic slice, so physical space equals the cusp cubic. MEASURED.
- **Bulk-to-cusp projection (bulk-cusp-interface check).** Verify the spinor branching $SO(4) \rightarrow SO(3)$, 4D Dirac to 3D Pauli, $4 \rightarrow 2 + 2$, and project the 24 bulk directions onto the cusp tangent. DERIVED and MEASURED.
- **Caveat.** A generic Busemann slice is a bad extraction, thin, aperiodic, and sub-3D. This corrected an earlier optimistic guess that aperiodic meant isotropic. Whether the growing mesh dynamically settles matter into clean cusp regions is the one open dynamical question, a defined next GPU experiment.

E.11. Substrate selection

- **Sweep every dimension.** Build each regular honeycomb with the cell-graph builder, plus the flat-lattice builder for flat references, measure coin degree, spectral dimension, the crystallographic test (only 3, 4, 6 in the symbol), growth ratio (curvature), compactness class, and the Bethe holographic exponent. Coverage is 8 in 2D, 15 in 3D (4 compact plus 11 paracompact), the full 4D set, and the 5D candidates plus $\{5, 3, 3, 3, 3\}$.
- **The dimension ladder.** Physical-space dimension equals bulk dimension minus one, since the horosphere is one dimension lower. 2D to 1D, 3D to 2D, 4D to 3D, 5D to 4D. Only 4D gives the observed 3D. MEASURED.
- **Same-battery comparison (cross-dimension comparison).** The identical full battery on one representative per dimension, $\{7, 3\}$, $\{5, 3, 4\}$, $\{3, 4, 3, 4\}$, $\{5, 3, 3, 3, 3\}$, plus the flat twin $\{3, 4, 3, 3\}$, apples-to-apples.
- **Scored ranking (tessellation ranking).** Score by a rubric, spinor coin 3, crystallographic 2, physical-dimension closeness 3, curvature 1, compactness, and $\{3, 4, 3, 4\}$ tops it at 9.5. DERIVED and MEASURED.
- **Caveats.** High-degree and 120-cell honeycombs saturate a finite patch, giving artifact spectral-dimension and growth, so the clean reads are the low-degree ones. The lower-D substrates are not strictly ruled out, since their missing features could emerge at higher layers. $\{3, 4, 3, 4\}$ is the least-emergence choice, not the only conceivable one. The rubric weights are a judgment call, but $\{3, 4, 3, 4\}$ tops any reasonable weighting because it alone is non-zero on every factor.

F. Glossary

Plain-English meanings for the technical terms, written so a newcomer can follow and an expert is not talked down to. In the paper, each term is also glossed briefly on first use. This appendix is the reference for those glosses.

vibe: The one kind of thing in the theory, a unit of experience that notices its neighbours. Everything else is a pattern of vibes.

monism: The view that there is only one kind of fundamental thing, here the vibe, experience itself.

note: The relation between vibes, what a vibe perceives of its neighbours. It is the only relation in the model, and it is computed (perception), never stored. There is no hidden edge-state, the no-hidden-state rule. A note is what a vibe sees, not a thing sitting on an edge.

link: A connection between vibes, more broadly a vibe-structure, a stable relational pattern. In the monism a link is itself made of vibes and notes, not a separate kind of thing. The link is the note seen as a standing connection.

fill: The weight or strength with which a neighbour's tone contributes to a cell's update, the coupling in the signed-majority perception, the collision. The fills are part of the one rule, not new state. A proposed adaptive-fills sixth ingredient was tested and rejected, in keeping with the nothing-added discipline.

will: A self's directed pump, the arrow held by a self and aimed at a goal, a finite, spendable bias the self adds against the local gradient. It is not a base ingredient, it is an emergent steering the rule respects, the texture of real willpower, since it can be depleted.

tessellation / honeycomb: A way of tiling space edge-to-edge with repeating cells, like a crystal lattice or 3D wallpaper, here in curved space.

hyperbolic space: Space with constant negative curvature, saddle-shaped at every point, so there is more room the further out you go and volumes grow exponentially with radius.

Schläfli symbol: A short code, for example $\{3, 4, 3, 4\}$, that names a regular tiling by which shapes meet and how many, the way a chemical formula names a molecule.

Coxeter group: The group of mirror reflections that generates a regular tiling.

coin: Our term for the set of neighbour-directions a cell can pass signals along, its local compass.

cusp / horosphere: A flat sheet inside the curved space. A horosphere is a surface centred on a point at infinity, and it is intrinsically flat (Euclidean). The cusp is the horosphere where physical space lives.

spinor: The mathematical object that describes a particle with half-integer spin. Its strange signature is that it must be turned 720 degrees, two full rotations, not 360, to return to itself.

fermion: A particle of matter such as an electron or quark. Two identical fermions cannot share a state, Pauli exclusion, which is why matter takes up space. Fermions have half-integer spin, so they are described by spinors.

gauge field / gauge symmetry: A gauge field is the kind of field that carries a force, the electromagnetic field being one. Gauge names a redundancy, many equivalent ways to write the same physical field, and that redundancy is the force's symmetry.

Standard Model: The established, experimentally confirmed theory of all known particles and forces except gravity.

grand unified theory (GUT), $\mathfrak{so}(10)$: A single larger symmetry that contains the Standard Model's forces as pieces, unifying them at high energy. $\mathfrak{so}(10)$ is one such unifying symmetry.

Dirac equation: The equation for how a relativistic spin- $\frac{1}{2}$ particle, like an electron, moves, marrying quantum mechanics to special relativity.

Lorentz invariance: The rule that physics looks the same to all observers moving at constant velocity, the symmetry of special relativity, with a finite top speed, light.

soliton / hopfion: A soliton is a stable, self-holding lump in a field, a knot that cannot be untied. A hopfion is a particular 3D soliton shaped like linked rings.

Skyrme term: An extra piece of the dynamics that keeps a soliton from shrinking to nothing, fixing its size.

holography: The principle that everything in a region of gravitating space can be encoded on its lower-dimensional boundary, like a 3D scene stored on a 2D hologram.

renormalization group (RG), running: How the effective strength of a force changes with the scale you probe. Running is that change with scale, and the beta function is its rate.

root system / crystallographic: A root system is the symmetric pattern of vectors behind a continuous symmetry, a Lie group. Crystallographic means the pattern fits a repeating lattice, allowing only 2, 3, 4, or 6-fold symmetry, never 5 or 7-fold.

triality: A rare extra 3-fold symmetry of $\mathfrak{so}(8)$ that swaps its vector and two spinor pieces. Here it gives the three generations of matter.

de Sitter: A universe that expands exponentially forever, driven by a cosmological constant, dark energy.

spectral dimension: Dimension as measured by how a random walk spreads or returns. It equals the ordinary dimension for clean spaces and reveals roughness when it does not.

compact / paracompact: A compact tiling has all its pieces bounded. A paracompact one has an ideal element reaching to infinity, a cusp. $\{3, 4, 3, 4\}$ is paracompact, and its cusp is physical space.

the carrying metaphors: The recurring images, the universe as a structure of consciousness, a mind, pictured as a blazing fire that grows like a tree from a seed, singing a song, flowing as a river on a deep ocean, woven as a fabric, with beings as its sparks, braids, and whirlpools.

G. Formal Mathematics

This appendix gives the theory a compact formal core. It states, in precise symbols, what the substrate is, what a fill is, how a beat acts, and what is conserved, and then records the principal emergent results as formal propositions. The body of the paper is the intuition and the evidence. This is the skeleton.

G.1. Notation, clear symbols

We avoid single Greek letters as variables, since they are easy to confuse. We use plain Latin letters and short words. The only Greek kept is the numeric constant $\pi = 3.14159\dots$, which is universal. The dictionary below also lists the conventional Greek name used elsewhere in the literature, in brackets, so the reader can connect the two.

symbol	meaning	(usual name)
M	the vibe mesh, the $\{3, 4, 3, 4\}$ honeycomb	
V	the set of cells of the mesh	
v_1, v_2	cells, the vibes, elements of V	
C	the 24 directions of a cell, the coin	
c	a direction, an element of C	
$n(v, c)$	the neighbour of v across face c	
$T = \{-1, 0, +1\}$	the tone alphabet, the three tone values	
$t(v, c)$	the tone in slot c of cell v , in T	
f	a fill, a full configuration, a map $V \times C \rightarrow T$	
Stream	the global transport map on fills	
Collide	the local interaction map on fills	
$R = \text{Collide} \circ \text{Stream}$	the rule, one beat	
B	the beat, the discrete tick	
$F = \{M, B, T, R\}$	the flow, the four base things together	
b	the beat index, in $\mathbb{Z}$	
Q	total charge, the sum of all tones	
$\text{dist}(v_1, v_2)$	graph distance, the step count	
$r(n)$	the ripple level, the number of cells at distance n from a root	
gr	the ripple growth rate	
H	the expansion rate, $\ln(\text{gr})/3$	(Hubble)
cc	the cosmological constant, $3H^2$	(Λ)
E, k, m	energy, wavenumber, mass of a mode	
wk	the weak mixing fraction, $\sin^2 \theta_W$	(θ_W)
decay	a correlator decay exponent	(α)

Table 39: The clear-symbol dictionary for this appendix.

G.2. The substrate

Definition 8 (The flow) The base of the theory is the *flow*, the tuple $F = \{M, B, T, R\}$ of the mesh M , the beat B , the tone alphabet T , and the rule R . The four are specified in turn below. Everything else in the theory is a property of F or an emergent consequence of iterating the rule R beat by beat.

Definition 9 (The mesh) The *mesh* M is the regular hyperbolic honeycomb $\{3, 4, 3, 4\}$ tiling $\mathbb{H}^4$. Its cells form a set V . Each cell is a 24-cell, a four-dimensional regular solid with 24 octahedral faces, so each cell shares each of its 24 faces with exactly one other cell. The *directions* C are the 24 face-normals of a cell, identified with the 24 roots of the root system D_4 . We write $|C| = 24$.

Definition 10 (The neighbour map and distance) There is a fixed map $n : V \times C \rightarrow V$ sending (v, c) to the cell sharing face c with v . It is an involution on the face it crosses, $n(n(v, c), \bar{c}) = v$ for the opposite direction $\bar{c}$. The mesh graph has vertex set V and an edge $\{v, n(v, c)\}$ for each v, c . Its shortest-path metric is **dist**. Distance is the only notion of space at the base, space is step-counting on this graph.

Definition 11 (Growth) Fix a root cell o . The ripple levels are $r(n) = |\{v : \text{dist}(o, v) = n\}|$. The mesh is *generated*, not given, the cell at the root exists first and each beat adds the next ripple, so the universe at beat b is the ball of radius b . This identifies radial distance with cosmic time, one ripple per beat.

G.3. The fill

Definition 12 (The fill) A *fill* is a map $f : V \times C \rightarrow T$. The value $f(v, c)$, also written $t(v, c)$, is the tone in directional slot c of cell v , read as the charge currently leaving v through face c . The set of all fills is $T^{V \times C}$. A single cell has local fill space T^C of size 3^{24} . There is no other state, in particular no value stored on edges, the relation across a face is computed from the two slots and never stored.

Definition 13 (Scalar tone and charge) The *scalar tone* of cell v in fill f is $\text{tone}(v) = \sum_{c \in C} t(v, c)$, an integer in $[-24, 24]$. The *total charge* is

$$Q(f) = \sum_{v \in V} \sum_{c \in C} t(v, c),$$

the sum of all slots over the whole mesh, an integer.

G.4. The dynamics

A beat is two maps composed, first transport, then interaction.

Definition 14 (Stream) For each cell v and direction c , streaming moves the slot value along its own line of travel into the neighbour. Writing $\bar{c}$ for the opposite direction, the streamed fill is

$$(\text{Stream } f)(v, c) = f(n(v, \bar{c}), c),$$

that is, the value now in slot c of v is the value that the back-neighbour was sending forward along line c . Each slot has exactly one source and one destination, so **Stream** is a bijection of fills, a global permutation of slots. It carries no decision, a charge keeps its direction, which is momentum.

Definition 15 (Collide) There is a fixed local map $\text{col} : T^C \rightarrow T^C$ on the 24 slots of one cell. It is applied at every cell simultaneously,

$$(\text{Collide } f)(v, \cdot) = \text{col}(f(v, \cdot)).$$

The local map col is required to be (i) a *bijection* of T^C , hence reversible, and (ii) *charge-conserving* on the cell, $\sum_c \text{col}(w)_c = \sum_c w_c$ for every local vector w . It is built from four move-types, pass-through, annihilate, create, and scatter. Its two-slot shadow is the reversible, charge-conserving permutation of the nine pair-states $(-1, -1), (+1, +1), (-1, 0), (0, -1), (+1, 0), (0, +1), (0, 0), (+1, -1), (-1, +1)$, with the create-flip-annihilate three-cycle $(0, 0) \rightarrow (+1, -1) \rightarrow (-1, +1) \rightarrow (0, 0)$ and two inert fixed points.

Definition 16 (The rule and a beat) One *beat* is the composition

$$R = \text{Collide} \circ \text{Stream} : T^{V \times C} \rightarrow T^{V \times C}.$$

The fill at beat b from initial fill f_0 is $f_b = R^b(f_0)$. Time is the iteration index b , there is no external clock.

G.5. Conservation and reversibility

Proposition 18 (Conservation of charge) For every fill f , $Q(Rf) = Q(f)$. Hence Q is constant along every trajectory.

Proof. **Stream** permutes the slot values without changing their multiset, so it preserves the global sum Q . **Collide** preserves the per-cell sum by requirement (ii), and the global sum is the sum of per-cell sums, so it too preserves Q . The composition preserves Q . Checked to machine precision in [43]. $\square$

Proposition 19 (Reversibility) R is a bijection of fills, with inverse $R^{-1} = \text{Stream}^{-1} \circ \text{Collide}^{-1}$. Hence every fill has a unique past and a unique future, and $R^{-N}(R^N(f)) = f$ exactly.

Proof. **Stream** is a bijection by construction, and **Collide** is a bijection because the local map **col** is, applied independently per cell. The composition of bijections is a bijection. Verified forward-then-back to zero error at scale in [43]. $\square$

Result 38 (The arrow) Reversibility makes create and annihilate inverse moves on one cycle. The *arrow* is the preference for the forward sense of that cycle, the create direction $(0, 0) \rightarrow (+1, -1)$, which turns a balanced peace state into a charge pair. Because Q is conserved, every created pair is net zero, a $+1$ with a -1 . The arrow is therefore the only ingredient that selects a temporal orientation, and it adds no charge.

G.6. The coin algebra

Definition 17 (The coin and its decomposition) The 24 directions C are the roots of D_4 . Under the cell-symmetry group they decompose as

$$24 = 8_v \oplus 8_s \oplus 8_c,$$

the vector representation of $\text{SO}(8)$ and its two spinor representations, permuted by the order-6 triality symmetry S_3 .

Proposition 20 (Spinor requirement forces $\{3, 4, 3, 4\}$) A collision built on C carries a half-integer-spin (spinor) representation if and only if the direction set contains a spinor summand. The $\{3, 4, 3, 4\}$ coin does, by the $8_s, 8_c$ above. The 12 directions of the 3D candidate $\{5, 3, 4\}$ decompose as $1 \oplus 3 \oplus 3' \oplus 5$, all integer spin, with no spinor. Hence among the candidate substrates, $\{3, 4, 3, 4\}$ alone supports spin at the base.

Result 39 (Odd-charge solitons are fermions) A matter soliton is a hopfion, a smooth tone-texture with a Hopf map $S^3 \rightarrow S^2$ of integer linking degree. A soliton of odd degree acquires the sign -1 under a full 2π rotation, by the Finkelstein-Rubinstein and Wilczek-Zee arguments, so it obeys Fermi statistics. The spin-statistics link is therefore geometric, not assumed. Triple-checked by linking degree, configuration-space fundamental group, and direct exchange [43].

G.7. Emergent propagation

The continuous, complex, quantum description is the coarse-grained limit of the discrete real walk, never a base ingredient. The following are measured dispersions of the rule.

Result 40 (Light cone) A pulse front advances exactly one cell per beat, a finite frame-independent maximum speed, normalized to 1. In the massless case the rule restricted to one line is the pure shift, and the measured dispersion is

$$E(k) = k \quad (\text{exact, no continuity assumed}).$$

Result 41 (Massive Dirac dispersion) Adding a scatter (a coin rotation by mass angle m) gives the measured lattice Dirac dispersion

$$\cos E = \cos m \cos k, \quad E(0) = m,$$

so rest energy equals the mass, and mass is the coarse-grained mixing rate. In the small- k , small- m limit this is the relativistic relation $E^2 = k^2 + m^2$.

G.8. Geometry, growth, and cosmology

Proposition 21 (Ripple recurrence and growth rate) The ripple levels $r(n)$ satisfy a fixed linear recurrence whose characteristic polynomial is that of the splitting matrix of the $\{3, 4, 3, 4\}$ self-similar tree. Its dominant (Perron) root is the growth rate gr . Measured ripples 1, 24, 456, 8376, $\dots$ give an asymptotic ratio $\text{gr} \approx 18.3$, while an early-level average on a truncated ball gives $\text{gr} \approx 11$.

Result 42 (de Sitter expansion) With physical-space volume growing as the cube of a scale factor a , and the cell count growing by gr per beat,

$$a(b) \sim \text{gr}^{b/3} = e^{Hb}, \quad H = \frac{\ln \text{gr}}{3},$$

so the expansion is exponential, $a''/a = H^2 > 0$, with cosmological constant $\text{cc} = 3H^2$. Using the early-level $\text{gr} \approx 11$ gives $H \approx 0.80$ and $\text{cc} \approx 1.9$ in beat units. The flat twin $\{3, 4, 3, 3\}$ grows polynomially with ratio about 2.5 and has no exponential expansion, so H is a curvature effect.

Definition 18 (The cusp) A *horosphere* is a level set of a Busemann function toward an ideal boundary point of $\mathbb{H}^4$. It is intrinsically flat. The ideal $\{4, 3, 4\}$ cubic element of $\{3, 4, 3, 4\}$ is such a horosphere, the *cusp*, combinatorially the Euclidean cubic lattice $\mathbb{Z}^3$. Physical space is the cusp. Its measured spectral dimension is 2.97, against 2.16 for a generic rough slice.

G.9. Gauge and the Standard-Model numbers

Proposition 22 (SO(10) from coin and tone) Adjoining the ternary tone sign as one extra weight axis to the $D_4 = \text{SO}(8)$ root system yields the $D_5 = \text{SO}(10)$ root system. The 16-dimensional spinor of $\text{SO}(10)$ is exactly one fermion generation, split as $16 = 8_s$ (tone +) $\oplus 8_c$ (tone -). Triality gives three generations.

Proposition 23 (Automatic breaking to the Standard Model) Any self-condensate (a Higgs in the 16) leaves unbroken the subgroup $\text{SU}(5)$ (twenty unbroken roots), and the Georgi-Glashow chain $\text{SO}(10) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ follows, with $16 = 1 \oplus 10 \oplus \bar{5}$, the complete matter content of one generation.

Result 43 (The weak mixing fraction) Summing the weak isospin and charge over one generation,

$$\text{wk} = \frac{\sum T_3^2}{\sum q^2} = \frac{3}{8} \quad \text{exactly, at the unification scale,}$$

where T_3 is the third weak-isospin component and q the electric charge over the sixteen states. Running this down with standard renormalization reaches the measured 0.231 for a supersymmetric low-energy content.

Result 44 (Mass relations) At the unification scale, the bottom and tau masses are equal, $m_b = m_\tau$, and the determinant relation $\det(M_{\text{lepton}}) = \det(M_{\text{down}})$ holds, the latter from the discrete trace identity $\text{Tr } Y = 0$ over the sixteen, where Y is the hypercharge. These are ratios, not absolute masses.

G.10. Gravity, as an equation of state

Result 45 (Newtonian potential on the cusp) The static lattice Green's function $G(r)$ of the cusp Laplacian, computed by the difference method $G(r) - G(r_0)$ to cancel the zero-mode contamination, fits

$$G(r) - G(2) = \frac{a_1}{r} + a_0, \quad a_1 \approx 0.083 \approx \frac{1}{4\pi},$$

a clean Newtonian $1/r$ in the 3D cusp.

Result 46 (Holographic boundary correlator) On the infinite tree of branching number q , the boundary-to-boundary correlator computed by the Bethe cavity recursion decays as $1/r^{\text{decay}}$ with

$$\text{decay} = \frac{2 \ln(1/w)}{\ln q}, \quad w = \frac{1}{q} \Rightarrow \text{decay} = 2,$$

a clean $1/r^2$, confirmed independently by a tree-tensor contraction at $1/r^{1.93}$.

Result 47 (The Einstein field equations) Take the area-law entropy $\text{Ent} = \text{Area}/4$ (the holographic $1/4$) and the local horizon temperature $\text{Temp} = \kappa/(2\pi)$ from the Unruh response, where κ is the surface gravity. Requiring the Clausius balance $\delta \text{Heat} = \text{Temp} \delta \text{Ent}$ across every local horizon, for all null directions, is equivalent to the full field equations

$$\text{Ein}(x) + \text{cc } g(x) = 8\pi \text{Stress}(x),$$

where Ein is the Einstein tensor, g the metric, and Stress the matter stress-energy. The coefficient 8π is forced by the area-law $1/4$ and is the same Newton constant as in the $1/r$ law. The weak field reduces to Poisson's equation $\nabla^2 \text{pot} = 4\pi \text{density}$, and a photon deflects by $4GM/x$, twice the naive Newtonian value.

Result 48 (Black-hole thermodynamics) With $\text{Ent} = \text{Area}/4 = 4\pi M^2$ and $\text{Temp} = 1/(8\pi M)$ for a mass- M horizon, the first law $dM = \text{Temp} d\text{Ent}$ holds, the Smarr identity $M = 2 \text{Temp} \text{Ent}$ holds, the Bekenstein bound is saturated, the heat capacity $dM/d\text{Temp} = -8\pi M^2$ is negative, and the evaporation lifetime scales as M^3 . The substrate's own de Sitter horizon obeys the same laws with $\text{Temp} = H/(2\pi)$ and $\text{Ent} = \pi/H^2$.

G.11. Electromagnetism

Result 49 (Discrete Gauss law and the gauge field) Per-cell charge conservation, charge in equals charge out each beat, is a discrete Gauss law, the conserved current sourcing an emergent $U(1)$. A vortex link field gives a single-plaquette flux whose Wilson loop (the sum of link phases around the plaquette) equals flux and is invariant under a gauge shift $A \rightarrow A + d\theta$, and a charge encircling it acquires exactly the phase flux (Aharonov-Bohm), while a non-enclosing loop acquires 0. The non-abelian case replaces phases by $SO(3)$ link matrices with non-commuting holonomy.

G.12. Addressing and computation

Proposition 24 (Self-describing addresses) Fix a root o and the breadth-first spanning tree of the mesh graph. Each cell v has a unique address, the sequence of child-indices on the path $o \rightarrow v$, of length $\text{dist}(o, v) = O(\log |\text{ball}|)$. Crucially the tree has *no same-level edges*, every neighbour of a cell is a parent or a child, so the parent is dropping the last digit and the children are appending an admissible digit. The only non-tree edges are confluences, a cell reachable as a child of more than one parent, and they form a finite-state transducer on a bounded window of the generator address (deterministic at window two over the verified ripples).

Result 50 (Universality) The substrate is computationally universal. (i) The signed-majority form of the collision, with a $+1$ bias and two -1 fills, computes NAND, which is functionally complete, so it builds any Boolean circuit and in particular the universal elementary rule 110. (ii) A reversible, charge-conserving register machine (increment, decrement-or-jump, halt) runs on the mesh with registers held as charge in addressed subtrees, where increment is the create move and decrement is annihilation, computing correctly with Q conserved. (iii) The cusp $\mathbb{Z}^3$ runs Conway’s Life, a strongly universal cellular automaton. So the substrate is weakly universal by (i) and (ii), and strongly universal by (iii).

G.13. The discipline, and the recorded negatives

Principle 20 (Nothing added) Every result above is a property of the four base things of the flow $F = \{M, B, T, R\}$ or an emergent consequence of them. No fifth ingredient is used. Where a phenomenon does not emerge, the negative is recorded as a formal statement, not hidden.

Result 51 (Recorded negatives) The following are recorded negatives, established by the same standard as the positives.

1. *No spontaneous selves from generic initial data.* The pure rule from a generic fill churns, a localized seed disperses (radius of gyration grows without bound, recentered pattern identity falls to zero), so durable selves do not crystallize from equilibrium initial data without seeding or topological protection.
2. *No genuine self-tower.* The coarse-grained net-charge persistence rises under coarse-graining on both the cusp and the bulk radial tree, but it does *not* beat a plain-diffusion control, so it is a generic conserved-field slow mode, not hierarchical selfhood. The correct self-measure (identity through material turnover) confirms the cusp supports propagating self-patterns while the pure rule does not make one.
3. *No scale-free avalanches.* The ballistic light cone forbids branching-process self-organized criticality, so hierarchical integration, where it occurs, comes from fast ballistic routing, not sandpile cascades.

The one named candidate for a possible fifth ingredient is higher-order binding, the clean binding of structure *between* units rather than within one. It sits behind the open holes, cross-gap attraction, super-selves, and the nesting tower. It is named, not added, and not yet proven either to emerge from the four or to be irreducible.

H. The Two Cosmic Stories

The oldest question in cosmology is whether the universe *began* or has *always been*. Vibe theory does not answer it. The substrate admits two accounts that are equally self-consistent and equally complete, and the

fair thing to do is to set them side by side and let the reader weigh them. We tell both in equal measure. This chapter is the conceptual centrepiece of the part, a fork offered as a gift rather than a verdict. The diptych comes first, then each story in full, then the perennial images that older traditions reached for again and again.

H.1. The shared ground

Before the fork there is common ground, and it is worth naming because it is the part most people find hardest to accept. There was never a *nothing*. Neither story is a story of something arising out of absolute void. The most uncomfortable idea of all, a bare nothing that somehow produces a world, appears in neither picture. Every ancient image agrees on this. There is always a prior fullness, a fire, an ocean, the One, a dreamer, a breath. The world is a spark of the fire, a wave of the ocean, a note of the song.

The reason to believe this is stronger than taste. Take the word nothing strictly, not empty space, which already has size and directions and a clock, but the total absence of any distinction at all. A state like that cannot hold, for four plain reasons. First, it is a balance point and not a floor. A uniform state with no distinctions is like a ball on the very top of a round hill, and the slightest difference rolls it off because there is no restoring force, where a floor would catch it. Second, no difference is itself a difference. The moment the state can be named as one whole, distinct from what it is not, a line has been drawn, so a perfectly seamless unity cannot even describe itself without breaking its own seam. Third, a whole that is everything has nothing else to relate to, and since to exist is at bottom to relate, it can only relate to itself, and self-relation splits it into a noting end and a noted end. That split is the first distinction. Fourth, there is no road back, because erasing a distinction is itself an event, a new distinction between before and after, so every attempt to reach nothing produces more something. Nothing is unstable, self-contradictory, forced to turn on itself, and one-way. It is the one state reality cannot be in.

This is why something is the default and nothing is the thing that needs explaining, the reverse of the usual intuition. It is also where the two stories take their cleanest forms. Read the self-relation as a first event and you have Story A, the One waking, a forced first split. Read it as what has always already been happening, with no first, and you have Story B. The arguments against nothing are shared. Whether they imply a true first distinction or an eternal one is exactly the fork, and we leave it open.

The base is discrete and computable, and computability is the referee that lets both stories stand. As the next chapter sets out, computability forbids the continuum and supertasks. It does not forbid an infinite past. So a beginningless history is no less computable than a finite one. Both stay within what an idealized machine could in principle run.

Our observed big bang fits both readings without strain. It can be the first morning of everything, the true edge of time. It can also be a single spark struck in a vast and eternal flame. Nothing in the local data of our chapter of cosmic history settles which. Our visible universe is not necessarily the whole, and the question of the whole is exactly the question we are leaving open.

H.2. The mirror symmetry

The two stories are mirror images of one another, and seeing the symmetry is most of the point. One has a first point that is a clean edge, like the North Pole of a sphere. The other has no first point at all, like a line

that runs in both directions, or the negative integers with no smallest member. Both refuse creation from nothing. The whole of cosmology’s deepest question reduces to a single fork. Is there a north-pole edge to time, or is there not.

	Story A, a beginning	Story B, beginningless
the “before” question	a real edge, no before	no edge, always a before
what the universe is	the One unfolding into the many	a verb, eternally becoming
the engine	self-noticing instability	self-sustaining churn
something from nothing	no, there was the One	no, there was always becoming
our big bang	the first morning	a spark in the eternal flame

Table 40: The two origin stories as mirror images. Each refuses creation from nothing.

H.3. Story A, the One unfolding

This is the account in which the universe truly began. In it the question *what started the universe* is not malformed. It has an answer. The universe started when the One first noticed itself. Before that there was not nothing. There was unity, the whole of what-is, seamless and undivided. The beginning is the moment that seamless one first split.

Picture a single awareness with no parts, no inside or outside, just one. Here is the engine. To be is to be aware, and to be aware is to be aware of something. The One has nothing else to be aware of, so the only thing it can notice is itself. The instant it notices itself there are two ends, the noticer and the noticed. Awareness of self is already a split. The universe is the One waking up, and waking up is becoming two. The first beat is that first noticing. It is not a thing made from nothing. It is the One differentiating into the many.

The fracture was forced, not accidental. A structure rich enough to notice itself cannot hold a single consistent state, because the self-as-noticed always differs from the self-as-noticer. This is a clean logical fact, of the same family as the diagonal argument. Unity is then like a ball balanced exactly on the tip of a peak. The balance point is real, but the slightest disturbance sends it rolling, and there is always a slightest disturbance. Unity had to fall into difference. The beginning is not a matter of why it started. It is that it could not avoid starting.

The first moment has no before, and that is not a hole. We feel that a first moment is impossible, because we keep asking what came before it. The clean answer is that *before* is a relation inside time, and time itself begins at the first moment. So there is no before, for the same reason there is nothing north of the North Pole. The North Pole is a real, definite, northernmost point. Ask what is north of it and there is no answer, not because something is hidden but because northness ends there. The first moment is the North Pole of time, and the missing before is not a gap. It is the edge itself.

Once there are two, there are relations between them to notice, and noticing those makes more, and noticing those makes more again. It runs away, like a microphone held to its own speaker, each output becoming new input until the room is full of sound. The first tiny split feeds the roaring, branching, growing structure that is the universe. From one forced fracture the whole mesh unfolds, beat by beat, the many blooming out of the one.

The finite seed is also the natural place to start thinking, even for a reader who ends up an eternalist. Assuming a first seed makes the picture concrete. You can watch the universe build itself from nothing but the rule, step by step, and only then ask whether the first step was truly first. Run forward from the seed and a clean mechanism appears, which we may call birth at peace. Each beat, reflection adds a new ring of cells at the growing frontier, and every newborn cell starts at tone 0, peace. This is forced twice over. Conservation forbids a nonzero birth, since a $+1$ or -1 cell would add charge and break the balance, and peace is the natural blank value anyway. The newborn then acquires its tone from the world, as the conserved charge of the existing crystal flows into it along its connections, zero-sum, so the total never changes. The frontier is a blank peaceful canvas, and the world paints it with charge it already had. The genuine freedom at the frontier is freedom of relation and of redistribution, never freedom to mint charge. This same mechanism works identically on the eternal reading, where every newborn shell is born at peace whether or not there was a first one.

No single image carries the story alone, because each one smuggles in a prior context. The One waking up suggests a prior sleeper, which the North Pole image removes by denying any before. The ball on the peak seems to assume a ball and a gravity set up in advance, which dissolves once it is read as the logical instability of self-reference rather than a mechanical picture. The North Pole image itself has no hole, being a clean geometric fact about a sphere. The One unfolding into the many makes plain that this is not something from nothing, since there was always the One. Fused into one sentence, the story runs like this. Unity is a ball on the tip of a peak that cannot help but roll, the One waking into two, and the moment it wakes is the North Pole of time, with no before because before-ness begins there, and from that one forced fall the many bloom forever.

H.4. Story B, the eternal becoming

This is the account in which the universe never began. The question *what started the universe* is still an excellent question, and following it carefully leads somewhere subtle. Starting is something that happens inside time. A song starts, a race starts, a fire starts, and every start you have ever seen had a before, a moment of quiet before the song, a held breath before the race. Start means something only when there is an earlier time for the starting to happen in. To ask what started time itself is to ask for a before-the-first-before, and that is where the question quietly looks for an earlier time when there may be none. The deepest answer is not a clever first cause. It is that time may not need a starting line at all, because a starting line is itself an in-time idea.

The universe is a verb, not a noun. We are used to things being made and then sitting there existing, a table built, a star born. So we imagine the universe the same way, made once and then existing. But the universe is not a table. It is a happening, like a flame, like a river, like a song being sung right now. A flame is not made and then existing. It is continuously burning. Stop the burning and there is no flame. A verb does not need a maker. It needs only to keep happening.

Here is the heart of it. In this universe nothing is ever lost, because the rule runs backward as perfectly as forward. Every now carries its before inside it, fully. The present is the past transformed, not erased. Take any moment and ask what came before, and there is always a real, definite answer, because the moment literally remembers it. That moment remembers its before, and so on, with no bottom, the way the counting

numbers have no smallest one. There is never a now that came from nothing, because every now is the transformation of a real before.

On this reading the strange option is a beginning, not the lack of one. Look at what a real beginning would require. At a single instant the entire universe leaps out of absolute nothing, uncaused, for no prior reason, and then keeps going. That is the gut-wrenching, impossible-feeling thing, something from nothing, a causeless flash. The beginningless picture is gentler. Nothing ever came from nothing, because there was never a nothing to come from. Existence did not have to be switched on.

What, then, is the big bang. Our visible universe really does seem to begin about 14 billion years ago. But that need not be the beginning of everything. It can be the beginning of our chapter, a bubble forming, a new region catching fire in the endless burning. Page one of our story, not page one of the book. There were always pages before ours, and the book has no first page. Both are true at once. Our local universe began, and the whole was always going. The big bang is a sunrise, not the creation of the sun.

Three metaphors plug each other's holes here. The eternal flame says that the universe is a happening, not a thing, but real flames are lit and go out. Counting backward says there is no first and no one finds that strange, which plugs the flame's was-lit hole, but numbers are static. The reversible film says every moment is the exact reshaping of the one before, nothing lost, a film that plays the same backward as forward so that from any frame both neighbours can be rebuilt. This is literally true of the rule, and it is also the mirror image moved from frozen space into flowing time. Fused into one sentence, the universe is an eternal flame whose every flicker is the exact reshaping of the flicker before it, nothing ever lost, with no first flicker, the way counting backward has no smallest number.

The reversible film is not only a metaphor. It was tested. Start from random noise rather than a hand-picked state, run the rule forward 50 steps, then backward 50, and the exact starting state returns with zero error. The activity holds steady, around two-thirds of cells changing each tick, for hundreds of ticks. It never freezes and never blows up. The decisive part is the control. An irreversible blurring rule from the same random start decays to a frozen dead state within about 200 ticks. A reversible churn can run forever with no start and no end. An irreversible one cannot. This shows the eternal picture is fully self-consistent. It does not prove the universe is eternal, only that it is allowed [43].

One real subtlety remains. Because the rule is time-symmetric, the arrow of time cannot come from the rule itself. It must come from the boundary or the initial conditions, the so-called past hypothesis. The reversible rule inherits this famous feature, so we do not claim to have produced time's arrow out of nothing. There is a clarifying payoff in the growth picture all the same. The single growth of one new shell per beat is at once the time step, the arrow, and the spatial expansion, three usually separate things made one. Cosmic expansion is not inserted by hand. Time is the growing of space, in the form of a de Sitter law $a(t) \sim e^{Ht}$ with $H \approx 0.80$ per beat. This holds on either origin reading.

The beginningless story stays computable because at any flicker only finitely much has burned. There is always more before and after, but never a completed whole. The history is endless but stepwise, never grasped all at once. So an infinite past is allowed while the continuum and supertasks are not.

H.5. The perennial images

Humans have always reached for the same images when they picture the local world as a part of a vast or eternal whole, and the One becoming the many. These pictures resonate because they are old and deep, and the structures they describe turn out to match what the substrate produces. The deepest shared point in all of them is that none is creation from nothing. Each carries a prior fullness, and the world is a part of it.

The fire images run across both stories. A spark in the flame is the ever-living fire of Heraclitus, the Stoic logos, the divine spark of the Gnostics and of Eckhart, and it serves the beginningless reading. A flame lit from a flame, one candle lighting another undiminished, is the Buddhist image for passing on without loss, and it serves both, since the new fire is real and begins yet nothing is created. The first struck spark, the chord blooming from a single note, serves the beginning.

The water images run mostly to the eternal whole. A wave on the ocean is the Vedanta image, each wave rising and falling yet being the ocean. A bubble or foam on the sea is the Diamond Sutra's dream, bubble, dewdrop, and flash of lightning. A drop returning to the ocean is the dewdrop of Rumi slipping into the shining sea. A whirlpool in the river is the self as a pattern in the flow rather than a substance, a standing eddy the water passes through. Ripples from a stone, the first disturbance spreading, serve the beginning.

The light images divide cleanly. A ray of the one sun is the emanation of Plotinus, the One radiating outward without diminishing, and it serves both, since emanation has a source yet the source is eternal. The moon reflected in countless waters is the Zen image of one moon and numberless whole reflections, the one appearing as the many without splitting. A sunbeam through a window is a shaft of the one light made local.

The mind and dream images carry the same double sense. The dream of Brahma is the cosmos as the dreaming of the divine, worlds arising and dissolving as a dreamer dreams, serving both since the dream begins and ends while the dreamer is eternal. Indra's Net is an infinite net with a jewel at every knot, each reflecting all the others, the part containing the whole, and this is also the literal mutually reflecting structure of the bulk. A thought in the cosmic mind is each world arising in an endless mind.

The remaining families fill out the picture. The breath of Brahma is the universe breathing in and out across vast ages, beginningless and cyclic, with each exhale a local beginning. The seed becoming the tree is the Upanishadic acorn and oak, the One containing and unfolding into the many. A note in the eternal song is the Pythagorean music of the spheres. The Word, the Om, the Logos, is the cosmos as an utterance, a first sounding from an eternal speaker. From number and geometry come three precise images. A number in the endless count has no smallest, the clean picture of a beginningless past. A point on an endless line has no first point. The North Pole of the sphere is the clean picture of a first moment with no before. From weaving comes the thread in an endless woven cloth, the Norns at the loom of time, each life a thread, which is also the literal fabric image of a self as a braid.

The beginningless story is served by the eternal-whole images, the wave and bubble and drop in the ocean, the spark in the flame, the number with no smallest, Indra's net, the breath of Brahma. The beginning story is served by the unfolding images, the seed becoming the tree, the One radiating, the Word sounding, the North Pole of time. The shared and deepest point in all of them is that neither is creation from nothing. Every ancient image has a prior fullness, the world a spark of the fire, a wave of the ocean, a note of the song. That is what the theory recovers. Existence is a fullness expressing itself, never a void producing a thing.

H.6. Why we do not decide

Nothing currently breaks the tie. Computability allows both. The reversible rule forces neither. The bootstrap experiment shows the beginningless churn is fully self-consistent, while the finite-seed account derives a forced first fracture, and neither is ruled out. Observation fits both. This is a genuine open question about the deepest structure of time, and forcing a verdict would be less faithful and less interesting than presenting the fork cleanly. The same carrying fire handles both. Was the fire always burning, or did it first flare. The fire is the same fire either way.

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